

Disorders of Acid-Base Balance

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Acid-base homeostasis is critical for normal health. However, a variety of both acute and chronic conditions commonly alter acid-base homeostasis and blood pH. This chapter will review normal acid-base homeostasis and various conditions causing disorders. The approach to acid-base homeostasis in this chapter uses measurements of arterial pH, PCO_2 , and $[\text{HCO}_3^-]$ plus an analysis of the electrolytes and anion gap. This is the most widely used and generally accepted approach used clinically by nephrologists, and the easiest model to understand.¹ Other approaches, such as that using “base excess” or “strong ion difference”, are used by some groups but will not be used here.

ACID-BASE HOMEOSTASIS

Acid-base homeostasis normally maintains systemic arterial pH within a narrow range, certainly between 7.35 and 7.45 pH units. This tight regulation is accomplished through 1) chemical buffering in each of the body fluid compartments and 2) control of CO_2 and HCO_3^- by the lungs and kidneys, respectively. These processes efficiently dispose of the physiologic daily load of acid in the forms of CO_2 (forming carbonic acid H_2CO_3) and nonvolatile acids, mainly derived from metabolism and dietary protein intake, and defend against the occasional addition of pathologic quantities of acid and alkali.

The intracellular H^+ concentration ($[\text{H}^+]_i$), or pH_i , is also regulated at a relatively stable value, usually less than plasma. Both cellular ion exchange mechanisms and intracellular buffers (hemoglobin, tissue proteins, organophosphate complexes, and bone apatite) participate in the blunting of changes in both $[\text{H}^+]_i$ and $[\text{H}^+]_e$. Extracellular and intracellular buffers provide the *first line of defense* against the addition of acid or base to the body. As a brief review,

the relationship between an undissociated acid (HA) and its conjugate, disassociated base (A^-) may be represented as follows:



Weak acids and their conjugate bases, and also weak bases and their conjugate acids, represent buffer pairs that are most effective when the pair are at the same concentration, e.g., $\text{HA} = \text{A}^-$; this occurs at a certain H^+ concentration or pH ($\text{pH} = -\log_{10}[\text{H}^+]$). This H^+ concentration also equals the equilibrium constant K_a of the reaction above:

$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]} \quad (16.2)$$

Buffer systems are critical to the physiology and pathophysiology of acid-base homeostasis because they attenuate the pH change in a solution or tissue by reversibly combining with or releasing H^+ . Thus, the pH change of a solution during the addition of acid or base equivalents is smaller in the presence of a buffer system than would have occurred if no buffer systems were present. The major buffer system in the body is the $\text{CO}_2/\text{HCO}_3^-$ system represented by the reversible reactions below, catalyzed by carbonic anhydrase:



This system is effective because of its abundance and especially because CO_2 and HCO_3^- can be independently modified by the lungs and kidneys, respectively. Note that because the reaction is reversible, changes in $[\text{H}^+]$ or pH can result from either an addition of an acid, e.g. H^+ , or loss of $[\text{HCO}_3^-]$ (or base), which would “pull” the reaction to the right. Physiologically, for pH considerations, gain of an acid is equivalent to the loss of a base or alkali.

Because the concentration of H_2CO_3 is low and proportional to the concentration of dissolved CO_2 , one can condense the equations above to:



and then derive:

$$K = \frac{[\text{H}^+][\text{HCO}_3^-]}{[\text{CO}_2][\text{H}_2\text{O}]} \quad (16.5)$$

The familiar Henderson-Hasselbach equation then can be derived and results when the solubility of CO_2 is considered (dissolved CO_2 in mmol/L = $0.03 \times \text{PCO}_2$ in mm Hg) and the pK of 6.1 is used:

$$\text{pH} = \text{pK}' + \log_{10} \frac{[\text{HCO}_3^-]}{\alpha_{\text{CO}_2} \text{PCO}_2} \quad (16.6)$$

Note that this equation illustrates that pH is a complex function of the ratio of HCO_3^- concentration to PCO_2 , and therefore pH can be altered by changes in either HCO_3^- or PCO_2 . However, if the HCO_3^- or PCO_2 changes, the ratio and pH can be returned toward or to normal by a similarly directed change in the other. For instance, if HCO_3^- is decreased by 50%, pH decreases; but then if PCO_2 decreases by 50% then pH is returned to normal.

If buffers, including the $\text{HCO}_3^-/\text{CO}_2$ system, are conceptualized as the first line of immediate defense of pH, the second line of defense can be conceptualized as the respiratory system. Pulmonary participation in acid-base homeostasis relies on the excretion of CO_2 by the lungs. Large amounts of CO_2 accumulate as metabolic end products of tissue metabolism (10 to 12 mmol/day). This CO_2 load is transported in the blood to the lungs as hemoglobin-generated HCO_3^- and hemoglobin-bound carbamino groups.² Because, under most circumstances, CO_2 excretion and CO_2 production are matched, the usual steady-state arterial PCO_2 , Paco_2 , is maintained at 40 mm Hg. Underexcretion of CO_2 produces hypercapnia, and overexcretion produces hypocapnia. Paco_2 is regulated primarily by neurorespiratory factors and is not subject to regulation by the rate of metabolic CO_2 production. Increases or decreases in PCO_2 represent derangements of control of neurorespiratory regulation or can result from compensatory changes in response to a primary alteration in the plasma HCO_3^- concentration. The respiratory response to acidemia or alkalemia blunts the change in blood pH that would occur otherwise. Such respiratory alterations that adjust blood pH toward normal are referred to as *secondary* or *compensatory* alterations, because they occur in response to primary metabolic changes (or changes in HCO_3^- discussed below). Ventilation can change within minutes of systemic pH changes and compensate for acid or base loads by adjustments in Paco_2 . Complete respiratory compensation can take hours. Primary changes in Paco_2 can also occur and result in acidosis or alkalosis, depending on whether CO_2 is elevated above or depressed below the normal value: 40 mm Hg. Such disorders are termed *respiratory acidosis* and *respiratory alkalosis*, respectively.

The third component of pH protection can be conceptualized as the plasma HCO_3^- concentration, which can be modulated by both metabolic and renal regulatory mechanisms. When discussed as a protective mechanism, renal induced changes in HCO_3^- can compensate for primary respiratory acid-base disorders (primary changes in PCO_2) or in response

to a “metabolic” acid or base load. “Metabolic” here refers to “non-volatile” acids or bases, namely non- CO_2 acids. As discussed below, the kidneys can regulate plasma HCO_3^- .

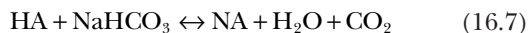
Metabolism generates a daily load of relatively strong acids (lactate, citrate, acetate, and pyruvate), which must be removed by other metabolic reactions. The oxidation of these organic acids in the Krebs cycle, for example, generates CO_2 , which must be excreted by the lungs. Nevertheless, the complete combustion of carbon involves the intermediate generation and metabolism of relatively strong organic acids, such as lactic acids, tricarboxylic acids, ketoacids, or other acids, depending on the type of fuel consumed. These organic acids do not accumulate in the body under most circumstances, with concentrations remaining in the low millimolar range. If production and consumption rates become mismatched, however, these organic acids can accumulate (e.g., lactic acid accumulation with strenuous exertion). Correspondingly, the HCO_3^- in the extracellular fluid (ECF) will decline as the organic acid concentration increases. During recovery, the organic acids reenter metabolic pathways to CO_2 production, removal of H^+ , and generation of HCO_3^- . Nevertheless, if the organic anions are excreted (e.g., ketonuria), these entities are no longer available for regeneration of HCO_3^- . Considered in this way, metabolizable organic anion may be viewed as “potential bicarbonate”. The metabolism of some body constituents such as proteins, nucleic acids, and small fractions of lipids and certain carbohydrates generate specific organic acids that cannot be burned to CO_2 (e.g., uric, oxalic, glucuronic, hippuric acids). In addition, the inorganic acids H_2SO_4 and H_3PO_4 , derived respectively from sulfur-containing dietary amino acids and organophosphates, must be excreted by the kidneys or the gastrointestinal tract. Persons eating a typical Western diet have a daily acid challenge, typically 0.5–1 mEq/kg body weight. The amount of net nonvolatile acid produced by diet and metabolism is defined as *endogenous acid production*. Net endogenous acid production is, therefore, dependent in part on diet, especially the relative amount of animal-derived dietary protein.³

Metabolic *regulation* of HCO_3^- is of minor importance in terms of overall physiologic regulation of acid-base balance. Nevertheless, regulatory enzymes, whose activity may be pH sensitive, may catalyze metabolic reactions that either generate or consume organic acids. Such a process constitutes a negative feedback regulatory system. The best example is phosphofructokinase, the pivotal enzyme in the glycolytic pathway. Phosphofructokinase is a kinase enzyme that phosphorylates fructose 6-phosphate in glycolysis. The activity of phosphofructokinase is inhibited by low pH and enhanced by high pH. Thus, an increase in pH_i accelerates glycolysis and generates pyruvate and lactate. It follows, therefore, that the generation of lactic acid in patients with lactic acidosis, and the generation of ketoacids in patients with ketoacidosis, are impeded by acidemia. Conversely, generation of lactic acid in patients with lactic acidosis, and the generation of ketoacids in patients with ketoacidosis, are augmented by reversal of acidemia.

RENAL REGULATION

Although temporary relief from changes in the pH of body fluids may be accomplished by chemical buffering or

respiratory compensation, the ultimate defense against the addition of nonvolatile acid or of alkali is the responsibility of the kidneys. The addition of a strong acid (HA) to the ECF titrates or consumes plasma HCO_3^- :



The CO_2 is expired by the lungs, and body HCO_3^- buffer stores are diminished. This process occurs constantly as endogenous metabolic acids are generated. To maintain a normal plasma HCO_3^- in the face of constant production of metabolic acids, predominantly as a result of dietary protein metabolism, the kidneys must 1) reabsorb virtually all of the HCO_3^- present in glomerular filtrate, and 2) regenerate the HCO_3^- consumed by reaction with metabolic acids (Equation 16.7). The first process (HCO_3^- reabsorption) is accomplished predominantly in the proximal tubule, with an additional contribution by the loop of Henle and a lesser, but critical, contribution by more distal nephron segments (Fig. 16.1). Under most circumstances, the filtered load of HCO_3^- is absorbed almost completely, especially during an acid load. "Acid production" in biologic systems is represented by the milliequivalents (mEq) of protons (H^+) added to body fluids. Ordinarily on a Western diet, especially animal-based proteins, there is a net daily endogenous acid load which consumes HCO_3^- . Conversely, base addition to the extracellular fluid via the GI tract is derived primarily from dietary fruits and vegetables. If less acid is generated or when, in the face of an alkali load, the plasma HCO_3^- concentration increases above the normal value of 25 mEq/L, HCO_3^- will be excreted into the urine. Therefore, the kidney must efficiently excrete any excess in alkali added to the ECF as well as regenerate the bicarbonate lost when net endogenous acid production is significant. The difference between endogenous acid production and the input of alkali absorbed by the gastrointestinal

system, i.e. the difference in acid production and base generation, is known as *net endogenous acid production*. Since a Western diet is high in animal-based protein, net endogenous production is positive and consumes bicarbonate; therefore, the kidney must regenerate the bicarbonate consumed by dietary protein intake.

The process of HCO_3^- regeneration, is represented by the renal output of acid or *net acid excretion*:

$$\text{Net Acid Excretion} = \text{NH}_4^+ + \text{Titrateable Acid} - \text{HCO}_3^- \quad (16.8)$$

The mechanisms of net acid excretion are discussed below, but include two components: titrateable acid and ammonium NH_4^+ (Fig. 16.2). Titrateable acid (TA) is the acid added to tubular fluid along the nephron to lower fluid pH to that of the final urine. The final pH of urine will depend on the amount of urinary buffer, principally phosphate, and the amount of acid added. TA can be measured by titrating urine (after removing ammonia) back to plasma pH or can be calculated using the urine and plasma pH and the buffer concentration (as a first approximation phosphate) and its pKa. NH_4^+ represents acid excretion since, as described in another chapter, for each NH_4^+ produced and excreted the carbon skeleton of glutamine from which it derives has been converted to HCO_3^- (Fig. 16.3).

On balance, each milliequivalent of net acid excreted corresponds to 1 mEq of HCO_3^- returned to the ECF (i.e., regeneration of HCO_3^-). This process of HCO_3^- regeneration is necessary to replace the HCO_3^- lost by the endogenous acid entry into the ECF or, less commonly, the HCO_3^- excreted in stool or urine. Because a typical Western diet, rich in animal-based proteins, generates fixed acids at 35 to 70 mEq/day, net acid excretion must approximate 35 to 70 mEq/day to match net acid production, and avoid metabolic acidosis.

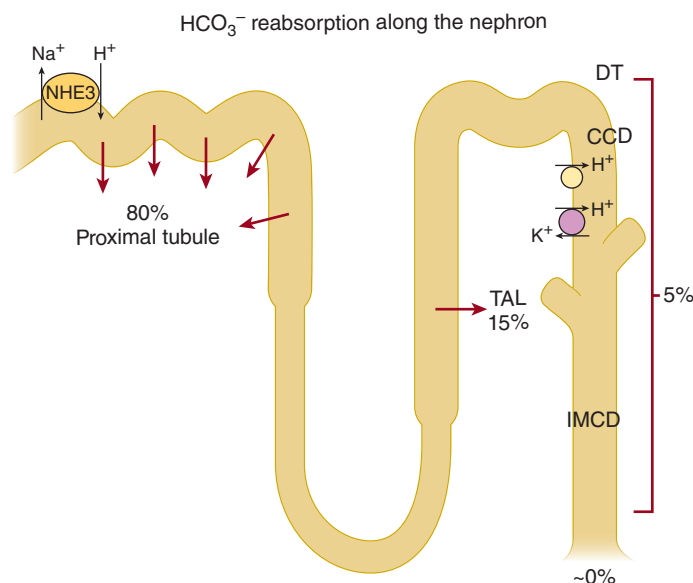


Fig. 16.1 HCO_3^- reabsorption along the nephron. The proximal tubule reabsorbs approximately 80% of the filtered HCO_3^- . Substantial amounts are also reabsorbed in the thick ascending limb. The remaining amounts (~ 5%) are reabsorbed in the collecting duct, which is critical for final regulation of acid excretion. The mechanisms are described in another chapter but the main apical transporters (Na-H exchange and H^+ -ATPase) are illustrated. Little HCO_3^- (~0 %) remains in the final urine.

Relative roles of titratable acid and ammonia in net acid excretion

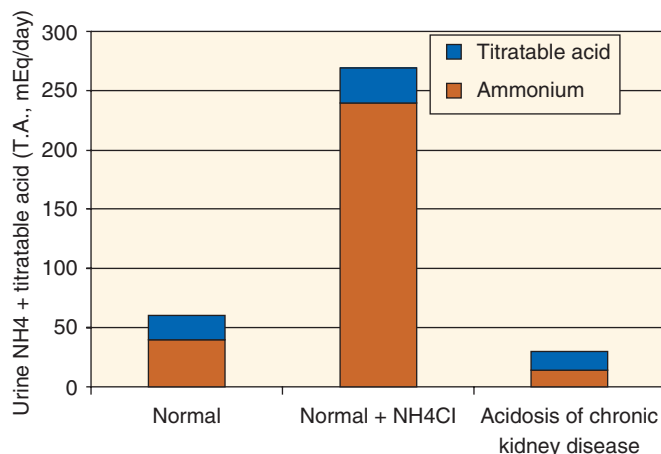


Fig. 16.2 Net acid excretion with typical Western diets, with acid loading, and with acidosis in chronic kidney disease (CKD). Ammonium excretion has the capability for the most increase in acidosis, and is also the component that is most compromised in CKD. T.A., Urine titratable acidity. (From Gauthier P, Simon EE, Lemann J, Jr. Acidosis of Chronic Renal Failure. In: DuBose TD, Hamm LL, editors. *Acid-Base And Electrolyte Disorders*. Philadelphia: Saunders; 2002. pp. 207-216.)

SYSTEMIC RESPONSE TO RESPIRATORY ACIDOSIS OR ALKALOSIS

ACUTE RESPONSE: GENERATION OF RESPIRATORY ACIDOSIS OR ALKALOSIS

Some conditions independent of pH stimulate a primary increase in ventilation, which lowers systemic PaCO_2 , and thus raises pH – respiratory alkalosis. These conditions include hypoxemia, fever, anxiety, central nervous system disease, acute cardiopulmonary processes, septicemia, liver failure, pregnancy, and drugs (e.g., salicylates).⁴ Conversely, PaCO_2 increases if the respiratory system is depressed by suppression of the respiratory control center or of the respiratory apparatus itself (neuromuscular, parenchymal, and airway components⁵) – pH is lowered, respiratory acidosis. In both kinds of acute respiratory disorders, CO_2 is added to or subtracted from the body until the PaCO_2 assumes a new steady state so that pulmonary CO_2 excretion equals CO_2 production. The acute accumulation or loss of CO_2 causes changes in blood pH within minutes. The plasma HCO_3^- decreases slightly as the PaCO_2 is reduced in acute respiratory alkalosis and increases slightly in acute respiratory acidosis.^{2,4-6} The small changes in HCO_3^- concentration are due to buffering by nonbicarbonate buffers.^{2,4-6} The estimated change in blood HCO_3^- concentration is approximately equal to 0.1 mEq/L of $[\text{HCO}_3^-]$ increase for each millimeter of mercury (mm Hg) increase in PCO_2 and 0.25 mEq/L decrease for each mm Hg decrease in PCO_2 .⁵ Acute alterations in PCO_2 in either direction within the physiologic range do not acutely

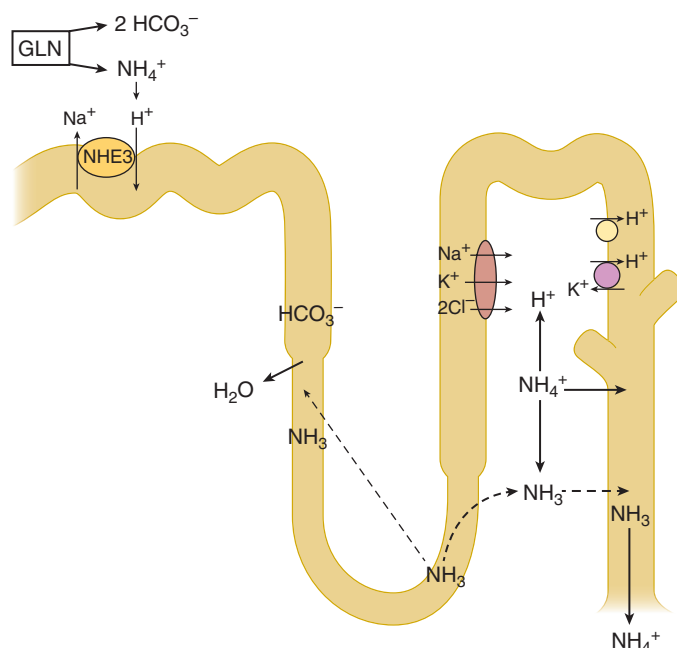


Fig. 16.3 Synchrony of regulation of ammonium production (from glutamine [GLN] precursors, and excretion). Process allows generation of “new” HCO_3^- by the kidney. NH_4^+ excretion is regulated in response to changes in systemic acid-base and K^+ balance. Contributing segments include the proximal convoluted tubule, proximal straight tubule, thin descending limb, thick ascending limb, and medullary collecting duct. Upregulated by acidosis and hypokalemia. Inhibited by hyperkalemia.

change the blood HCO_3^- concentration by more than a total of about 4 to 5 mEq/L from normal.

CHRONIC RESPONSE

Although the blood pH is relatively poorly defended during acute changes in PaCO_2 , during chronic changes, the kidneys excrete or retain HCO_3^- and return blood pH toward normal. The persistence of hypocapnia reduces renal HCO_3^- absorption to achieve a further decrease in the plasma HCO_3^- concentration. Hypocapnia decreases renal HCO_3^- reabsorption⁴ by inhibiting acidification in both the proximal⁶ and the distal nephrons. The resulting decrease in plasma HCO_3^- concentration is equal to about 0.4 to 0.5 mEq/L for each mm Hg decrease in PCO_2 .⁶ Thus the arterial pH falls toward but not completely back to normal.

With chronic hypocapnia, several hours to days are required for full expression of the renal response,^{5,6} which includes a reduction in the rate of H^+ secretion, an increase in urine pH, a decrease in NH_4^+ and titratable acid excretion, and a modest bicarbonaturia. An increase in blood Cl^- concentration occurs simultaneously by means of several mechanisms: a shift of Cl^- out of red blood cells, ECF volume contraction, and enhanced Cl^- reabsorption.

During chronic hypercapnia, the increase in generation of HCO_3^- by the kidney takes several days for completion. The mechanism of HCO_3^- retention involves increased H^+ secretion by both proximal and distal nephron segments, regardless of sodium bicarbonate or sodium chloride intake, mineralocorticoid levels, or K^+ depletion.^{2,5-7} Chronic hypercapnia results in sustained increases in renal cortical PCO_2 , and the increase in renal cortical PCO_2 that occurs with chronic hypercapnia stimulates acidification.⁶ The increased PCO_2 enhances distal H^+ secretion so that increased NH_4^+ excretion occurs even with a low-salt diet or with hypoxemia. These responses are blunted in AKI or chronic kidney disease (CKD). For instance, if hyperkalemia ensues or is present initially, the renal adaptation to chronic hypercapnia is blunted significantly.⁷ With time, an adaptation occurs in the proximal nephron: HCO_3^- reabsorption is stimulated after several days of hypercapnia.⁸ In summary, although primary alterations in systemic PaCO_2 cause relatively marked changes in blood pH, renal homeostatic mechanisms allow the blood pH to return toward normal over a sufficient period. The renal response to chronic hypercapnia is manifest primarily by an increase in net acid excretion and HCO_3^- absorption, which is accomplished by augmented H^+ secretion in both proximal and distal nephron segments.⁹ In chronic hypercapnia, the blood HCO_3^- concentration increases about 0.25 to 0.50 mEq/L for each mm Hg elevation in PaCO_2 .^{5,6}

SYSTEMIC RESPONSE TO ADDITION OF METABOLIC ACIDS

In addition to generating large quantities of CO_2 , the metabolic processes of the body produce a smaller quantity of nonvolatile acids or metabolic acids. The lungs readily excrete CO_2 , and this process can respond rapidly to changes in production. In contrast, the kidneys must excrete nonvolatile acids through a much slower adaptive response, taking a few days for full effect.

SOURCES OF ENDOGENOUS ACIDS

Pathologically, acid loads may be derived from endogenous acid production (e.g., generation of ketoacids and lactic acids) or loss of base (e.g., diarrhea) or from exogenous sources (e.g., ammonium chloride or toxin ingestion). Under normal physiologic circumstances, a daily input of acid derived from the diet and metabolism confronts the body with an acid challenge. The net result of these processes amounts to the entry of about 0.5 to 1.0 mEq of new H^+ per kilogram per day into the ECF.^{2,6}

Sulfuric acid is formed when organic sulfur from methionine and cysteine residues of proteins are oxidized to SO_4^{2-} . The metabolism of sulfur-containing amino acids is the primary source of acid in the usual Western diet, especially animal-derived proteins, accounting for approximately 50%. The quantity of sulfuric acid generated is equal to the SO_4^{2-} excreted in the urine.

Organic acids are derived from intermediary metabolites formed by partial combustion of dietary carbohydrates, fats, and proteins as well as from nucleic acids (uric acid). Organic acid generation contributes to net endogenous acid production when the conjugate bases are excreted in the urine as organic anions. If full oxidation of these acids can occur, however, H^+ is reclaimed and eliminated as CO_2 and water. The net amount of H^+ added to the body from this source can be estimated by the quantity of organic anions excreted in the urine.

Phosphoric acid can be derived from hydrolysis of PO_4^{3-} esters in proteins and nucleic acids if it is not neutralized by mineral cations (e.g., Na^+ , K^+ , and Mg^{2+}). The contribution of dietary phosphates to acid production is dependent on the kind of protein ingested. Some proteins generate phosphoric acid, whereas others generate only neutral phosphate salts.^{2,6}

Potential sources of bases are also found in the diet (e.g., acetate, lactate, citrate), primarily from fruits and vegetables, and can be absorbed to neutralize partially the H^+ loads from the three sources just mentioned. These potential base equivalents may be estimated by subtracting the unmeasured anions in the stool ($\text{Na}^+ + \text{K}^+ + \text{Ca}^{2+} + \text{Mg}^{2+} - \text{Cl}^- - 1.8 \text{ P}$) from those measured in the diet. The net base absorbed by the gastrointestinal tract is derived from the anion gap (AG) of the diet minus that of the stool.

In summary, diets contain many sources of acids and bases. The usual North American diet, which is replete in animal-derived proteins, represents a daily source of acid generation for which the body must compensate constantly.

HEPATIC AND RENAL ROLES IN ACID-BASE HOMEOSTASIS

The generation of acid by protein catabolism is balanced by the generation of new HCO_3^- through renal NH_4^+ and titratable acid excretion (or, in sum, net acid excretion). Hepatic catabolism of proteins, with the exception of sulfur- and PO_4^{3-} -containing amino acids, can be considered a neutral process. The products of these neutral reactions are HCO_3^- and NH_4^+ . Most of the NH_4^+ produced by metabolism of amino acids reacts with HCO_3^- and forms urea and thus has no impact on acid-base balance. A portion of this NH_4^+ is diverted to glutamine synthesis, the amount of which is

regulated by pH. Acidemia promotes and alkalemia inhibits glutamine synthesis. Glutamine enters the circulation and reaches the kidney, where it is deaminated to form glutamate initially; glutamate can also be deaminated, releasing another NH_4^+ and forming α -ketoglutarate. Renal glutamine deamination results in NH_4^+ production and initiates a metabolic process that generates new HCO_3^- through α -ketoglutarate. Glutamine deamination in the kidney is also highly regulated by systemic pH, so that acidemia augments and alkalemia inhibits NH_4^+ and HCO_3^- production. The ultimate control, however, resides in the renal excretion of NH_4^+ , because the NH_4^+ must be excreted to escape entry into the hepatic urea synthetic pool. Hepatic urea synthesis would negate the new HCO_3^- realized from α -ketoglutarate in the kidney. Hepatic regulation of NH_4^+ metabolic pathways appears to facilitate glutamine production when NH_4^+ excretion is stimulated by acidemia or, conversely, blunts glutamine production when excretion is inhibited by alkalemia.¹⁰

RESPIRATORY RESPONSE TO ACIDEMIA

A critically important response to an acid load is the control of ventilation. A fall in systemic arterial pH is sensed by the chemoreceptors that stimulate ventilation and, therefore, reduce PaCO_2 . The fall in blood pH that would otherwise occur in uncompensated metabolic acidosis is therefore blunted. The pH is not restored to normal; however, with normal lungs PaCO_2 declines by an average of 1.25 mm Hg for each 1.0 mEq/L drop in HCO_3^- concentration. The clinical features are hyperpnea (increased tidal volume) and tachypnea (increased respiratory rate), which is termed Kussmaul respiration. The appropriate PaCO_2 in steady-state metabolic acidosis can be estimated from the

prevailing HCO_3^- concentration according to the following expression¹¹:

$$\text{PaCO}_2 = 1.5 [\text{HCO}_3^-] + 8 (\pm 2 \text{ mm Hg}) \quad (16.9)$$

Other formulas to predict respiratory compensation are also available. Because the PaCO_2 cannot fall below about 10 to 12 mm Hg, the blood pH is less well defended by respiration after very large reductions in the plasma HCO_3^- concentration (Table 16.1). Approximately 12 to 24 hours is required to achieve full respiratory compensation for metabolic acidosis (see Fig. 16.4).

RENAL EXCRETION

As already discussed, the kidneys eliminate the acid that is produced daily by metabolism and diet and have the capacity to increase urinary net acid excretion (and, hence, HCO_3^- generation) in response to endogenous or exogenous acid loads. Acidosis enhances proximal HCO_3^- absorption, decreasing delivery of HCO_3^- out of the proximal tubule, and enhances distal acidification. The enhanced distal acidification serves to reabsorb any remaining HCO_3^- and facilitate titratable acid excretion and NH_4^+ excretion. Increased HCO_3^- reabsorption per se in the proximal tubule does not facilitate acid excretion since little HCO_3^- is excreted into the urine even in normal circumstances. Net acid excretion is increased by stimulation of NH_4^+ production and excretion. Renal excretion of acid is usually matched to the net production of metabolic and dietary acids, often about 35 to 70 mEq/day, so little disturbance in systemic pH or HCO_3^- concentration occurs.

As an acid load is incurred, the kidneys respond to restore balance primarily by increasing NH_4^+ excretion

Table 16.1 Acid-Base Abnormalities and Appropriate Compensatory Responses for Simple Disorders

Primary Acid-Base Disorders	Primary Defect	Effect on pH	Compensatory Response	Expected Range of Compensation	Limits of Compensation
Respiratory acidosis	$\uparrow \text{PCO}_2$ Alveolar hypoventilation	$\downarrow$	$\text{HCO}_3^- \uparrow$ Chronic: $\uparrow$ Renal reabsorption & production HCO_3^-	Acute $\Delta [\text{HCO}_3^-] = +1 \text{ mEq/L}$ for each $\uparrow \Delta \text{PCO}_2$ of 10 mm Hg Chronic $\Delta [\text{HCO}_3^-] = +4 \text{ mEq/L}$ for each $\uparrow \Delta \text{PCO}_2$ of 10 mm Hg	Acute $[\text{HCO}_3^-] = 38 \text{ mEq/L}$ Chronic $[\text{HCO}_3^-] = 45 \text{ mEq/L}$
Respiratory alkalosis	$\downarrow \text{PCO}_2$ Alveolar hyperventilation	$\uparrow$	$\text{HCO}_3^- \downarrow$ Chronic: $\downarrow$ Renal HCO_3^- reabsorption	Acute $\Delta [\text{HCO}_3^-] = -2 \text{ mEq/L}$ for each $\downarrow \Delta \text{PCO}_2$ of 10 mm Hg Chronic $\Delta [\text{HCO}_3^-] = -5 \text{ mEq/L}$ for each $\downarrow \Delta \text{PCO}_2$ of 10 mm Hg	Acute $[\text{HCO}_3^-] = 18 \text{ mEq/L}$ Chronic $[\text{HCO}_3^-] = 15 \text{ mEq/L}$
Metabolic acidosis	$\downarrow \text{HCO}_3^-$ Loss of HCO_3^- or gain of H^+	$\downarrow$	$\downarrow \text{PCO}_2$ Alveolar hyperventilation	$\text{PCO}_2 = 1.5[\text{HCO}_3^-] + 8 \pm 2$ $\text{PCO}_2 = \text{last 2 digits of pH} \times 100$ $\text{PCO}_2 = 15 + [\text{HCO}_3^-]$	$[\text{PCO}_2] = 15 \text{ mm Hg}$
Metabolic alkalosis	$\uparrow \text{HCO}_3^-$ Gain of HCO_3^- or loss of H^+	$\uparrow$	$\uparrow \text{PCO}_2$ Alveolar hypoventilation	$\text{PCO}_2 = +0.6 \text{ mm Hg}$ for $\Delta [\text{HCO}_3^-]$ of 1 mEq/L $\text{PCO}_2 = 15 + [\text{HCO}_3^-]$	$[\text{PCO}_2] = 55 \text{ mm Hg}$

PCO_2 , Carbon dioxide pressure.

Adapted from Bidani A, Tauzon DM, Heming TA. Regulation of whole body acid-base balance. In DuBose TD, Hamm LL, editors: *Acid-base and electrolyte disorders: a companion to Brenner and Rector's the kidney*, Philadelphia: Saunders; 2002, pp 1-2.

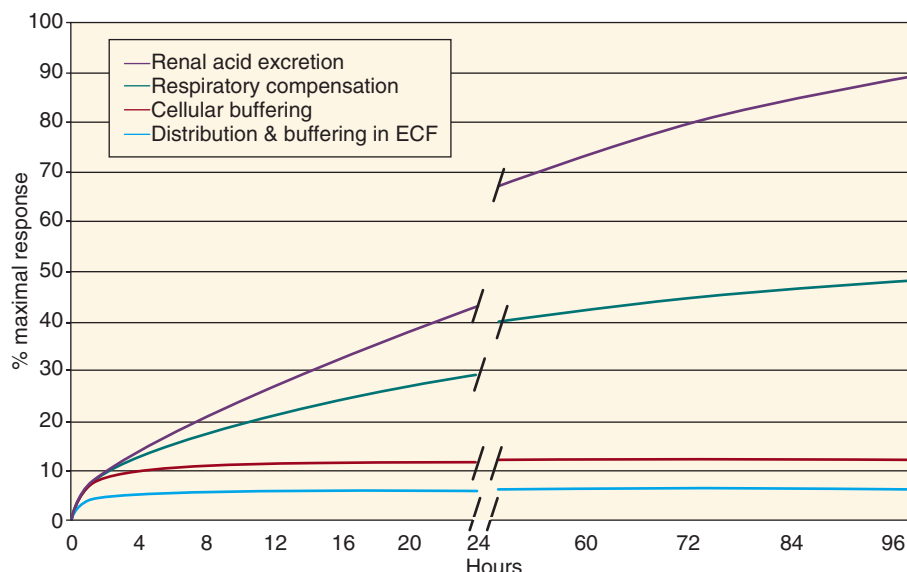


Fig. 16.4 Conceptual illustration of time course of acid-base compensatory mechanisms in response to a metabolic acid load. Component processes include rapid distribution and extracellular buffering mechanisms and cellular buffering events. But these most rapid mechanisms have limited capacity. Slower are respiratory and renal regulatory processes. ECF, Extracellular fluid.

since titratable acid excretion has limited capacity for upregulation (Fig. 16.2). With continued acid loading, renal net acid excretion increases over the course of 3 to 5 days (see Fig. 16.4). Thus, the renal response to an acid load requires 1) reabsorption of the filtered HCO_3^- by the proximal tubule, and 2) augmentation of NH_4^+ production and excretion by the distal nephron. In this way, the kidneys efficiently retain all filtered base and attempt to generate enough new base to restore the arterial pH toward normal.

There is some controversy about whether the kidneys *completely* excrete higher than usual acid loads, particularly in patients with CKD. Certainly patients with reduced plasma HCO_3^- have a chronic metabolic acidosis. But some patients with CKD and normal plasma HCO_3^- levels may have positive acid balance (e.g., acid retention), perhaps buffered by bone carbonate and reflected in an acid renal interstitium.^{3,12} There is also growing evidence (discussed more below) that not only does acidosis accelerate progression of CKD but that higher ammonium production and excretion may be harmful.¹³ On the other hand, in some studies higher urinary acid and ammonia excretion may be beneficial and/or be present in patients with slower progression.¹⁴ Both the mechanistic understandings and clinical implications are under active investigation. Certainly, current recommendations are to maintain plasma $[\text{HCO}_3^-] > 22 \text{ mEq/L}$ in patients with CKD with either alkali supplements (see below) or diets with less acid and more fruits and vegetables (more plant-based proteins versus animal-based proteins). Whether to treat patients with CKD with normal plasma HCO_3^- levels is currently under consideration.

SYSTEMIC RESPONSE TO GAIN OF ALKALI

Base or alkali loads are somewhat less common but rely on the same three systems outlined for defense of an acid challenge,

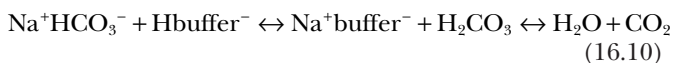
namely, cellular buffering and distribution within the ECF, respiratory compensation, and renal excretion.

DISTRIBUTION AND CELLULAR BUFFERING

Ninety-five percent of a base load in the form of HCO_3^- is distributed in the ECF within about 25 minutes^{2,6,10,15} (see Fig. 16.4). The cellular buffering over the next few hours against an alkaline load is somewhat less effective than the defense against an acid load. There is also poorer stabilization of intracellular pH in the alkaline than in the acid range.^{2,15}

RESPIRATORY COMPENSATION

The pulmonary response to an acute increase in HCO_3^- concentration is biphasic. Neutralization of sodium bicarbonate by buffers (H^+ buffer $^-$) results in CO_2 liberation and an increase in PCO_2 :



A clinical corollary would be the acute increase in PCO_2 observed in patients given intravenous NaHCO_3 infusions. The increased PCO_2 stimulates ventilation acutely to return PCO_2 toward normal. If the pulmonary system is compromised or the ventilation rate is controlled artificially, increased CO_2 production from infused sodium bicarbonate can lead to hazardous hypercapnia.^{2,6,15}

About an hour after an abrupt increment in the HCO_3^- concentration, when the increased generation of CO_2 subsides, stimulation of respiration is transformed into suppression of respiration by elevated pH, and PCO_2 increases. This secondary hypercapnic response takes several hours and partially compensates for the elevated HCO_3^- concentration so that arterial pH is returned toward (although not completely to) normal (see Fig. 16.4).

The hypercapnic response to metabolic alkalosis is more difficult to reliably predict than other acid-base disturbances.

Most studies have found that an increase in PCO_2 regularly occurs in response to alkalosis. The hypoventilatory response can lead to borderline or even frank hypoxemia in patients with chronic lung disease.¹⁵ In general, the increase in Paco_2 can be predicted to equal 0.75 mm Hg per 1.0 mEq/L increase in plasma HCO_3^- ; or more simply, add the value of 15 to the measured plasma $[\text{HCO}_3^-]$ ¹⁶ to predict the expected Paco_2 (see Table 16.1).

RENAL EXCRETION

WITH EXTRACELLULAR VOLUME EXPANSION

The addition of sodium bicarbonate to the body results in prompt cellular buffering and respiratory compensation. However, as with an acid load, the kidneys have the ultimate responsibility for the disposal of base and restoration of base stores to normal. The speed and efficiency with which HCO_3^- can be excreted by the kidneys are such that it is difficult to render a patient with normal renal function more than mildly alkalotic on a long-term basis, even when as much as 24 mEq/kg/day of sodium bicarbonate is ingested for several weeks.¹⁵ This efficiency results from the fact that ordinarily the kidneys actively reabsorb some 4000 mEq of HCO_3^- per day; to excrete excess HCO_3^- , the kidneys only need to reabsorb less.

The proximal tubule is responsible principally for HCO_3^- excretion when the blood HCO_3^- concentration increases. Absolute proximal HCO_3^- reabsorption does not increase in proportion to HCO_3^- load because of suppression of proximal acidification by alkalemia so that HCO_3^- delivery to the distal nephron increases. The limited capacity of the distal nephron to secrete H^+ can be overwhelmed easily, and bicarbonaturia increases progressively. NH_4^+ and titratable acid excretion decline in response to the increasing systemic and urine pH.^{7,16}

The type B intercalated cell in the collecting tubule also secretes HCO_3^- through the activity of the $\text{HCO}_3^-/\text{Cl}^-$ exchanger pendrin. In the face of an alkaline systemic pH this exchanger is responsible for net bicarbonate secretion. Accordingly, HCO_3^- secretion by the type B intercalated cell prevents a more severe alkalosis and participates in the HCO_3^- excretory response.

In summary, when kidney function and ECF volume are both normal, an acute base load is excreted entirely, and the blood HCO_3^- concentration is returned to normal within 12 to 24 hours because of depression of fractional proximal HCO_3^- reabsorption. In addition to suppression of reabsorption of the filtered HCO_3^- load, direct HCO_3^- secretion in the cortical collecting tubule (CCT) has been proposed as another mechanism for mediating HCO_3^- disposal during metabolic alkalosis.¹⁶

The increased delivery of HCO_3^- out of the proximal tubule in response to an increased blood HCO_3^- concentration (and, hence, filtered HCO_3^- load) in the setting of ECF expansion facilitates HCO_3^- excretion and the return of blood pH toward normal. However, other factors may independently enhance distal H^+ secretion sufficiently to prevent HCO_3^- excretion and thus counterbalance the suppressed proximal HCO_3^- reabsorption. Under these circumstances, the alkalosis is maintained. For example, in the setting of primary hyperaldosteronism, despite the expanded ECF, a stable mild alkalotic condition persists because of augmented collecting

duct H^+ secretion.¹⁶ In such cases, concurrent hypokalemia facilitates the generation and maintenance of metabolic alkalosis by enhancing NH_4^+ production and excretion.^{7,16} Moreover, chronic hypokalemia dramatically enhances the abundance and functionality of the H^+-K^+ -adenosine triphosphatase (H^+-K^+ -ATPase) in the medullary collecting tubule, thus increasing rather than decreasing distal bicarbonate absorption.¹⁶⁻¹⁹

WITH EXTRACELLULAR VOLUME CONTRACTION AND POTASSIUM DEFICIENCY

The renal response to an increase in plasma HCO_3^- concentration can be modified significantly in the presence of ECF contraction and K^+ depletion. Na^+ , Cl^- , and ECF volume depletion are almost always present together and for most clinical purposes are nearly synonymous. Effective ECF and K^+ stores are critical in modifying net HCO_3^- reabsorption. Deficiency of both Cl^- and K^+ is common in metabolic alkalosis because of renal and/or gastrointestinal losses that occur concurrently with the generation of the alkalosis.^{18,20} With Cl^- depletion alone, the normal bicarbonaturic response to an increase in plasma HCO_3^- is prevented and metabolic alkalosis can develop. K^+ depletion, even without mineralocorticoid administration, can cause metabolic alkalosis. When Cl^- and K^+ depletion coexist, severe metabolic alkalosis may develop.

Two general mechanisms serve to maintain increased plasma HCO_3^- in the setting of Cl^- and K^+ depletion: 1) decreased glomerular filtration rate (GFR) and therefore diminished HCO_3^- filtration, and 2) increased HCO_3^- reabsorption and increased new HCO_3^- generation by acid excretion.^{16-19,21} That extracellular and plasma volume depletion decrease GFR is well documented. GFR can also be decreased by K^+ depletion, possibly the result of increased production of the vasoconstrictors angiotensin II and thromboxane B_2 .^{17,19} By reducing filtered HCO_3^- (compared to what it would be with elevated plasma HCO_3^- in metabolic alkalosis), all of the filtered HCO_3^- can be reabsorbed without any increase in proximal or distal HCO_3^- reabsorption. But as to the second mechanism, Cl^- deficiency and/or K^+ deficiency do increase overall renal HCO_3^- reabsorption in the setting of a normal GFR and high filtered HCO_3^- load. Overall renal HCO_3^- reabsorption and, therefore, acidification is increased. An increase in renal acidification occurs as a result of an increase in H^+ secretion by the proximal and the distal nephron.¹⁶⁻¹⁸ The increase in HCO_3^- absorption in the proximal tubule is due, at least in part, to an increase in the delivered load of HCO_3^- . The augmented HCO_3^- absorption in distal nephron segments appears to be due to a primary increase in H^+ secretion that is independent of the HCO_3^- load delivered. Chronic hypokalemia dramatically enhances the abundance and function of the H^+ , K^+ -ATPase in the medullary collecting tubule. Therefore, upregulation of the H^+ , K^+ -ATPase by hypokalemia may be a significant factor in the maintenance of chronic metabolic alkalosis.^{17,22,23}

The maintenance of a high plasma HCO_3^- concentration by the kidney can be repaired by repletion of Cl^- .²⁴ The mechanism by which Cl^- repairs metabolic alkalosis may include normalization of the low GFR that was induced by ECF repletion. In addition, Cl^- repletion results in a decrease in proximal HCO_3^- reabsorption and an increase in HCO_3^- secretion by the distal nephron.

In summary, the normal physiologic response by the kidney to a base load associated with volume *expansion* is to excrete the base. In the setting of K^+ and/or Cl^- deficiency, base is instead retained by a decreased GFR, an increased proximal HCO_3^- reabsorption, and enhanced distal HCO_3^- reabsorption, acid excretion.

STEPWISE APPROACH TO THE DIAGNOSIS OF ACID-BASE DISORDERS

Acid-base derangements are important for their own physiologic consequences but also as indicators of underlying processes, which sometimes may not be otherwise apparent. Therefore, analysis of acid-base status is important in many clinical circumstances, particularly in the critically ill. An acid-base disorder may be suspected from the clinical circumstance, from an abnormal arterial blood gas (ABG; i.e., abnormal pH, $Paco_2$, or HCO_3^- concentration), or from abnormal electrolytes even in the absence of a blood gas. It is important to stress that relying only on serial oxygen saturation levels in a critically ill patient is inadequate when assessing the acid-base status of a patient.

The four simple acid-base disorders and the predicted compensatory responses are summarized in Table 16.1. The complete diagnosis can best be determined by a stepwise approach presented below (Table 16.2).²⁵ Before initiating direct examination of acid-base parameters, certain parameters such as history and other electrolytes should be considered.

CLINICAL AND LABORATORY PARAMETERS IN ACID-BASE DISORDERS

For correct diagnosis of a simple or mixed acid-base disorder, it is imperative that a careful history be obtained. Patients with pneumonia, sepsis, or cardiac failure frequently have a respiratory alkalosis, and patients with chronic obstructive pulmonary disease or a sedative drug overdose often display respiratory acidosis. *The clinical setting of an acid-base disorder may be much more revealing than complex formulas.* The patient's drug history assumes importance because patients taking

loop or thiazide diuretics may have metabolic alkalosis and patients receiving acetazolamide frequently have metabolic acidosis. Physical findings are often helpful as well. Tetany may occur with alkalemia, cyanosis with respiratory acidosis, and volume contraction with metabolic alkalosis.

The plasma K^+ value is often useful but should be considered only in conjunction with the HCO_3^- concentration and blood pH. It is generally appreciated that the serum K^+ value can be altered by primary acid-base disturbances as a result of shifts of K^+ either into the extracellular compartment or into the intracellular compartment.²⁶ Metabolic acidosis frequently leads to hyperkalemia. Some suggest that for each decrease in blood pH of 0.10 pH unit, the K^+ concentration should increase by 0.6 mEq/L. Thus, a patient with a pH of 7.20 would be expected to have a plasma K^+ value of 5.2 mEq/L. However, considerable variation in this relationship has been reported in several conditions due to endogenous acid production, especially diabetic ketoacidosis (DKA) and lactic acidosis, which are often associated with K^+ depletion. The lack of correlation between the degree of acidemia and the plasma K^+ level is a result of several factors, including the nature and cellular permeability of the accompanying anion, the magnitude of the osmotic diuresis, the level of renal function, the presence or absence of preexisting changes in K^+ homeostasis, and the degree of catabolism. It is important to appreciate that the relationship between arterial blood pH and plasma K^+ is complex and therefore often variable. Nevertheless, the failure of a patient with severe acidosis to exhibit hyperkalemia or, conversely, the failure of a patient with severe metabolic alkalosis to exhibit hypokalemia raises the possibility of a significant derangement of body K^+ homeostasis. The combination of a low plasma K^+ and elevated HCO_3^- suggests metabolic alkalosis (or the action of aldosterone or the like, see below), whereas the combination of an elevated plasma K^+ and low HCO_3^- suggests metabolic acidosis.

It is helpful to compare the serum Cl^- concentration with the Na^+ concentration. In the absence of an exogenous source of concentrated sodium (e.g., hypertonic saline or $NaHCO_3$), the serum Na^+ concentration changes only as a result of changes in hydration or total body water. The Cl^- concentration changes for two reasons: 1) changes in hydration and 2) changes in acid-base balance. Thus, changes in Cl^- not reflected by proportional changes in Na^+ suggest the presence of an acid-base disorder. For example, consider a patient with a history of vomiting, volume depletion, a Cl^- concentration of 85 mEq/L, and an Na^+ concentration of 130 mEq/L. In this case, both Na^+ and Cl^- are reduced, but the reduction in Cl^- is proportionally greater (15% versus 7%). A disproportionate decrease in Cl^- suggests metabolic alkalosis or respiratory acidosis, and a disproportionate increase in Cl^- suggests metabolic acidosis or respiratory alkalosis.

The four simple acid-base disorders and the predicted compensatory responses are summarized in Table 16.1. The complete diagnosis, including the possibility of complex mixed disturbances, can best be determined by a stepwise approach (Table 16.2).

STEP 1: MEASURE ARTERIAL BLOOD GAS AND ELECTROLYTE VALUES

For a complete diagnosis of acid-base disorders, one needs both an ABG and a simultaneous set of electrolytes. For a

Table 16.2 Systematic Method for Diagnosis of Simple and Mixed Acid-Base Disorders

1. Measure arterial blood gas and electrolyte concentrations simultaneously.
2. Determine whether the compensation is appropriate for a simple acid-base disorder; if inappropriate, a mixed acid-base disorder is present (see Table 16.1 and Fig. 16.5).
3. Calculate the anion gap (corrected for albumin) to determine the presence of a high AG metabolic acidosis. Appreciate the four major categories of high anion gap acidoses: ketoacidosis, lactic acidosis, renal failure acidosis, toxin- or poison-induced acidosis. Appreciate the two major causes of non-AG acidosis: gastrointestinal loss of HCO_3^- , renal loss of HCO_3^- .
4. Compare the relative changes of HCO_3^- and AG (ΔHCO_3^- and ΔAG , respectively) to look for mixed disorders (see text).

simple single disorder, the diagnosis is easy: if the pH is low, a respiratory acidosis (with high PCO_2 and somewhat high HCO_3^-) or metabolic acidosis (with low HCO_3^- and secondarily low PCO_2) is present. The opposite situations occur with high pH. Or with electrolytes, if the HCO_3^- concentration is low and the Cl^- concentration is high, either chronic respiratory alkalosis or hyperchloremic metabolic acidosis is present. The ABG determination serves to differentiate the two conditions. Although both have a decreased PaCO_2 , the pH is high with a primary respiratory alkalosis and low in a metabolic acidosis. Chronic respiratory acidosis and metabolic alkalosis are both associated with high HCO_3^- and low Cl^- concentration in plasma. Again, a pH measurement distinguishes the two conditions.

But neither ABG nor electrolytes alone can necessarily give a complete picture. For instance, an ABG alone may miss mixed disorders, which are only revealed by an analysis of the anion gap (see later discussion). And while electrolyte abnormalities may suggest an acid-base disorder (e.g., high anion gap metabolic acidosis), the actual pH and the complete diagnosis with compensatory response requires an ABG. Of note, the calculated value for HCO_3^- or (total CO_2) reported with the ABG should be within ± 2 mEq/L of the measured HCO_3^- concentration (total CO_2) obtained on the electrolyte panel. A greater difference may imply a laboratory error or non-simultaneous measurements.

STEP 2: DETERMINE THE COMPENSATION TO DISTINGUISH SIMPLE FROM MIXED ACID-BASE DISORDERS

In many clinical situations, a mixture of acid-base disorders may exist. Diagnosis of these disturbances requires additional information and a more complex analysis of data.

A convenient, but not always reliable, approach is an acid-base map, such as the one displayed in Fig. 16.5, which defines the 95% confidence limits of simple acid-base disorders.^{26,27} If the arterial acid-base values fall within one of the blue bands in Fig. 16.3, one may assume that a simple acid-base disturbance is present, and a tentative diagnostic category can be assigned. There can be complex exceptions. Values that fall outside the blue areas imply, but do not prove, that a mixed disorder exists.

The two broad types of acid-base disorders are metabolic and respiratory. Metabolic acidosis and alkalosis are disorders characterized by primary disturbances in the concentration of HCO_3^- in plasma, whereas respiratory disorders involve primarily alteration of PaCO_2 . The most commonly encountered clinical disturbances are simple acid-base disorders, that is, one of the four cardinal acid-base disturbances—metabolic acidosis, metabolic alkalosis, respiratory acidosis, or respiratory alkalosis—occurring in a pure or simple form. More complicated clinical situations, especially in severely ill patients, may give rise to *mixed acid-base disturbances*.²⁷

To appreciate and recognize a mixed acid-base disturbance, it is important to understand the physiologic compensatory responses that occur in the simple acid-base disorders. Primary respiratory disturbances invoke secondary metabolic responses, and primary metabolic disturbances evoke a predictable respiratory response (see Table 16.1 and Fig. 16.3). *These predictable compensations are defined as being part of the primary simple acid-base disorder.* To illustrate, metabolic acidosis as a result of gain of endogenous acids (e.g., lactic acid or ketoacidosis) lowers the concentration of HCO_3^- in the ECF and thus extracellular pH. As a result of *acidemia*, the medullary chemoreceptors are stimulated and invoke an increase in ventilation. As a result of the hypocapnic response, the ratio of HCO_3^- to PaCO_2 and the subsequent pH are returned

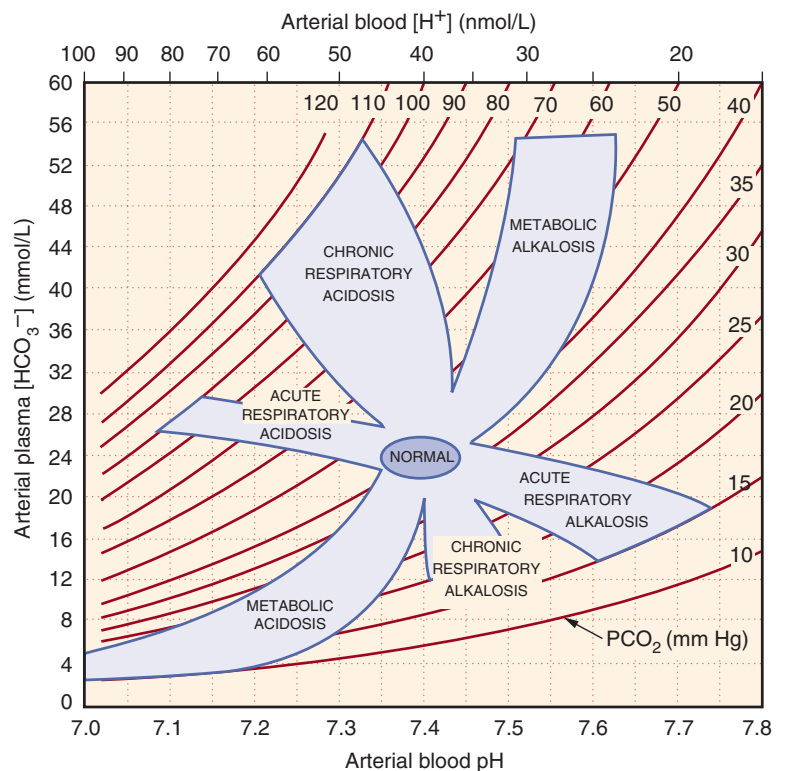


Fig. 16.5 Acid-base nomogram (map). Shaded areas represent the 95% confidence limits of the normal respiratory and metabolic compensations for primary acid-base disturbances. Data falling outside the shaded areas denote a mixed disorder if a laboratory error is not present (see text).

toward, but not completely to, normal. Thus, a patient with metabolic acidosis and a plasma HCO_3^- concentration of 12 mEq/L would be expected to have a Paco_2 between 24 and 28 mm Hg. Values of Paco_2 below 24 or higher than 28 mm Hg define a *mixed metabolic-respiratory disturbance* (metabolic acidosis and respiratory alkalosis or metabolic acidosis and respiratory acidosis, respectively). Therefore, by definition, an inappropriate compensation, too little or too much, identifies one form of a mixed acid-base disturbance.

Compensation is a predictable physiologic consequence of the primary disturbance and does not represent a secondary acidosis or alkalosis (see Fig. 16.5 and Table 16.1). The recognition of mixed disturbances may alert one to an additional clinical disorder that may require immediate attention or additional therapy (e.g., inadequate respiratory response to metabolic acidosis defining a respiratory acidosis).

The limits of compensation should also be recognized (see Table 16.1). For example, the plasma HCO_3^- concentration rarely falls below 12 to 15 mEq/L as a result of compensation for respiratory alkalosis and rarely exceeds 45 mEq/L as a result of compensation for respiratory acidosis.²⁷

STEP 3: CALCULATE THE ANION GAP TO DETERMINE THE PRESENCE OF A HIGH AG METABOLIC ACIDOSIS

All evaluations of acid-base disorders should include a simple calculation of the AG since an elevated AG usually implies a metabolic acidosis. The AG is calculated from the serum electrolytes and is defined as follows:

$$\text{AG} = \text{Na}^+ - (\text{Cl}^- + \text{HCO}_3^-) = 10 \pm 2 \text{ mEq/L} \quad (16.11)$$

The AG represents the unmeasured anions normally present in plasma and unaccounted for by the serum electrolytes exclusive of K^+ that are measured on the electrolyte panel. Normal values for AG vary with the laboratory and analyte measurement techniques, but in general have declined with more precise measurement of serum electrolytes by ion selective electrodes. The normal value for AG ranges from 8–12 mEq/L, but the clinician should know the normal value for the AG in clinical laboratories used in their practice. Because of this range of normal values for the AG and for convenience, the following computations will use the value of 10 mEq/L as the “normal” anion gap. The unmeasured anions that contribute to this value are normally present in serum and include anionic proteins (principally albumin and, to lesser extent, α - and β -globulins), PO_4^{3-} , SO_4^{2-} , and organic anions. As already emphasized, interpretation of the anion gap requires either a normal serum albumin, or correction of the AG to a normal plasma albumin. In general, reduction in the serum albumin level by 1 g/dL from the normal value of 4.5 g/dL decreases the AG by 2.5 mEq/L. When acid anions, such as acetoacetate and lactate, are produced endogenously in excess and accumulate in ECF, the AG increases above the normal value. This is referred to as a *high anion gap acidosis*.^{27,28} Additionally, in a simple high AG metabolic acidosis, for each milliequivalent per liter increase in the corrected AG, there should be an equal decrease, measured as milliequivalent per liter, in the plasma HCO_3^- concentration.

A number of conditions other than metabolic acidosis can occasionally change the AG up or down (Table 16.3). An

Table 16.3 Anion Gap = $\text{Na}^+ - (\text{Cl}^- + \text{HCO}_3^-) = 9 \pm 3 \text{ mEq/L}$ (Assumes Normal Albumin)^a

Increased Anion Gap	Decreased Anion Gap
Increased Anions (Not Cl^- or HCO_3^-)	Increased Cations (Not Na^+)
↑ Albumin	↑ Ca^{2+} , Mg^{2+}
Alkalosis	↑ Li^+
↑ Inorganic anions	↑ Immunoglobulin G
Phosphate	
Sulfate	Decreased Anions (Not Cl^- or HCO_3^-)
↑ Organic anions	Hypoalbuminemia^a
L-Lactate	Acidosis
D-Lactate	
Ketones	Laboratory Error
Uremic	
↑ Exogenously supplied anions	Hyperviscosity
Salicylate	Bromism
Paraldehyde	
Ethylene glycol	
Propylene glycol	
Methanol	
Toluene	
Pyroglutamic acid (5-oxyprolene)	
Other toxins	
Uremic	
Hyperosmolar, nonketotic states	
Myoglobinuric acute renal failure	
Decreased Cations (Not Na^+)	
↓ Ca^{2+} , Mg^{2+}	

^aFor each decline in albumin by 1 g/dL from normal (4.5 g/dL), the anion gap decreases by 2.5 mEq/L.

The items in bold are most common or notable.

increase in the AG may be due to a decrease in unmeasured cations or an increase in unmeasured anions. Combined severe hypocalcemia and hypomagnesemia represent a decrease in the contribution of unmeasured cations. In addition, the AG may increase secondary to an increase in anionic albumin, as a consequence of either an increased albumin concentration or alkalemia.^{27,28} The increased AG in severe alkalemia can be explained in part by the effect of alkaline pH on the electrical charge of albumin.

A decrease in the AG can be generated by an increase in unmeasured cations or a decrease in unmeasured anions (see Table 16.3). A decrease in the AG can result from 1) an increase in unmeasured cations (Ca^{2+} , Mg^{2+} , K^+), or 2) the addition to the blood of abnormal cations, such as Li^+ (Li^+ intoxication) or cationic immunoglobulins (immunoglobulin G as in plasma cell dyscrasias). Because albumin is the major unmeasured anion, the AG will also decrease if the quantity of albumin is low (e.g., nephrotic syndrome, protein malnutrition, capillary leak).²⁹ Since with each decline in the serum albumin level by 1 g/dL from the normal value of 4.5 g/dL, the AG will decline by 2.5 mEq/L. When hypoalbuminemia exists, it is possible to underestimate the AG and even miss an increased AG unless correction for the low albumin and its effect on the AG is taken into account. For example, in a patient with an albumin level of 1.5 g/dL

and an uncorrected AG of 10 mEq/L, the “corrected” AG is 17.5 mEq/L.

Laboratory errors can create a falsely low AG. Hyperviscosity and hyperlipidemia lead to an underestimation of the true Na^+ concentration, and bromide (Br^-) intoxication causes an overestimation of the true Cl^- concentration.²⁷

In the presence of a normal serum albumin level, elevation of unmeasured anions is usually due to addition of non- Cl^- -containing acid anion (e.g., lactate). *Thus, in most clinical circumstances, a high AG indicates that a metabolic acidosis is present.* The anions accompanying such acids include inorganic (PO_4^{3-} , SO_4^{2-}), organic (ketoacids, lactate, uremic organic anions), exogenous (salicylate or ingested toxins with organic acid production), or unidentified anions.²⁷ The preexisting Cl^- concentration is unchanged when the new acid anion is added to the blood. Therefore, the high AG acidosis exhibit normochloremia as well as a high gap. If the kidney does not excrete the anion, the magnitude of the decrement in HCO_3^- concentration will match the increment in the AG. If the retained anion can be metabolized to HCO_3^- directly or indirectly (e.g., ketones or lactate, after successful treatment), normal acid-base balance is restored as the AG returns toward the normal value. Alternatively, if the anion is excreted, hyperchloremic acidosis may emerge as the anion is excreted and the AG disappears.

The presence of a significant elevation of the AG (unexplained by laboratory error or the unusual conditions noted) implies the presence of a metabolic acidosis process whether or not acidemia (low pH) exists.³⁰ For instance, a coexisting respiratory alkalosis or metabolic alkalosis could dominate the net pH even in the presence of a metabolic acidosis.

Utility of identifying an elevated AG derives from knowledge of the four causes of a high AG metabolic acidosis: 1) ketoacidosis, 2) lactic acidosis, 3) renal failure acidosis, and 4) toxin-induced metabolic acidosis (Table 16.4). In contrast, if the AG is normal in the face of metabolic acidosis, a hyperchloremic or non-AG acidosis exists. The specific causes of hyperchloremic acidosis that must be appreciated are outlined in a later section. Table 16.4 displays the directional changes in pH, PCO_2 , and HCO_3^- for the four simple acid-base disorders. With this stepwise approach, in subsequent sections, the specific causes of the major types of acid-base disorders are reviewed in detail.

STEP 4: COMPARE RELATIVE CHANGES OF HCO_3^- AND AG (DELTA/DELTA)

By definition, a high AG acidosis has two identifying features: a low HCO_3^- concentration and an elevated AG. The elevated AG will remain present even if another disorder coincides to modify the HCO_3^- concentration independently. Simultaneous metabolic acidosis of the high AG variety plus either metabolic alkalosis or chronic respiratory acidosis is an example of such a situation. The resulting HCO_3^- concentration may be normal or even high in such a setting. However, the AG will be high, and the Cl^- concentration relatively depressed. Consider a patient with chronic obstructive pulmonary disease with compensated respiratory acidosis (PaCO_2 of 65 mm Hg and HCO_3^- concentration of 40 mEq/L initially) in whom acute bronchopneumonia and respiratory decompensation develop. If this patient has an HCO_3^- concentration of 24 mEq/L, Na^+ of 145 mEq/L, K^+ of 4.8 mEq/L,

Table 16.4 Clinical Causes of High Anion Gap and Normal Anion Gap Acidosis

High Anion Gap Acidosis	Non-Anion Gap Acidosis
Ketoacidosis	Gastrointestinal loss of HCO_3^- (negative urine anion gap)
Diabetic ketoacidosis	Diarrhea
Alcoholic ketoacidosis	External fistula
Starvation ketoacidosis	Renal loss of HCO_3^- or failure to excrete NH_4^+
Lactic acidosis	Positive urine anion gap = low net acid excretion
L-Lactic acidosis (types A and B)	Proximal renal tubular acidosis (RTA) (low serum K^+)
D-Lactic acidosis	Classical distal renal tubular acidosis (low serum K^+)
Renal failure acidosis	Generalized distal renal tubular defect (high serum K^+)
Toxin-induced acidosis	Drugs that cause RTA
Ethylene glycol	Carbonic anhydrase inhibitors (mixed proximal-distal RTA)
Methyl alcohol	Amphotericin B (“gradient” classical distal RTA)
Salicylate	Miscellaneous
Propylene glycol	NH_4Cl ingestion
Pyroglutamic acid (5-oxoproline)	Sulfur ingestion
	Dilutional acidosis

RTA, Renal tubular acidosis.

The items in bold are most common or notable.

and Cl^- of 96 mEq/L, it would be incorrect to assume that this “normal” HCO_3^- concentration represents improvement in acid-base status toward normal. Indeed, the arterial pH would probably be low (7.19), as a result of a more serious degree of hypercapnia than observed previously (e.g., if the PCO_2 increased from 65 to 80 mm Hg as a result of pneumonia). Even without blood gas measurements, prompt recognition that the AG was elevated to 25 mEq/L should suggest that a life-threatening lactic acidosis is superimposed on a preexisting chronic respiratory acidosis, which necessitates immediate therapy. In this example, the change in AG, frequently referred to as ΔAG , is computed as 25–10, or patient’s calculated AG minus the normal AG, and is equal to 15 mEq/L.

Similarly, a normal arterial HCO_3^- concentration, PaCO_2 , and pH do not ensure the absence of an acid-base disturbance. For example, an alcoholic patient who has been vomiting may develop a metabolic alkalosis with a pH of 7.55, HCO_3^- concentration of 40 mEq/L, PCO_2 of 48 mm Hg, Na^+ of 135 mEq/L, Cl^- of 80 mEq/L, and K^+ of 2.8 mEq/L. If such a patient were then to develop a superimposed alcoholic ketoacidosis (AKA) with a β -hydroxybutyrate concentration of 15 mmol/L, the arterial pH would fall to 7.40, HCO_3^- concentration to 25 mEq/L, and PCO_2 to 40 mm Hg. Although the blood gas values are normal, the AG (assuming no change in Na^+ or Cl^-) is elevated (25 mEq/L), and the ΔAG is 15 mEq/L, which indicates the existence of a mixed metabolic acid-base disorder (mixed metabolic alkalosis and metabolic acidosis). The combination of metabolic acidosis and metabolic alkalosis is not uncommon and is most easily

recognized, as in this case, when the ΔAG is elevated, but the HCO_3^- concentration and pH are near normal ($\Delta AG > \Delta HCO_3^-$, or 15 vs. 0 mEq/L).

Although a variety of formulaic approaches have been proposed to compare the ΔAG and ΔHCO_3^- , a conceptual approach as above with attention to the clinical ingredients is most useful. With an elevated AG, if the HCO_3^- is higher than expected based on the ΔAG , then suspect a coexisting metabolic alkalosis or chronic respiratory acidosis (which also elevates the HCO_3^-). With an elevated AG, if the HCO_3^- is lower than expected (or Cl⁻-higher) based on the ΔAG , suspect a coexisting hyperchloremic metabolic acidosis or chronic respiratory alkalosis.

MIXED ACID-BASE DISORDERS

Mixed acid-base disorders—defined as independently coexisting disorders, not merely compensatory responses—are often seen in patients in critical care units and can lead to dangerous extremes of pH. Patients with underlying pulmonary disease may not respond to metabolic acidosis with an appropriate ventilatory response because of insufficient respiratory reserve. Such imposition of respiratory acidosis on metabolic acidosis can lead to severe acidemia and a poor outcome. Note that failure of an appropriate ventilator response to metabolic acidosis represents a combined metabolic acidosis and respiratory acidosis. A patient with DKA (metabolic acidosis) may develop an independent respiratory problem, leading to respiratory acidosis or alkalosis. When metabolic acidosis and metabolic alkalosis coexist in the same patient, the pH may be normal or near normal. When the pH is normal, an elevated AG denotes the presence of a metabolic acidosis. A discrepancy in the ΔAG (prevailing minus normal AG of 10 mEq/L) and the ΔHCO_3^- (normal, 25 mEq/L, minus prevailing HCO_3^-) indicates the presence of a mixed high gap acidosis–metabolic alkalosis (see example later).

Even more complex are triple acid-base disturbances. For example, patients with metabolic acidosis due to alcoholic ketoacidosis may develop metabolic alkalosis due to vomiting and superimposed respiratory alkalosis due to the hyperventilation of hepatic dysfunction or alcohol withdrawal. Conversely, when hyperchloremic acidosis and metabolic alkalosis occur concomitantly, the increase in Cl⁻ is out of proportion to the change in HCO_3^- concentration ($\Delta Cl^- > \Delta HCO_3^-$).²⁷

In summary, an AG exceeding that expected for a patient's albumin concentration, i.e. > 10 mEq/L, denotes the existence of either a simple high AG metabolic acidosis or a complex acid-base disorder in which an organic acidosis is superimposed on another acid-base disorder.

RESPIRATORY DISORDERS

RESPIRATORY ACIDOSIS

Respiratory acidosis occurs as the result of severe pulmonary disease, respiratory muscle disorders, or depression in ventilatory control. An increase in $PaCO_2$ owing to reduced alveolar ventilation is the primary abnormality leading to acidemia. In acute respiratory acidosis, there is an immediate small compensatory elevation in HCO_3^- (due to cellular buffering mechanisms) of 1 mEq/L for every 10 mm Hg increase in

$PaCO_2$. In chronic respiratory acidosis (>24 hours), renal adaption is achieved and the HCO_3^- increases by 4 mEq/L for every 10 mm Hg increase in $PaCO_2$. The serum bicarbonate concentration usually does not increase above 38 mEq/L, however.

The clinical features of respiratory acidosis vary according to the underlying disease, the severity, duration, and presence or absence of accompanying hypoxemia. A rapid increase in $PaCO_2$ may result in anxiety, dyspnea, confusion, psychosis, and hallucinations and may progress to coma. More subtle dysfunction in chronic hypercapnia may include sleep disturbances, loss of memory, daytime somnolence, and personality changes. Coordination may be impaired, and motor disturbances such as tremor, myoclonic jerks, and asterixis may develop. The sensitivity of the cerebral vasculature to the vasodilating effects of CO_2 can cause headaches and other signs that mimic increased intracranial pressure, such as papilledema, abnormal reflexes, and focal muscle weakness.

Causes of respiratory acidosis are listed in Table 16.5. A reduction in ventilatory drive from depression of the respiratory center by a variety of drugs, injury, or disease can produce respiratory acidosis. Acutely, this may occur with general anesthetics, sedatives, narcotics, alcohol, and head trauma. Chronic causes of respiratory center depression include sedatives, alcohol, intracranial tumors, and the syndromes of sleep-disordered breathing, including the primary alveolar and obesity-hypoventilation syndromes. Neuromuscular disorders involving abnormalities or disease in the motor neurons, neuromuscular junction, and skeletal muscle can cause hypoventilation. Although a number of diseases should be considered in the differential diagnosis, drugs and electrolyte disorders should always be ruled out.

Mechanical ventilation, may result in respiratory acidosis when not properly adjusted and supervised or when complicated by barotrauma or displacement of the endotracheal tube. This occurs if carbon dioxide production rises (e.g., fever, agitation, sepsis, overfeeding) or if alveolar ventilation falls because of worsening pulmonary function. High levels of positive end-expiratory pressure in the presence of reduced cardiac output may cause hypercapnia as a result of large increases in alveolar dead space.

Permissive hypercapnia may be utilized in the critical care setting since lower tidal volumes may reduce the incidence of the barotrauma associated with high airway pressure and peak airway pressures in mechanically ventilated patients with respiratory distress syndrome.³¹ Acute hypercapnia of any cause can lead to severe acidemia, neurologic dysfunction, and death. However, when CO_2 levels are allowed to increase gradually, the resulting acidosis is less severe, and the elevation in arterial PCO_2 is tolerated more readily. Although the resulting hypercapnia is not the goal of this approach, but secondary to the attempt to limit airway pressures, the arterial pH will decline. The magnitude of the acidemia associated with permissive hypercapnia may be augmented if superimposed on metabolic acidosis, such as lactic acidosis. This combination is not uncommon in the critical care unit. Bicarbonate infusion may be indicated with mixed metabolic acidosis–respiratory acidosis, but the goal of therapy with alkali is to not increase the bicarbonate and pH to normal. With low tidal volume ventilation, a reasonable therapeutic

Table 16.5 Respiratory Acid-Base Disorders**Alkalosis**

Central nervous system stimulation

- Pain
- Anxiety
- Psychosis
- Fever
- Cerebrovascular accident
- Meningitis
- Encephalitis
- Tumor
- Trauma

Hypoxemia or tissue hypoxia

- High-altitude acclimatization
- Pneumonia, Pulmonary edema
- Aspiration
- Severe anemia

Drugs or hormones

- Pregnancy (progesterone)
- Salicylates
- Nikethamide

Stimulation of chest receptors

- Hemothorax
- Flail chest
- Cardiac failure
- Pulmonary embolism

Miscellaneous

- Septicemia
- Hepatic failure
- Mechanical hyperventilation
- Heat exposure
- Recovery from metabolic acidosis

Acidosis

Central nervous system depression

- Drugs (anesthetics, morphine, sedatives)
- Stroke
- Infection

Airway obstruction

- Asthma

Parenchyma disease

- Emphysema/chronic obstructive pulmonary disease
- Pneumoconiosis
- Bronchitis
- Adult respiratory distress syndrome
- Barotrauma

Mechanical ventilation

- Hypoventilation
- Permissive hypercapnia

Neuromuscular

- Poliomyelitis
- Kyphoscoliosis
- Myasthenia gravis
- Muscular dystrophies
- Multiple sclerosis

Miscellaneous

- Obesity
- Hypoventilation

target for arterial pH is approximately 7.25.³¹ Moreover, with hypercapnia in the range of 60 mm Hg, a larger amount of bicarbonate will be necessary to achieve this goal. Bicarbonate administration will further increase the PCO_2 , especially in patients with fixed rates of ventilation, and will add to the magnitude of the hypercapnia. Use of a continuous bicarbonate infusion in this setting may be necessary, but frequent monitoring of arterial blood gases, electrolytes, and the volume status of the patient is necessary.

Disease and obstruction of the airways, when severe or long-standing, causes respiratory acidosis. Acute hypercapnia follows sudden occlusion of the upper airway or the more generalized bronchospasm that occurs with severe asthma, anaphylaxis, and inhalational burn or toxin injury. Chronic hypercapnia and respiratory acidosis occur in end-stage obstructive lung disease.⁵ Restrictive disorders involving either the chest wall or the lungs can cause acute and chronic hypercapnia. Rapidly progressing restrictive processes in the lung can lead to respiratory acidosis, because the high cost of breathing causes ventilatory muscle fatigue. Intrapulmonary and extrapulmonary restrictive defects present as chronic respiratory acidosis in their most advanced stages.

The diagnosis of respiratory acidosis requires, by definition, the measurement of arterial Paco_2 and pH. Detailed history and physical examination often provide important diagnostic clues to the nature and duration of the acidosis. When a diagnosis of respiratory acidosis is made, its cause should be investigated. Chest radiography is an initial step. Pulmonary function studies, including spirometry, diffusion capacity, lung volumes, and arterial Paco_2 and oxygen saturation usually provide adequate assessment of whether respiratory acidosis is secondary to lung disease. Workup for nonpulmonary causes should include a detailed drug history, measurement of hematocrit, and assessment of upper airway, chest wall, pleura, and neuromuscular function.^{4,5}

The treatment of respiratory acidosis depends on the underlying disease, the severity and rate of onset. Acute respiratory acidosis can be life-threatening, and measures to reverse the underlying cause should be undertaken simultaneously with restoration of adequate alveolar ventilation to relieve severe hypoxemia and acidemia. Temporarily, this may necessitate tracheal intubation and assisted mechanical ventilation. Oxygen level should be carefully titrated in patients with severe chronic obstructive pulmonary disease and chronic CO_2 retention who are breathing spontaneously. When oxygen is used injudiciously, these patients may experience progression of the respiratory acidosis when ventilation is driven by oxygen pressure (Pao_2) and not the normal parameters of Paco_2 and pH. Aggressive and rapid correction of hypercapnia should be avoided, because the falling Paco_2 may provoke the same complications noted with acute respiratory alkalosis (i.e., cardiac arrhythmias, reduced cerebral perfusion, and seizures). It is advisable to lower the Paco_2 gradually in chronic respiratory acidosis, with the aim of restoring the Paco_2 to baseline levels while at the same time providing sufficient chloride and potassium to enhance the renal excretion of bicarbonate.⁵

Chronic respiratory acidosis is frequently difficult to correct, but general measures aimed at maximizing lung function, including cessation of smoking; use of oxygen, bronchodilators, corticosteroids, etc. and/or diuretics; and physiotherapy can help some patients and can forestall further deterioration.

Various respiratory stimulants used in the past are seldom utilized presently.

RESPIRATORY ALKALOSIS

Alveolar hyperventilation decreases the PaCO_2 and increases the $\text{HCO}_3^-/\text{PaCO}_2$ ratio, thus increasing pH (alkalemia). Nonbicarbonate cellular buffers respond by titrating HCO_3^- down. Hypocapnia develops whenever a sufficiently strong ventilatory stimulus causes CO_2 output in the lungs to exceed the metabolic production of CO_2 by tissues. Plasma pH and HCO_3^- concentration vary approximately proportionately with PaCO_2 over a range from 40 to 15 mm Hg. The arterial $[\text{HCO}_3^-]$ will decrease acutely approximately 2 mEq/L for each 10 mm Hg decrease in PCO_2 . The relationship between pH and PaCO_2 is about 0.01 pH unit/mm Hg.⁴

Beyond 2 to 6 hours, sustained hypocapnia is further compensated by renal response, a decrease in renal ammonium and titratable acid excretion and a reduction in filtered HCO_3^- reabsorption. The full expression of renal adaptation may take several days and depends on a normal volume status and renal function. The kidneys appear to respond directly to the lowered PaCO_2 rather than to the alkalemia per se, although both pH and PCO_2 may be factors. With chronic respiratory alkalosis, a 1 mm Hg fall in PaCO_2 causes a 0.4 to 0.5 mEq/L drop in HCO_3^- and a 0.003 unit rise in pH, or the $[\text{HCO}_3^-]$ will decrease 4 mEq/L for each 10 mm Hg decrease in PaCO_2 .⁴ Chronic respiratory alkalosis is an exception to the general rule that physiologic compensation is never 100% efficient, since some patients with this acid-base disorder may exhibit a normal arterial pH and are, therefore, fully compensated.

The effects of respiratory alkalosis vary according to its duration and severity and the underlying disease. Acute respiratory alkalosis causes intracellular shifts of sodium, potassium, and phosphate and reduces ionized calcium by increasing the protein-bound fraction of calcium based on the acute pH changes. A rapid decline in PaCO_2 may cause dizziness, mental confusion, and seizures, even in the absence of hypoxemia, as a consequence of reduced cerebral blood flow. The cardiovascular effects of acute hypocapnia in the awake human are generally minimal, but in the anesthetized or mechanically ventilated patient, cardiac output and blood pressure may fall because of the depressant effects of anesthesia and positive pressure ventilation on heart rate, systemic resistance, and venous return. Cardiac rhythm disturbances may occur in patients with coronary artery disease as a result of changes in oxygen unloading by blood from a left shift in the hemoglobin-oxygen dissociation curve (Bohr effect). Hypocapnia-induced hypokalemia is usually minor.⁴

Respiratory alkalosis is among the most common acid-base disturbances in critically ill patients (often as a component of a mixed disorder) and, when severe, portends a poor prognosis. Many cardiopulmonary disorders manifest respiratory alkalosis in the early to intermediate stages. Hyperventilation usually results in hypocapnia. Respiratory alkalosis is a common occurrence during mechanical ventilation. In those with respiratory disorders or distress, the finding of normocapnia (particularly after prior hypocapnia) and increasing hypoxemia may herald the onset of rapid respiratory failure and should prompt an assessment to determine whether the patient is becoming fatigued.

The causes of respiratory alkalosis are summarized in Table 16.5. The hyperventilation syndrome may mimic a number of serious conditions and may be disabling. Paresthesias, circumoral numbness, chest wall tightness or pain, dizziness, inability to take an adequate breath, and, rarely, tetany may be themselves sufficiently stressful to perpetuate a vicious circle. ABG analysis demonstrates an acute or chronic respiratory alkalosis, often with hypocapnia in the range of 15 to 30 mm Hg and no hypoxemia. Central nervous system diseases or injury can produce several patterns of hyperventilation with sustained arterial PaCO_2 levels of 20 to 30 mm Hg. Salicylates, the most common cause of drug-induced respiratory alkalosis, stimulate the medullary chemoreceptor directly. The methylxanthine drugs theophylline and aminophylline stimulate ventilation and increase the ventilatory response to CO_2 . High progesterone levels increase ventilation and decrease the arterial PaCO_2 by as much as 5 to 10 mm Hg. Thus, chronic respiratory alkalosis is an expected feature of pregnancy. Respiratory alkalosis is a prominent feature in liver failure, and its severity correlates well with the degree of hepatic insufficiency and mortality. Respiratory alkalosis is common in patients with gram-negative septicemia, and it is often an early finding, before fever, hypoxemia, and hypotension develop. It is presumed that some bacterial product or toxin acts as a respiratory center stimulant, but the precise mechanism remains unknown.

The diagnosis of respiratory alkalosis requires measurement of arterial pH and PaCO_2 (higher and lower than normal, respectively). The plasma K^+ concentration is often reduced, and the serum Cl^- concentration increased. In the acute phase, respiratory alkalosis is not associated with increased renal HCO_3^- excretion, but within hours, net acid excretion is reduced. In general, the HCO_3^- concentration falls by 2.0 mEq/L for each 10 mm Hg decrease in PaCO_2 acutely. Chronic hypocapnia reduces the serum bicarbonate concentration by 4 to 5 mEq/L for each 10 mm Hg decrease in PaCO_2 . It is unusual to observe a plasma bicarbonate concentration below 12 mEq/L as a result of a pure respiratory alkalosis. When a diagnosis of hyperventilation or respiratory alkalosis is made, the cause should be investigated. The diagnosis of hyperventilation syndrome is made by exclusion. In some cases, it may be important to rule out other conditions such as pulmonary embolism, coronary artery disease, pneumothorax, and hyperthyroidism.

The treatment of respiratory alkalosis is primarily directed toward alleviation of the underlying disorder. Respiratory alkalosis is rarely life-threatening. Direct measures to correct respiratory alkalosis will be unsuccessful if the underlying stimulus remains unchecked. If respiratory alkalosis complicates ventilator management, changes in dead space, tidal volume, and frequency can minimize the hypocapnia. Patients with hyperventilation syndrome may benefit from reassurance, rebreathing into a paper bag (with oxygenation monitoring) during symptomatic attacks, and attention to underlying psychologic stress. Antidepressants and sedatives are not recommended, although in a few patients, β -adrenergic blockers may help to ameliorate distressing peripheral manifestations of the hyperadrenergic state. The relationship of (and treatment of) recurrent hyperventilation syndrome and psychological disorders and panic attacks can be explored.

METABOLIC DISORDERS

METABOLIC ACIDOSIS

Metabolic acidosis occurs mechanistically as a result of: 1) a marked increase in endogenous production of acid (such as L-lactic acid and ketoacids) such that the ability of the kidneys to respond is exceeded, 2) loss of HCO_3^- or potential HCO_3^- salts (diarrhea or renal tubular acidosis [RTA]), or 3) inability of the kidneys to excrete acid resulting in progressive accumulation of endogenous acids. However, empirically for diagnostic purposes, metabolic acidosis is frequently classified according to the anion gap (AG): high anion gap metabolic acidosis or non-anion gap acidosis (or probably more appropriately, hyperchloremic metabolic acidosis). The AG, which can be corrected for the prevailing albumin concentration,²⁷ serves a useful role in the diagnosis of the metabolic acidoses and should always be considered. Metabolic acidosis with a normal AG (hyperchloremic or non-AG acidosis) suggests that HCO_3^- has been effectively replaced by Cl^- . Thus, the AG does not change.

In contrast, a high anion gap usually indicates accumulation of an anion other than hydrochloric acid in the ECF. Metabolic acidosis with a high AG (see Table 16.3) reflects that an anion replaces titrated HCO_3^- without disturbing the Cl^- concentration. Hence, the acidosis is normochloremic and the AG increases. In some cases, a disease process may cause an acidosis that is either hyperchloremic (non-anion gap) or high-anion gap; early CKD and diabetic ketoacidosis in well-hydrated persons are the best examples. The excretion rate of the accompanying anion (e.g., acetoacetate in ketoacidosis) is the determining factor in which type acidosis is found in a given person.^{24,28}

HYPERCHLOREMIC (NON-ANION GAP) METABOLIC ACIDOSES

The diverse clinical disorders that may result in hyperchloremic or non-gap metabolic acidosis are outlined in Table 16.6. Because a reduced plasma HCO_3^- and elevated Cl^- concentration may also occur in chronic respiratory alkalosis, it is often important to confirm the acidemia by measuring arterial pH. Normal anion gap metabolic acidosis occurs most often as a result of loss of HCO_3^- from the gastrointestinal tract (diarrhea) or as a result of a renal acidification defect. The majority of disorders in this category can be attributed to one of two major causes: 1) loss of bicarbonate from the gastrointestinal tract (diarrhea) or from the kidney (proximal RTA), or 2) inappropriately low renal acid excretion (classical distal RTA [cDRTA], type 4 RTA, or renal failure). Hypokalemia may accompany both gastrointestinal loss of HCO_3^- and proximal RTA and cDRTA. Therefore, a major challenge in some cases is distinguishing these causes to be able to determine whether the response of renal tubular function to the prevailing acidosis is appropriate (gastrointestinal origin) or inappropriate (renal origin).

Diarrhea results in the loss of large quantities of HCO_3^- or potential HCO_3^- (organic acid anions such as propionate and butyrate which could be metabolized to HCO_3^-) and therefore, metabolic acidosis frequently results. Volume depletion and hypokalemia often coexist due to sodium and

Table 16.6 Differential Diagnosis of Non-Anion Gap (Hyperchloremic) Metabolic Acidosis

Gastrointestinal Bicarbonate Loss

Diarrhea

External pancreatic or small bowel drainage

Uterosigmoidostomy, jejunal loop

Drugs

Calcium chloride (acidifying agent)

Magnesium sulfate (diarrhea)

Cholestyramine (bile acid diarrhea)

Renal Acidosis

Hypokalemia

Proximal RTA (type 2)

Distal (classical) RTA (type 1)

Drug-induced acidosis

Acetazolamide and topiramate (proximal RTA)

Amphotericin B (distal RTA)

Ifosfamide

Hyperkalemia

Generalized distal nephron dysfunction (type 4 RTA)

Mineralocorticoid deficiency (e.g., **low renin/aldosterone in diabetic nephropathy**)

Mineralocorticoid resistance (PHA-1 autosomal dominant)

Voltage defects (PHA-1, autosomal recessive)

PHA-2

↓ Na^+ delivery to distal nephron

Tubulointerstitial disease

Drug-induced acidosis

Potassium-sparing diuretics (amiloride, triamterene, spironolactone)

Trimethoprim

Pentamidine

Angiotensin-converting enzyme inhibitors

Angiotensin II receptor blockers

Nonsteroidal antiinflammatory drugs

Cyclosporine, Tacrolimus

Normokalemia

Chronic kidney disease (stage 3-4)

Other

Acid loads (ammonium chloride, hyperalimentation)

Loss of potential bicarbonate: ketosis with ketone excretion

Dilution acidosis (rapid saline administration)

Hippurate

Cation exchange resins

PHA, Pseudohypoaldosteronism; RTA, renal tubular acidosis.

Bold = common in adults.

K^+ losses. Hypokalemia is additionally from volume depletion causing secondary hyperaldosteronism, which enhances renal K^+ secretion by the collecting duct.

Although a low urine pH may be found with diarrhea, consistent with metabolic acidosis and appropriate renal response, in some cases a urine pH of 6.0 or more may be found. This occurs because chronic metabolic acidosis and hypokalemia increase renal NH_4^+ synthesis and excretion,

which can increase urine pH. Therefore, the urine pH, when 6.0 or higher, may erroneously suggest a renal cause of acidosis. Nevertheless, metabolic acidosis caused by gastrointestinal losses with a high urine pH can be differentiated from renal causes by measuring or estimating urinary NH_4^+ , as described below. Urinary NH_4^+ excretion is typically low in patients with renal causes of acidosis and high in patients with diarrhea.^{32,33} Urine acidification and pH may also be low with marked volume depletion, which impairs delivery of distal sodium delivery which in turn limits distal acidification.

The level of urinary NH_4^+ excretion (not usually measured by clinical laboratories) in metabolic acidosis can be assessed indirectly³⁴ by calculating the urine anion gap (UAG) or urine osmolal gap. The UAG is:

$$\text{UAG} = [\text{Na}^+ + \text{K}^+]_u - [\text{Cl}^-]_u \quad (16.12)$$

where u denotes the urine concentration of these electrolytes. The rationale for using the UAG as a surrogate for ammonium excretion is that, in chronic metabolic acidosis, ammonium excretion should be elevated if renal tubular function is intact. Because ammonium is a cation, it should balance part of the negative charge of chloride in the previous expression, assuming there is not a lot of HCO_3^- in the urine as in an alkaline urine. Therefore, the UAG should become progressively negative as the rate of ammonium excretion increases in response to acidosis or to acid loading.^{24,32} NH_4^+ can be assumed to be present if the sum of the major cations ($\text{Na}^+ + \text{K}^+$) is less than the concentration of Cl^- in urine. A negative UAG (more than -20 mEq/L) implies that appropriate NH_4^+ is present in the urine with the acidosis, as might occur with an extrarenal origin of the hyperchloremic acidosis, e.g., diarrhea. Conversely, urine estimated to contain little or no NH_4^+ has more $\text{Na}^+ + \text{K}^+$ than Cl^- (UAG is positive),³² which indicates a renal mechanism for the hyperchloremic acidosis, such as in cDRTA (with hypokalemia) or hypoaldosteronism with hyperkalemia. Note that this qualitative test is useful only in the differential diagnosis of a non-gap metabolic acidosis. If the patient has ketonuria, drug anions (penicillins or aspirin), or toluene metabolites in the urine, the test is not reliable and should not be used. And the test is controversial in the sense that its reliability and utility remain in question.

The urinary ammonium ($\text{U}_{\text{NH}_4^+}$) may also be estimated more reliably from the urine osmolal gap, which is the difference in measured urine osmolality (U_{osm}), and the urine osmolality calculated from the urine $[\text{Na}^+ + \text{K}^+]$ and the urine urea and glucose (all expressed in mmol/L):

$$\text{U}_{\text{NH}_4^+} = 0.5 (\text{U}_{\text{osm}} - [2 \text{Na}^+ + \text{K}^+]_u + \text{urea}_u + \text{glucose}_u) \quad (16.13)$$

Urinary ammonium concentrations of 75 mEq/L or more would be anticipated if renal tubular function is intact and the kidney is responding to a prevailing metabolic acidosis by increasing ammonium production and excretion. Conversely, values below 25 mEq/L in the presence of acidosis denote inappropriately low urinary ammonium concentrations. In addition to the UAG and osmolal gap, the fractional excretion of Na^+ may be helpful and would be expected to be low ($<1\%$ to 2%) in patients with HCO_3^- loss from the gastrointestinal tract but usually exceeds 2% to 3% in patients with RTA.^{33,35}

Gastrointestinal HCO_3^- loss, as well as proximal RTA (type 2) and cDRTA (type 1), results in ECF contraction and

stimulation of the renin-aldosterone system, which leads typically to hypokalemia. The serum K^+ concentration therefore serves to distinguish the previous disorders, which have a low K^+ , from either generalized distal nephron dysfunction (e.g., type 4 RTA), in which the renin-aldosterone-distal nephron axis is abnormal and hyperkalemia exists, or the acidosis of progressive chronic kidney disease, in which normokalemia or mild hyperkalemia is common (see later).

In addition to gastrointestinal tract HCO_3^- loss, external loss of pancreatic and biliary secretions, as well as cholestyramine, calcium chloride, and magnesium sulfate ingestion can all cause a non-gap acidosis (see Table 16.6), especially in patients with renal insufficiency. Coexistent lactic acidosis is common in severe diarrheal illnesses and noted by an increase in the AG.

Severe non-gap or hyperchloremic metabolic acidosis with hypokalemia may occur in patients with ureteral diversion procedures. Because the ileum and the colon are both endowed with $\text{Cl}^-/\text{HCO}_3^-$ exchangers, when the Cl^- from the urine enters the gut, or pouch, the HCO_3^- concentration increases as a result of the exchange process.²⁴ Moreover, K^+ secretion is stimulated, which, together with HCO_3^- loss, can result in a hyperchloremic hypokalemic metabolic acidosis. This defect is particularly common in patients with uretero-sigmoidostomies and is more common with this type of diversion because of the prolonged transit time of urine caused by stasis in the colonic segment.

Dilutional acidosis, acidosis caused by exogenous acid loads and the posthypocapnic state, can usually be excluded by the history. When isotonic saline is infused rapidly, particularly in patients with temporary or permanent renal functional impairment, the serum HCO_3^- declines reciprocally in relation to Cl^- ,²⁴ in part due the dilutional effect of the fluid. Hyperchloremic acidosis can also be caused by administration of acid or acid equivalents such as infusion of arginine or lysine hydrochloride during parenteral hyperalimentation or ingestion of ammonium chloride.

Hyperchloremic metabolic acidosis may also occur in some settings of ketoacidosis. In mild, chronic ketoacidosis if GFR is maintained with adequate ECFV expansion and renal ketone excretion is high, the continued titration of HCO_3^- with Cl^- retention and excretion of potential base (ketones) may result in hyperchloremic acidosis. A similar situation could contribute to continued acidosis during recovery from typical high anion gap ketoacidosis when the sodium salts of ketones may be excreted and lost as potential HCO_3^- .³⁶ A similar mechanism for hyperchloremic acidosis in early CKD may occur with metabolism of sulfur to sulfuric acid and excretion of SO_4^{2-} with Cl^- retention.

Loss of functioning renal parenchyma in progressive kidney disease is known to be associated with metabolic acidosis. Typically, the acidosis is a hyperchloremic, non-gap type when the GFR is between 20 and 50 mL/min but may convert to the typical high AG acidosis of uremia with more advanced renal failure, that is, when the GFR is less than 20 mL/min.³⁷ It is generally assumed that such progression is observed more commonly in patients with tubulointerstitial forms of renal disease, but non-anion gap metabolic acidosis can also occur with advanced glomerular disease. The acidoses of CKD with an anion gap and hyperkalemic, hyperchloremic metabolic acidosis are discussed subsequently below. The principal defect in acidification of advanced renal failure is

that ammoniogenesis is reduced in proportion to the loss of functional renal mass. In addition, medullary NH_4^+ accumulation and trapping in the outer medullary collecting tubule may be impaired.³⁷ Because of adaptive increases in K^+ secretion by the collecting duct and colon, the acidosis of chronic renal insufficiency is typically normokalemic.³⁷

Non-anion gap metabolic acidosis accompanied by hyperkalemia is almost always associated with a generalized dysfunction of the distal nephron.^{33,35} However, K^+ -sparing diuretics (amiloride, triamterene), as well as pentamidine, cyclosporine, tacrolimus, nonsteroidal antiinflammatory drugs (NSAIDs), angiotensin-converting enzyme (ACE) inhibitors, angiotensin II receptor blockers (ARBs), beta-blockers, and heparin may mimic or cause this disorder, resulting in hyperkalemia and a non-gap metabolic acidosis.^{33,35} Because hyperkalemia augments the development of acidosis by suppressing urinary net acid excretion, discontinuing these agents while reducing the serum K^+ allows ammonium production and excretion to increase, which will help repair the acidosis.

PROXIMAL RENAL TUBULAR ACIDOSIS

Physiology

The proximal tubule reabsorbs 80% of the large amount of filtered HCO_3^- so that defects in the proximal tubule cause initial profound loss of HCO_3^- into the urine.⁶ When HCO_3^- that is not absorbed floods the distal nephron, the HCO_3^- delivery exceeds the distal nephron's capacity to reabsorb HCO_3^- , and bicarbonaturia ensues. Enhanced Cl^- reabsorption, stimulated by ECF volume contraction, also results in a hyperchloremic metabolic acidosis. As serum HCO_3^- levels fall, less will be filtered, eventually reaching a level of distal delivery of HCO_3^- such that the distal nephron can reabsorb the remaining HCO_3^- , and the distal nephron can acidify the urine (to $\text{pH} < 5.5$) (Fig. 16.1). A steady state with lower serum HCO_3^- is reached in which acid excretion is normal. The serum HCO_3^- concentration usually reaches a nadir of 15 to 18 mEq/L, so that systemic acidosis is not progressive. With bicarbonate administration, the amount of bicarbonate in the urine increases dramatically when blood levels increase with the fractional excretion of bicarbonate ($\text{FE}_{\text{HCO}_3^-}$) rising to 10% to 15%, and the urine pH becomes alkaline.³³

PATHOGENESIS – INHERITED AND ACQUIRED FORMS

Proximal RTA can present in one of three ways: isolated HCO_3^- defect, generalized proximal tubule defect, and as a part of a mixed proximal/distal RTA (type 3). Inheritance patterns for isolated proximal RTA include autosomal recessive and autosomal dominant. Isolated, recessive proximal RTA with accompanying ocular abnormalities has been most studied and has been attributed to missense mutations of the gene *SLC4A4* that encodes for the basolateral transporter, NBCe1. A rare variant, inherited as an autosomal dominant trait with short stature, has been described and appears to be a mutation of the gene that encodes the apical Na^+/H^+ exchanger, NHE-3.

Familial disorders associated with proximal RTA and generalized proximal defects, referred to as Fanconi's syndrome, include cystinosis, tyrosinemia, hereditary fructose intolerance, galactosemia, glycogen storage disease type 1, Wilson's disease, and Lowe's syndrome. These have defects

in reabsorption of many substances reabsorbed in the proximal tubule: glucose, amino acids, phosphate, uric acid, citrate. These substances spill into the urine. Numerous defects of the proximal tubule have been proposed and verified experimentally as feasible. Certain drugs (e.g., aminoglycosides, cisplatin, ifosfamide, tenofovir, valproic acid), heavy metal intoxications, and myeloma proteins can also cause Fanconi's syndrome.

Development of Fanconi's syndrome by intracellular PO_4^{3-} depletion has also been proposed in hereditary fructose intolerance, in which ingestion of fructose leads to accumulation of fructose 1-phosphate in the proximal tubule. Because these patients lack the enzyme fructose 1-phosphate aldolase, fructose 1-phosphate cannot be further metabolized, and intracellular PO_4^{3-} is sequestered in this form.^{33,38}

Vitamin D deficiency and proximal RTA with aminoaciduria and urinary phosphate wasting are also associated. Correction of vitamin D deficiency has allowed correction of the proximal tubule dysfunction.³⁹ Similar results have been obtained in patients with vitamin D-dependent and vitamin D-resistant rickets treated with dihydrotachysterol.³³ The mechanisms involved in the proximal tubule dysfunction are not yet clear.

Additionally, features of both proximal RTA (bicarbonate wasting), and distal acidification abnormalities are evident in patients with autosomal recessive RTA (mixed proximal and distal, or type 3 RTA) that has been attributed to a defect in CA2 that encodes for carbonic anhydrase II (CAII), an intracellular form of the enzyme distributed to the proximal tubule, TALH, DCT, CCD, and MCD.³³ The phenotype includes osteopetrosis, cerebral calcifications, and ocular abnormalities (Guibaud-Vainsel syndrome). Sly and associates, who worked out much of the pathophysiology, found no other evidence for proximal tubule dysfunction, and intact carbonic anhydrase IV.⁴⁰ Carbonic anhydrase II is present in the cytoplasm of renal cells, and thus an acidification defect occurring in association with its deficiency is not unexpected.

Clinical Spectrum

In general, proximal RTA is more common in children due to the genetic diseases noted above. In adults, the most common causes of acquired proximal RTA are multiple myeloma, in which increased excretion of immunoglobulin light chains injures the proximal tubule epithelium, and chemotherapeutic drug injury of the proximal tubule (e.g., ifosfamide). The light chains that cause injury in multiple myeloma often have a variable domain that is resistant to degradation by proteases in lysosomes in proximal tubule cells. Accumulation of the variable domain fragments may be responsible for the impairment in tubular function. Carbonic anhydrase inhibitors cause pure bicarbonate wasting but not Fanconi's syndrome. A comprehensive list of the disorders associated with proximal RTA is presented in Table 16.7.³³ Some of the entities on this list are no longer seen and are of only historic interest. For example, application of sulfanilamide to the skin of patients with large-surface-area burns is no longer practiced in most centers, but sulfanilamide, a carbonic anhydrase inhibitor, is absorbed from burned skin. Topiramate, widely used in the prevention of migraine headaches and treatment of seizure disorders, is a potent carbonic anhydrase inhibitor and an important cause of non-gap metabolic acidosis. As many as 15% to 25% of patients on topiramate will manifest a stable non-gap metabolic acidosis. Since the enzyme carbonic

Table 16.7 Proximal Renal Tubular Acidosis**Isolated Pure Bicarbonate Wasting (Unassociated With Fanconi's Syndrome)**

- Primary
 - Inherited autosomal recessive with ocular abnormalities (missense mutations of *SLC4A4*)
 - Autosomal dominant with short stature (mutation of *SLC9A3/NHE3*)
 - Carbonic anhydrase deficiency, inhibition, or alteration
 - Drugs: acetazolamide, topiramate, sulfanilamide, mafenide acetate
 - Carbonic anhydrase II deficiency with osteopetrosis (mixed proximal and distal RTA –type 3)

Generalized (Associated With Fanconi's Syndrome)

- Primary (without associated systemic disease)
 - Genetic
 - Sporadic
- Genetically transmitted systemic diseases
 - Cystinosis
 - Lowe's syndrome
 - Wilson's syndrome
 - Tyrosinemia
 - Galactosemia
 - Hereditary fructose intolerance (during fructose ingestion)
 - Metachromatic leukodystrophy
 - Pyruvate carboxylase deficiency
 - Methylmalonic acidemia
- Dysproteinemic states
 - Multiple myeloma, monoclonal gammopathy
- Secondary hyperparathyroidism with chronic hypocalcemia
 - Vitamin D deficiency or resistance, vitamin D dependency
- Drugs or toxins
 - Ifosfamide, lead, outdated tetracycline, 3-methylchromone, streptozotocin, amphotericin B (historic)
- Tubulointerstitial diseases
 - Sjögren's syndrome
 - Medullary cystic disease
 - Renal transplantation
- Other renal and miscellaneous diseases
 - Nephrotic syndrome, amyloidosis, paroxysmal nocturnal hemoglobinuria

anhydrase II is present in both the proximal and distal tubules, topiramate appears to cause a mixed form of renal tubular acidosis (RTA), having features of both proximal and distal RTA (type 3 RTA). This manifestation subsides when topiramate is discontinued. Pharmaceutical manufacturing techniques have improved, and outdated tetracycline is no longer associated with proximal RTA. Some of the agents and disorders on this list—such as ifosfamide, Sjögren's syndrome, renal transplantation, and amyloidosis—also appear as causes of distal RTA (see Table 16.7).

Diagnosis

The diagnosis of proximal RTA relies initially on the documentation of a chronic hyperchloremic metabolic acidosis. In the steady state these patients generally show chronic metabolic acidosis, an acid urine pH (< 5.5), and a low fractional excretion of HCO_3^- (when plasma levels are low). With alkali therapy or slow infusion of sodium bicarbonate

intravenously, bicarbonaturia ensues, and the urine becomes alkaline. When the plasma bicarbonate concentration is increased with an intravenous infusion of sodium bicarbonate at a rate of 0.5 to 1.0 mEq/kg/hr, the urine pH, even if initially acid, will increase once the reabsorptive threshold for bicarbonate has been exceeded. Thus, the urine pH may exceed 7.5 and $\text{FE}_{\text{HCO}_3^-}$ will increase dramatically to 15% to 20%, making it very difficult to increase serum HCO_3^- levels to the normal range.

Hypokalemia is usually present. If bicarbonate administration has been high in an attempt to repair the acidosis, the bicarbonaturia will drive kaliuresis and the hypokalemia may be severe.³³ Patients with proximal tubule dysfunction exhibit intact distal nephron function (generate steep urine pH gradients and titrate luminal buffers) when the serum HCO_3^- concentration and hence distal HCO_3^- delivery are sufficiently reduced. A low HCO_3^- threshold exists. Below this plasma HCO_3^- concentration, distal acidification can compensate for defective proximal acidification, although at the expense of systemic metabolic acidosis. When the plasma HCO_3^- concentration is raised to normal values, a large fraction of the filtered HCO_3^- is inappropriately excreted because the limited reabsorptive capacity of the distal nephron cannot compensate for the reduced proximal nephron reabsorption.

Associated Clinical Features

K^+ excretion is typically high in patients with proximal RTA, especially during NaHCO_3 administration.³³ Kaliuresis is promoted by the increased delivery of a relatively impermeant anion, HCO_3^- , to the distal nephron in the setting of secondary hyperaldosteronism, which is due to mild volume depletion. Therefore, correction of acidosis in such patients leads to an exaggeration of the kaliuresis and K^+ deficiency.

If the acidification defect is part of a generalized proximal tubule dysfunction (Fanconi's syndrome), such patients will have hypophosphatemia, hyperphosphaturia, hypouricemia, hyperuricosuria, glycosuria, aminoaciduria, hypercitratemia, hypercalciuria, and proteinuria.

Although Ca^{2+} excretion may be high in patients with proximal RTA, nephrocalcinosis and renal calculi are rare. This may be related to the high rate of citrate excretion in patients with proximal RTA compared with that of most patients with acidosis from other causes. Osteomalacia, rickets, abnormal gut Ca^{2+} and phosphorus absorption, and abnormal vitamin D metabolism in children are common, although not invariably present. Adults tend to have osteopenia without pseudofractures.³³

The proximal reabsorption of filtered low-molecular-weight proteins may also be abnormal in proximal RTA. Lysozymuria and increased urinary excretion of immunoglobulin light chains can occur.³³

Treatment

The magnitude of the bicarbonaturia (>10% of the filtered load) at a normal HCO_3^- concentration requires that large amounts of HCO_3^- be administered for treatment. At least 10 to 30 mEq/kg/day of HCO_3^- or its metabolic equivalent (citrate) is required to maintain plasma HCO_3^- concentration at normal levels. Correcting the HCO_3^- to near normal values (22 to 24 mEq/L) is desirable in children to reestablish normal growth. Correction to this level is less desirable in

adults. Large supplements of K^+ are often necessary because of the kaliuresis induced by high distal HCO_3^- delivery when the plasma HCO_3^- concentration is normalized. Thiazides have proved useful in diminishing therapeutic requirements for HCO_3^- supplementation by causing ECF contraction to stimulate proximal absorption. However, K^+ wasting continues to be a problem, often requiring the addition of a K^+ -sparing diuretic.³³ Vitamin D and PO_4^{3-} may be supplemented and in some patients even improve the acidification defect. Fructose should be restricted in patients with fructose intolerance.³⁸

DISTAL RENAL TUBULE ACIDOSIS (CLASSIC WITH HYPOKALEMIA)

Pathophysiology

The mechanisms involved in the pathogenesis of hypokalemic distal RTA, or classic distal RTA, cDRTA (type 1 RTA) have been established for many cases by modern genetic and molecular methods. The inherited forms of cDRTA can be due to defects in the basolateral HCO_3^-/Cl^- exchanger (SLC4A1), or subunits of the H^+ -ATPase (ATP6V1B1 and ATP6V0a4).

Defects in these transport pathways and an increase in apical membrane permeability are displayed in Fig. 16.6, which depicts acid-base transporters of a type A intercalated cell in the medullary collecting duct and the possible abnormalities causing cDRTA. Although the classical feature of this entity is an inability to acidify the urine maximally (to a pH of <5.5) in the face of systemic acidosis, attention to urine ammonium excretion rather than urine pH alone is necessary to diagnose this disorder.^{32,33} The pathogenesis of the acidification defect in most patients has been shown previously by the response of the urine PCO_2 to sodium bicarbonate infusion. When normal subjects are given large infusions of sodium bicarbonate to produce a high HCO_3^- excretion, distal nephron H^+ secretion leads to the generation of a high PCO_2 in the renal medulla and final urine.⁴¹ The magnitude of the urinary PCO_2 (often referred to as the *urine minus blood PCO_2* or

U – B PCO_2) has been and can be used as an index of distal nephron H^+ secretory capacity.^{39,42} The *U – B PCO_2* is generally subnormal in classical hypokalemic distal RTA, with the notable exception of amphotericin B-induced distal RTA, which remains the most common example of the “gradient” or permeability defect.^{41–43} A leakage defect independent of amphotericin B has been reported or postulated in rare patients with cDRTA without exposure to the antibiotic, but has never been documented unequivocally.⁴⁴

Patients with impaired collecting duct H^+ secretion and cDRTA also exhibit uniformly low excretory rates of NH_4^+ when the degree of systemic acidosis is taken into account.^{7,32,33} Low NH_4^+ excretion equates with inappropriately low renal regeneration of HCO_3^- , which indicates that the kidney is responsible for causing or perpetuating the chronic metabolic acidosis. Low NH_4^+ excretion in classical hypokalemic distal RTA occurs because of the failure to trap NH_4^+ in the medullary collecting duct as a result of higher than normal tubule fluid pH in this segment and loss of the disequilibrium pH (pH > 6.0).⁴⁵ The high urine pH indicates impaired H^+ secretion.

Inherited and Acquired Defects in Type A Intercalated Cell Acid-Base Transporters Responsible for cDRTA. Recessive cDRTA with deafness is caused by loss-of-function mutations in either of two subunits of the H^+ -ATPase of type A intercalated cells (the B1 subunit of the V1 cytoplasmic ATPase, and the a4 subunit of the V0 transmembrane complex). Dominant and recessive forms of cDRTA are also caused by loss-of-function mutations in the basolateral membrane AE1 Cl^-/HCO_3^- exchanger of type A intercalated cells. Depending on the type, associated erythroid changes may be mild or severe.^{46,47} Karet and colleagues⁴⁸ have described two different mutations in the *ATP6V1B1* gene encoding the B1 subunit of H^+ -ATPase. One defect is associated with sensorineural deafness (*rdRTA1*) and the other with normal hearing (*rdRTA2*).⁴⁹ The former recessive disorder is manifest in the first year of life as a failure to thrive; bilateral sensorineural hearing deficits; hyperchloremic, hypokalemic metabolic acidosis; severe nephrolithiasis; nephrocalcinosis; and osteodystrophy. The H^+ -ATPase is critical for maintaining pH in the cochlea and endolymph, and its loss in this disorder explains the hearing deficit as well as the renal tubule acidification defect. An autosomal recessive form that is much less commonly associated with deafness is due to a defect in the *ATP6V0a4* gene that encodes for the H^+ -ATPase a4 subunit.^{50,51} The genetic and molecular basis of distal RTA is outlined in Table 16.8.

Acquired defects of H^+ -ATPase have been demonstrated in renal biopsy specimens of patients with Sjögren's syndrome with evidence of classical hypokalemic distal RTA.³³ These biopsy specimens revealed an absence of H^+ -ATPase protein in the apical membrane of type A intercalated cells.

Theoretically, abnormalities in H^+ , K^+ -ATPase could also result in both hypokalemia and metabolic acidosis. This has been experimentally supported by rat studies using vanadate. Of note, an unusually high incidence of hypokalemic distal RTA (endemic RTA) has been observed in northeastern Thailand.⁵² To date, no genetic linkages between H^+ - K^+ -ATPase genes and inherited forms of cDRTA have been documented. Nevertheless, such an abnormality has been suggested in an infant with severe metabolic acidosis and hypokalemia.³³

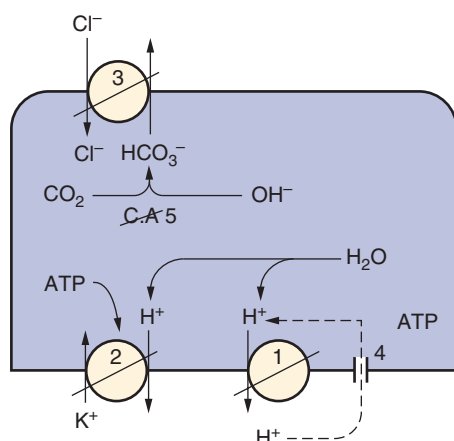


Fig. 16.6 Type A intercalated cell of the collecting duct displaying five pathophysiologic defects that could result in classical distal renal tubular acidosis: defective H^+ -ATPase, defective H^+ - K^+ -ATPase, defective HCO_3^-/Cl^- exchanger, H^+ leak pathway, and defective intracellular carbonic anhydrase (type II). ATP, Adenosine triphosphate.

Table 16.8 Genetic and Molecular Bases of Distal Renal Tubular Acidoses (RTA)**Classical Distal RTA****Inherited**

Autosomal dominant	Defect in AE1 gene encodes for missense mutation in the $\text{HCO}_3^-/\text{Cl}^-$ exchanger (band 3 protein) Transporter may be mistargeted to apical membrane
Autosomal recessive With deafness	Mutations in ATP6V1B1 encoding B1-subunit of the apical H^+ -ATPase in distal tubule (rdRTA1)
With normal hearing Carbonic anhydrase II	Mutations in ATP6V0a4 (rdRTA2) Defect in CA II in red blood cells, bone, kidney
Endemic (northeastern Thailand)	Possible abnormality in H^+ , K^+ -ATPase
Acquired	Reduced apical expression of H^+ -ATPase (Sjögren's syndrome)

Generalized Distal Nephron Dysfunction**Pseudohypoaldosteronism type 1**

Autosomal recessive	Loss-of-function mutations of ENaC; mutations of genes encoding three subunits of ENaC
Autosomal dominant	Heterozygous mutations of mineralocorticoid receptor gene
Pseudohypoaldosteronism type 2	Mutations in WNK1 and WNK4, and in cullin-RING ligase (CRL) proteins cullin 3 (Cul3) 4 and kelch-like 3 (KLHL3) constitutively activate NCCT, increasing NaCl absorption in distal convoluted tubule

ENaC, Epithelial sodium channel.

Clinical Spectrum and Associated Features

The phenotypic hallmark of classical hypokalemic distal RTA, when fully expressed, is the inability to acidify the urine appropriately during spontaneous or chemically induced metabolic acidosis, and is a disease of type A intercalated cells of the collecting duct. The defect in acidification by the collecting duct secondarily impairs NH_4^+ and titratable acid excretion and results in positive acid balance, hyperchloremic metabolic acidosis, and volume depletion.^{33,53–55} Moreover, medullary interstitial disease, which commonly occurs in conjunction with distal RTA, may impair NH_4^+ excretion by interrupting the medullary countercurrent system for NH_4^+ .^{32,33,53,54} The complete form of classical distal RTA is manifest by a non-anion gap metabolic acidosis. The clinical spectrum of complete cDRTA may include stunted growth, hypercalciuria, hypocitraturia, osteopenia, nephrolithiasis, and nephrocalcinosis, all as direct consequence of the chronic non-gap metabolic acidosis. The dissolution of bone is due to calcium resorption and mobilization from bone in response

to the acidosis³³ and through activation of the pH sensitive G-protein coupled receptor, OGR1 or *GPR68*, which resides in bone.⁵⁶ Interestingly OGR1/*GPR68* is also a pH- and shear stress-sensor on vascular endothelial cells. Other common electrolyte abnormalities, not due to acidosis include hypokalemia, hypernatremia and salt wasting, and polyuria due to nephrogenic diabetes insipidus. The hypokalemia, previously attributed to volume depletion and activation of the renin-angiotensin-aldosteronism system, may be due to a signaling pathway involving activation and release of PGE2 by beta intercalated cells that directly communicate to enhance sodium absorption and potassium secretion by activation of the epithelial sodium channel (ENaC) in collecting duct principal cells. Because chronic metabolic acidosis also increases proximal tubule reabsorption of citrate,^{7,32,33} the resulting hypocitraturia in combination with hypercalciuria creates an environment favorable for urinary stone formation and nephrocalcinosis. Nephrocalcinosis appears to be a reliable marker for cDRTA because nephrocalcinosis does not occur in proximal RTA or with generalized dysfunction of the nephron associated with hyperkalemia.^{32,33} Nephrocalcinosis probably aggravates further the reduction in net acid excretion by impairing the transfer of ammonia from the loop of Henle into the collecting duct. Pyelonephritis is a common complication of distal RTA, especially in the presence of nephrocalcinosis, and eradication of the causative organism may be difficult.³³ Distal RTA occurs frequently in patients with Sjögren's syndrome.⁵⁷

The clinical spectrum of cDRTA is outlined in detail in Table 16.9.^{32,33,55,57}

Treatment. Correction of chronic metabolic acidosis can usually be achieved readily in patients with cDRTA by administration of alkali in an amount sufficient to neutralize the production of metabolic acids derived from the diet and metabolism.³³ The goal is to correct the plasma HCO_3^- concentration to normal (25 mEq/L) and should be monitored frequently. In adult patients with distal RTA, this amount may be equal to no more than 1 to 3 mEq/kg/day.⁵⁸ In growing children, endogenous acid production is usually between 2 and 3 mEq/kg/day but may, on occasion, exceed 5 mEq/kg/day. Larger amounts of bicarbonate must be administered to fully correct the acidosis and maintain normal growth.^{32,33} The various forms of alkali replacement are outlined in Table 16.10.

In adult patients with distal RTA, correction of acidosis with alkali therapy reduces urinary K^+ excretion and typically corrects the hypokalemia and Na^+ depletion.³³ Therefore, in most adult patients with distal RTA, K^+ supplementation is usually not necessary once potassium has been corrected initially. Frank wasting of K^+ may occur in a minority of adult patients and in some children in association with secondary hyperaldosteronism despite correction of the acidosis by alkali therapy, so that K^+ supplementation is needed. If required, potassium can be administered as potassium bicarbonate (K-Lyte 25 or 50 mEq), potassium citrate (Urocit-K), or potassium citrate combination products (PolyCitra, K-Shohl's solution).^{32,33} Maintenance of a normal serum bicarbonate concentration with alkali therapy also raises urinary citrate level, reduces urinary calcium excretion, lowers the frequency of nephrolithiasis, and tends to correct bone disease and restore normal growth in children.^{58,59} Therefore, every

Table 16.9 Classical Distal Renal Tubular Acidosis**Primary****Familial**

Autosomal dominant

AE1 gene

Autosomal recessive

With deafness (rdRTA1 or ATP6V1B1 gene)

Without deafness (rdRTA2 or ATP6V0A4)

Sporadic

Endemic

Northeastern Thailand

Secondary to Systemic Disorders**Autoimmune Diseases: Sjögren's syndrome:**

Hyperglobulinemic purpura, cryoglobulinemia, thyroiditis, human immunodeficiency syndrome nephropathy, fibrosing alveolitis, chronic active hepatitis, primary biliary cirrhosis, polyarthritides nodosa

Hypercalciuria and Nephrocalcinosis: Primary

hyperparathyroidism, vitamin D intoxication, hyperthyroidism, idiopathic hypercalciuria, medullary sponge kidney, Wilson's disease, Fabry's disease, hereditary fructose intolerance, X-linked hypophosphatemia

Drug- and Toxin-Induced Disease: Amphotericin B, toluene, cyclamate, mercury, hepatic cirrhosis, vanadate, ifosfamide, lithium, classical analgesic nephropathy, foscarnet**Tubulointerstitial Diseases:** Balkan nephropathy, kidney transplantation, chronic pyelonephritis, leprosy, obstructive uropathy, jejunoileal bypass with hyperoxaluria, vesicoureteral reflux**Associated with Genetically Transmitted Diseases:**

Ehlers-Danlos syndrome, hereditary elliptocytosis, sickle cell anemia, Marfan's syndrome, medullary cystic disease, jejunal bypass with hyperoxaluria, hereditary sensorineural deafness, carnitine palmitoyltransferase deficiency, osteopetrosis with carbonic anhydrase II deficiency (mixed proximal-distal RTA - type 3)

Table 16.10 Forms of Alkali Replacement

Shohl's Solution: Na ⁺ epithelial sodium channel citrate 500 mg, citric acid 334 mg/5 mL	Each 1 mL contains 1 mEq sodium and is equivalent to 1 mEq of bicarbonate
NaHCO ₃ tablets	3.9 mEq/tablet (325 mg) 7.8 mEq/tablet (650 mg)
Baking soda	60 mEq/teaspoon
K-Lyte	25-50 mEq/tablet
Na ⁺ citrate and K ⁺ citrate (Virtrate-2 or Cytra-3) Na ⁺ citrate 500 mg, and K ⁺ citrate 550 mg/5 mL	Each 1 mL contains 1 mEq potassium and 1 mEq sodium and is equivalent to 2 mEq bicarbonate
K ⁺ citrate 550 mg, citric acid 334 mg/5 mL	
Polyitra-K crystals	Each packet contains 30 mEq potassium and is equivalent to 30 mEq bicarbonate
K ⁺ citrate 3300 mg, citric acid 1002 mg/packet	
Urocit-K tablets, K ⁺ citrate	5 or 10 mEq/tablet

chloremic metabolic acidosis are outlined in detail in [Table 16.11](#).^{32,33} The classification of these disorders varies with some experts including both the frequent hyporeninemic hypoaldosteronism (e.g., in diabetics with mild to moderate CKD) and so-called voltage defects in type 4 RTA; other experts separate the voltage defect disorders as separate from type 4 RTA. All of these disorders have hyperkalemia and hyperchloremic metabolic acidosis that is disproportionate to any decreased GFR.

The regulation of potassium excretion is primarily the result of regulation of potassium secretion, which responds to hyperkalemia, aldosterone, sodium delivery, acid-base status, and nonreabsorbable anions in the CCD. Therefore, a clinical estimate of K⁺ transfer into that segment can be helpful to recognize hyperkalemia of renal origin. An abnormally low excretion of potassium in the face of hyperkalemia implies hyperkalemia of renal origin. Calculation of the transtubular potassium gradient (TTKG) can be helpful in this regard. When the TTKG is low (<5), or the fractional excretion of potassium (FE_{K+}) is less than 25%, in a hyperkalemic patient, it reveals that the collecting tubule is not responding appropriately to the prevailing hyperkalemia and that potassium secretion is impaired. In contrast, in hyperkalemia of nonrenal origin, the kidney should respond by increasing K⁺ secretion, as evidenced by a sharp increase in the TTKG. The TTKG calculation assumes that there is no significant net addition or absorption of K⁺ between the CCD and the final urine, that CCD tubular fluid osmolality is approximately the same as plasma osmolality, that "osmoles" are not extracted between CCD and the final urine, and that plasma [K⁺] approximates peritubular fluid [K⁺]. It is important to note that under certain clinical conditions, some or none of these assumptions may be entirely correct. With high urine flow rates, for example, the TTKG underestimates K⁺ secretory capacity in the hyperkalemic patient.

Hyperkalemia should also be regarded as an important mediator of the renal response to acid-base balance. Potassium status can affect distal nephron acidification by both direct and indirect mechanisms. First, the level of potassium in systemic blood is an important determinant of aldosterone

attempt should be made to correct and maintain a near-normal serum [HCO₃⁻] in all patients with cDRTA.

Severe hypokalemia with flaccid paralysis, metabolic acidosis, and hypocalcemia may occur in some patients under extreme circumstances and require immediate therapy. Initial increasing systemic pH with alkali therapy may worsen the hypokalemia, potentially resulting in respiratory failure from respiratory muscle paralysis. Therefore, immediate intravenous potassium replacement should be achieved prior to alkali administration.

HYPERKALEMIC RENAL TUBULAR ACIDOSIS, GENERALIZED DISTAL NEPHRON DYSFUNCTION (TYPE 4 RTA)

The coexistence of hyperkalemia and hyperchloremic (non-gap) metabolic acidosis often indicates a generalized dysfunction in the cortical and medullary collecting tubules. In the differential diagnosis, it is important to evaluate the functional status of the renin-aldosterone system and ECF volume. The specific disorders causing hyperkalemic hyper-

Table 16.11 Disorders With Dysfunction of Renal Acidification—Generalized Abnormality of Distal Nephron With Hyperkalemia**Mineralocorticoid Deficiency****Primary Mineralocorticoid Deficiency**

Combined deficiency of aldosterone, deoxycorticosterone, and cortisol

Addison's disease, bilateral adrenalectomy, bilateral adrenal destruction, hemorrhage, or carcinoma

Congenital enzymatic defects

21-hydroxylase deficiency, 3 β -hydroxydehydrogenase deficiency, Desmolase deficiency

Isolated (selective) aldosterone deficiency

Chronic idiopathic hypoaldosteronism, heparin (low-molecular weight or unfractionated) administration in critically ill patient, familial hypoaldosteronism, corticosterone methyl oxidase deficiency types 1 and 2, primary zona glomerulosa defect, transient hypoaldosteronism of infancy, persistent hypotension, and/or hypoxemia

Angiotensin II converting enzyme inhibition

Endogenous, **Angiotensin-converting enzyme inhibitors and angiotensin II receptor antagonists**

Secondary Mineralocorticoid Deficiency

Hyporeninemic hypoaldosteronism

Diabetic nephropathy, tubulointerstitial nephropathies, nephrosclerosis, nonsteroidal antiinflammatory agents, acquired immunodeficiency syndrome, immunoglobulin M monoclonal gammopathy, **obstructive uropathy**

Mineralocorticoid Resistance

PHA-1—autosomal dominant (human mineralocorticoid receptor defect)

Renal Tubular Dysfunction (Voltage Defect)

PHA-1—autosomal recessive

PHA-2—autosomal dominant

Drugs that interfere with Na⁺ channel function in the CCT

Amiloride, triamterene, Trimethoprim, pentamidine

Drugs that interfere with Na⁺-K⁺-ATPase in the CCT

Cyclosporine, tacrolimus

Drugs that inhibit aldosterone effect on the CCT

Spirolactone, epleronone

Disorders associated with tubulointerstitial nephritis and renal insufficiency

Lupus nephritis, methicillin nephrotoxicity, obstructive nephropathy, kidney transplant rejection, sickle cell disease, Williams' syndrome with uric acid nephrolithiasis

elaboration, which is also an important determinant of distal H⁺ secretion. Chronic hyperkalemia can suppress ammoniogenesis.^{60,61} These changes in ammonium production may also affect medullary interstitial ammonium concentration and buffer availability.⁶¹ Hyperkalemia has no effect on ammonium transport in the superficial proximal tubule but markedly impairs ammonium absorption in the thick ascending limb of the loop of Henle (TAL), reducing inner medullary concentrations of total ammonia and decreasing secretion

Table 16.12 Effects of Hyperkalemia on Ammonium Excretion

Decrease in NH₄⁺ production

Decrease in NH₄⁺ absorption in thick ascending limb of loop of Henle

Decrease in interstitial NH₄⁺ concentration

Impaired countercurrent multiplication

Decrease in NH₃/NH₄⁺ secretion into outer and inner medullary collecting ducts

of NH₃ into the inner medullary collecting duct. Hyperkalemia may also decrease entry of NH₄⁺ into the medullary collecting duct through competition of NH₄⁺ and K⁺ for the K⁺-secretory site on the basolateral membrane Na-K-ATPase (Figs. 16.7 and 16.8).^{26,62}

In summary, hyperkalemia may have a dramatic impact on ammonium production and excretion (Table 16.12). Chronic hyperkalemia decreases ammonium production in the proximal tubule and whole kidney, inhibits absorption of NH₄⁺ in the mTAL, reduces medullary interstitial concentrations of NH₄⁺ and NH₃, and decreases entry of NH₄⁺ and NH₃ into the medullary collecting duct. This same series of events leads, in the final analysis, to a marked reduction in urinary ammonium excretion. The potential for development of a hyperchloremic metabolic acidosis is greatly augmented when a reduction in functional renal mass (GFR of <60 mL/min) coexists with hyperkalemia or when aldosterone deficiency or resistance is present.

Clinical Disorders

Type 4 RTA with a generalized distal nephron dysfunction is manifest as a hyperchloremic (non-gap), hyperkalemic metabolic acidosis in which urinary ammonium excretion is invariably depressed and renal function is often compromised. Although all patients with CKD may have some tendency to acidosis and hyperkalemia, these abnormalities are not usually severe, particularly in the early stages of CKD. Patients with type 4 RTA often have marked hyperkalemia (>5.5 mEq/L) that is disproportionate to the reduction in the GFR. The TTKG and/or the fractional excretion of K⁺ (FE_K⁺) is usually low in patients with this disorder. In such patients, often with diabetic nephropathy or tubulointerstitial disease, a unique dysfunction of potassium and acid secretion by the collecting tubule coexists and can be attributed to either mineralocorticoid deficiency, resistance to mineralocorticoid, or a specific type of renal tubular dysfunction (voltage defects). The clinical spectrum of generalized abnormalities in the distal nephron is summarized in Table 16.11.

Primary Mineralocorticoid Deficiency

Angiotensin II and plasma K⁺ are the principal determinants of production and secretion of aldosterone. Adrenocorticotrophic hormone (ACTH), endothelin, dopamine, acetylcholine, epinephrine, nitric oxide, renin, and plasma Mg²⁺ can also have some influence. Destruction of the adrenal cortex by hemorrhage, infection, ischemia, invasion by tumors, amyloid or autoimmune processes results in Addison's disease. This disorder is manifest by combined glucocorticoid and mineralocorticoid deficiency and is recognized clinically by hypoglycemia, anorexia, weakness, hyperpigmentation, and a

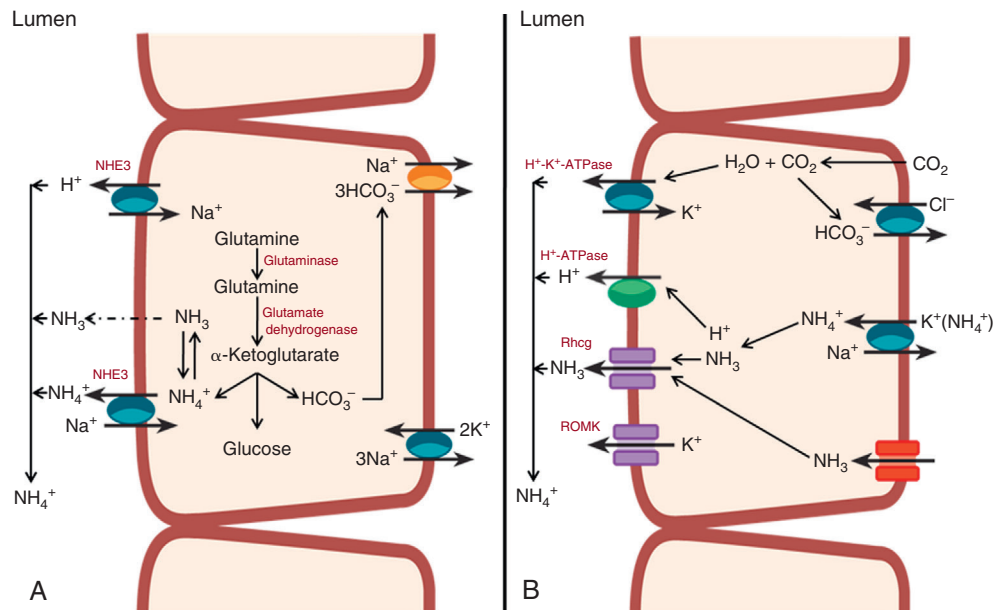


Fig. 16.7 Cell models of ammonia synthesis and excretion pathways. **A**, Proximal convoluted tubule. Ammonia is derived from glutamine precursors to produce two NH_4^+ and two HCO_3^- molecules through an enzymatic pathway activated by acidemia and hypokalemia and inhibited by alkalemia and hyperkalemia. **B**, Type A intercalated cell in collecting tubule. Ammonium entry across basolateral membrane through substitution of NH_4^+ for K^+ on K^+ conductance and secreted across apical membrane via ROMK or RhCG (see text). In both **A** and **B**, NH_3 diffusion coupled with H^+ secretion traps NH_4^+ in the tubule lumen.

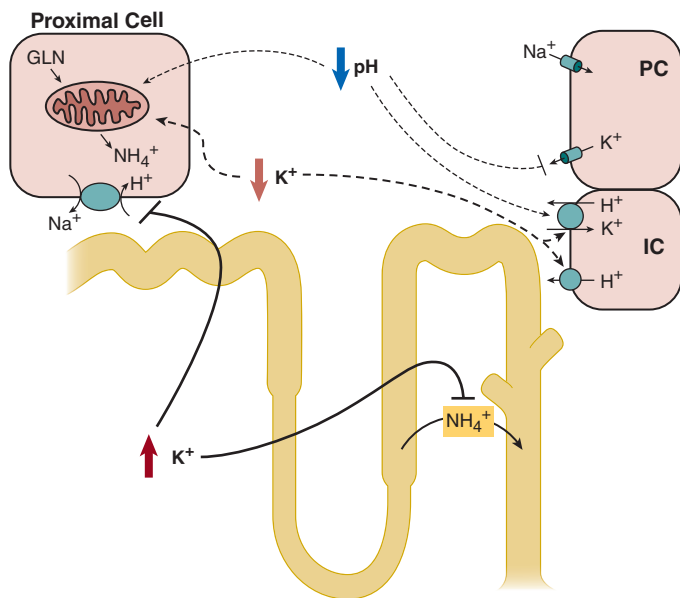


Fig. 16.8 Diagram of some of the renal interactions of acid-base status and K^+ . Proximal tubule cell, and a collecting duct principal cell (PC) and intercalated cell (IC) are shown. Acidosis ($\downarrow \text{pH}$) causes renal potassium retention by several actions including inhibiting secretory K^+ channels, stimulating $\text{H}^+-\text{K}^+-\text{ATPase}$ which reabsorbs K^+ , and increasing ammoniogenesis which may have secondary actions that block K^+ secretion. Alkalosis can have the reverse effects. The effects also work in the other direction: potassium depletion ($\downarrow \text{K}^+$) stimulates ammoniogenesis and H^+ secretion/bicarbonate reabsorption, and also distal acid secretion – all of which may cause metabolic alkalosis. Hyperkalemia can have some of the opposite effects; such as inhibiting thick ascending limb ammonium transport, inhibiting urinary ammonium excretion, an effect which may cause acidosis. (From Hamm LL, Hering-Smith KS, Nakhoul NL. Acid-base and potassium homeostasis. *Semin Nephrol.* 2013;333:257-64.)

failure to respond to stress. These defects can occur in association with renal salt wasting and hyponatremia, hyperkalemia, and metabolic acidosis, all associated with mineralocorticoid deficiency alone. The most common congenital adrenal defect in steroid biosynthesis is 21-hydroxylase deficiency, which is associated with salt wasting, hyperkalemia, and metabolic acidosis in a fraction of patients. Causes of Addison's disease include tuberculosis, autoimmune adrenal failure, fungal infections, adrenal hemorrhage, metastasis, lymphoma, acquired immunodeficiency syndrome (AIDS), amyloidosis, and drug toxicity (ketoconazole, fluconazole, phenytoin, rifampin, and barbiturates). These disorders are associated with low plasma aldosterone levels and high levels of plasma renin activity.³³ The metabolic acidosis of mineralocorticoid deficiency results from a decrease in hydrogen ion secretion in the collecting duct secondary to decreased H^+-ATPase activity. The accompanying hyperkalemia of mineralocorticoid deficiency decreases ammonium production and excretion.

Hyporeninemic Hypoaldosteronism

In contrast to patients with primary adrenal disorders, patients with hyporeninemic hypoaldosteronism exhibit low plasma renin activity, are usually older (mean age 65 years), and frequently have mild to moderate renal insufficiency (70%) and acidosis (50%) in association with chronic hyperkalemia in the range of 5.5 to 6.5 mEq/L (Table 16.13).³³ In this relatively common disorder, the metabolic acidosis and the hyperkalemia are out of proportion to the level of reduction in GFR. The most frequently associated renal diseases are diabetic nephropathy and tubulointerstitial disease. Additional disorders associated with hyporeninemic hypoaldosteronism include obstructive uropathy, systemic lupus erythematosus, and human immunodeficiency virus (HIV) infection. For 80% to 85% of such patients, there is a reduction in plasma renin activity that does not respond to the usual physiologic

Table 16.13 Hyporeninemic Hypoaldosteronism: Typical Clinical Features

Mean age 65 yrs
Asymptomatic hyperkalemia (75%)
Hyperchloremic metabolic acidosis (>50%)
Renal insufficiency (70%)
Diabetes mellitus (50%)
Hypertension (75%)
Weakness (25%)
Arrhythmia (25%)
Congestive heart failure (50%)

maneuvers. Because approximately 30% of patients with hyporeninemic hypoaldosteronism are hypertensive, the finding of a low plasma renin activity in such patients suggests a volume-dependent form of hypertension with physiologic suppression of renin elaboration. The mechanism(s) of renin suppression is not firmly established for all cases but possibilities include volume expansion, atrial natriuretic peptides, and direct effects of the various renal diseases.

Impaired ammonium excretion is the combined result of hyperkalemia, impaired ammoniogenesis, a reduction in nephron mass, reduced proton secretion, and impaired transport of ammonium by nephron segments in the inner medulla.^{32,33,63} Hyperchloremic metabolic acidosis occurs in approximately 50% of patients with hyporeninemic hypoaldosteronism but is often less of a clinical issue than the hyperkalemia. Drugs that may result in similar manifestations are reviewed later.

Isolated Hypoaldosteronism in Critically Ill Patients

Isolated hypoaldosteronism, which may occur in critically ill patients, particularly in the setting of severe sepsis or cardiogenic shock, is manifest by markedly elevated ACTH and cortisol levels in concert with a decrease in aldosterone elaboration in response to angiotensin II. This may be secondary to selective inhibition of aldosterone synthase as a result of hypoxia or in response to cytokines such as tumor necrosis factor- α or interleukin-1 or, alternatively, as a result of high circulating levels of atrial natriuretic peptide (ANP).^{32,33,64} ANP, a powerful suppressor of aldosterone secretion, may be elevated in congestive heart failure (CHF), with atrial arrhythmias, in subclinical cardiac disease, and in volume expansion. The tendency to manifest the features of hypoaldosteronism, including hyperkalemia and metabolic acidosis, is often potentiated by the administration of potassium-sparing diuretics, potassium loads in parenteral nutrition solutions, or heparin. The latter (both unfractionated and low molecular weight heparin) can suppress aldosterone synthesis in the critically ill patient.

Resistance to Mineralocorticoid and Voltage Defects: Hyperkalemic Distal Renal Tubular Acidosis

A generalized defect in distal tubule secretory function that results in hyperkalemic hyperchloremic metabolic acidosis has been designated as *hyperkalemic distal RTA* because of the coexistence of hyperkalemia, and an inability to acidify the urine (urine pH of >5.5) during spontaneous acidosis or following an acid load. The hyperkalemia is the result of

impaired renal K^+ secretion, and the TTKG or FE_{K^+} is invariably lower than expected for non-renal hyperkalemia. Urine ammonium excretion is reduced, but aldosterone levels may be low, normal, or even increased. Hyperkalemic distal RTA can be distinguished from selective hypoaldosteronism because plasma renin and aldosterone levels are usually high or normal. In selective hypoaldosteronism, the urine pH is typically low and the defect in urinary acid excretion can be attributed to the decrease in ammonium excretion.

Drug-Induced Renal Tubular Secretory Defects

Impaired Renin-Aldosterone Elaboration. Drugs may impair renin or aldosterone elaboration or cause mineralocorticoid resistance and produce effects that mimic the clinical manifestations of the acidification defect seen in the generalized form of distal RTA with hyperkalemia. COX inhibitors (NSAIDs or COX-2 inhibitors) can generate hyperkalemia and metabolic acidosis as a result of inhibition of renin release.⁶⁵ β -adrenergic antagonists cause hyperkalemia by altering potassium distribution and by interfering with the renin-aldosterone system. Heparin impairs aldosterone synthesis as a result of direct toxicity to the zona glomerulosa and inhibition of aldosterone synthase. ACE inhibitors and ARBs interrupt the renin-aldosterone system and result in hypoaldosteronism with hyperkalemia and acidosis, particularly in patients with advanced renal insufficiency or in patients with a tendency to develop hyporeninemic hypoaldosteronism (diabetic nephropathy). The combination of potassium-sparing diuretics and ACE inhibitors should be avoided in diabetic patients.

Inhibitors of Potassium Secretion in the Collecting Duct. Spironolactone and eplerenone act as competitive inhibitors of aldosterone and inhibit aldosterone action. These drugs may cause hyperkalemia and metabolic acidosis when administered to patients with significant renal insufficiency, patients with advanced liver disease, or patients with unrecognized renal hemodynamic compromise. Similarly, amiloride and triamterene may be associated with hyperkalemia, but through an entirely different mechanism. Both potassium-sparing diuretics occupy and thus block the apical Na^+ -selective channel (ENaC) in the collecting duct principal cell. Occupation of ENaC inhibits Na^+ absorption and reduces the negative transepithelial voltage, which alters the driving force for K^+ secretion.

Trimethoprim (TMP) and pentamidine are related structurally to amiloride and triamterene and inhibit ENaC.^{66–68} Hyperkalemia has been observed in 20% to 50% of HIV-infected patients receiving high-dose trimethoprim-sulfamethoxazole (TMP-SMX) or TMP-dapsone for the treatment of opportunistic infections and as many as 100% of patients with AIDS-associated infections (due to *Pneumocystis jirovecii*) receiving pentamidine for longer than 6 days.⁶⁸ Because both TMP and pentamidine decrease the electrochemical driving force for both K^+ and H^+ secretion in the CCT, metabolic acidosis may accompany the hyperkalemia even in the absence of severe renal failure, adrenal insufficiency, tubulointerstitial disease, or hypoaldosteronism. Whereas it has been assumed that such a “voltage” defect could explain the decrease in H^+ secretion, it is likely that, in addition, hyperkalemia plays a significant role in the development of metabolic acidosis by inhibition of ammonium production and excretion (see Fig. 16.8 and Table 16.12).

The calcineurin inhibitors cyclosporine A and tacrolimus may be associated with hyperkalemia in transplant recipients as a result of inhibition of the basolateral $\text{Na}^+\text{-K}^+\text{-ATPase}$ and the consequent decrease in intracellular $[\text{K}^+]$ and the transepithelial potential, which together reduce the driving force for K^+ secretion.⁶⁵ Additionally, studies indicate that calcineurin inhibitors may also inhibit K^+ secretion by directly interfering with the ROMK K^+ channel.⁶⁹ An additional explanation for the association of hyperkalemia, volume expansion and hypertension, a syndrome that resembles the phenotype of familial hyperkalemic hypertension or PHA-2, is enhanced activity of NCC in the DCT.⁷⁰

Secondary Renal Diseases Associated With Acquired Voltage Defects

Hyperkalemia that is out of proportion to the degree of renal insufficiency can also be observed in the nephropathies associated with sickle cell disease, HIV infection, systemic lupus erythematosus, obstructive uropathy, acute and chronic renal allograft rejection, hypoaldosteronism, multiple myeloma, and amyloidosis.^{33,71} Tubulointerstitial disease with hyperkalemia and hyperchloremic metabolic acidosis with or without salt wasting may be associated with analgesic abuse, nephrolithiasis, nephrocalcinosis, and hyperuricemia.³³

A variety of familial disorders are associated with hyperkalemia and hyperchloremic metabolic acidosis. Autosomal dominant pseudohypoaldosteronism type 1 (PHA-1) is an example of a voltage defect in the collecting tubule resulting from aldosterone resistance. This disorder, which is clinically less severe than the autosomal recessive form discussed later, is associated with hyperkalemia from impaired potassium secretion, renal salt wasting, elevated levels of renin and aldosterone, and relative hypotension. Physiologic mineralocorticoid replacement therapy does not correct the hyperkalemia. The autosomal dominant disorder has been shown to be the result of a mutation in the intracellular mineralocorticoid receptor in the collecting tubule.⁷² In contrast to the autosomal recessive disorder, this defect is not expressed in organs other than the kidney and becomes less severe with advancing age. Because of the decrease in mineralocorticoid actions, apical Na^+ absorption and the transepithelial voltage difference are reduced and K^+ secretion is secondarily impaired. Four hours after administration of fludrocortisone (0.1 mg orally) to patients with autosomal dominant PHA-1, the TTKG will *not* increase, which demonstrates that resistance to mineralocorticoid causes the hyperkalemia.

Autosomal recessive PHA-1 is a prototypical voltage defect in the CCT. This disorder is the result of a loss-of-function mutation of the gene that encodes one of the α -, β -, or γ -subunits of the ENaC.^{73–77} Children with this disorder have severe hyperkalemia and renal salt wasting because of impaired sodium absorption in principal cells of the CCT. In addition, the hyperchloremic metabolic acidosis may be severe and is associated with hypotension and marked elevations of plasma renin and aldosterone. These children also manifest vomiting, hyponatremia, failure to thrive, and respiratory distress. The latter is due to involvement of ENaC in the alveolus, which prevents Na^+ and water absorption in the lungs.^{76–78} Patients with this disease respond to a high salt intake and correction of the hyperkalemia. Unlike the autosomal dominant form, autosomal recessive PHA-1 persists throughout life.

Familial Hyperkalemic Hypertension. A number of additional adult patients have been reported with a rare form of autosomal dominant low-renin hypertension that is invariably associated with hyperkalemia, hyperchloremic metabolic acidosis, mild volume expansion, normal renal function, and low aldosterone levels. The acidosis in these patients is mild and can be accounted for by the hyperkalemia; the acidosis and renal potassium excretion are resistant to mineralocorticoid administration. Thiazide diuretics consistently correct the hyperkalemia and metabolic acidosis, as well as the hypertension, plasma aldosterone level, and plasma renin level. This syndrome has been recently designated *familial hyperkalemic hypertension* (FHH) but is also known as *pseudohypoaldosteronism type 2* (PHA-2)⁷⁹ or *Gordon's syndrome*.^{80,81} Both genes encode members of the WNK (with no lysine [K]) family of serine-threonine kinases. These kinases signal via serine/threonine protein kinase 39 (SPAK; STK39) and oxidative stress-response 1 (OxSR1 or OSR1; OXSR1), which phosphorylate and activate the thiazide sensitive Na-Cl cotransporter (NCC) in the distal convoluted tubule, among other transporters affected.^{81–83} Cul3 is part of an E3 ubiquitin ligase complex that regulates protein degradation. Loss of regulation of NCC by WNK4 results in a gain in NCC function, which, through enhanced absorption of Na^+ and Cl^- , causes volume expansion, decreased more distal delivery of Na^+ and therefore reduced K^+ secretion particularly in the connecting tubule.^{81,84} More recent studies have revealed that FHH is more commonly caused by mutations in the cullin-RING ligase (CRL) proteins cullin 3 (Cul3)⁴ and kelch-like 3 (KLHL3).^{85,86} Loss of inhibition of NCC by WNK4 results in a gain in NCC function, which, through enhanced absorption of Na^+ and Cl^- , causes volume expansion, decreased distal delivery of Na^+ and therefore reduced K^+ secretion, particularly in the connecting tubule.⁸⁷

Treatment. In patients with hyperkalemic hyperchloremic metabolic acidosis, ideally treatment follows delineation of the underlying cause. However, in certain clinical situations the cause can be easily surmised and treatment begun. For instance, hyporeninemic hypoaldosteronism is common in patients with diabetes and mild to moderate CKD; and certain drugs commonly cause hyperkalemic, hyperchloremic metabolic acidosis. A good history and assessment of the clinical setting are crucial. Contributing or precipitating factors should be considered, including low urine flow or decreased distal Na^+ delivery, a rapid decline in GFR (especially in AKI superimposed on chronic renal failure), hyperglycemia or hyperosmolality, and sources of exogenous K^+ intake. Assessment of the blood pressure and volume status is important. In cases in which aldosterone is low or suspected to be low, consideration and/or exclusion of combined cortisol deficiency (Addison's disease) is crucial. The workup may include evaluation of the TTKG or the fractional excretion of potassium and an estimate of renal ammonium excretion (UAG or urine osmolar gap, and urine pH).¹

¹The urine pH may be assessed under stimulated conditions with dietary salt restriction and furosemide-induced volume depletion, and measurement of the response of potassium excretion to furosemide and fludrocortisone. An increase in the TTKG to a value of more than 6 measured 4 hours after a single oral dose of fludrocortisone (0.05 mg) suggests that mineralocorticoid deficiency, but not resistance, is causative.

Since acidosis and hyperkalemia are frequently interrelated, treatment of one may aid the other. Treatment options are given in Table 16.14. The severity of the hyperkalemia often drives the treatment. Reduction in the serum potassium level will often improve the metabolic acidosis by increasing ammonium excretion. In hyporeninemic hypoaldosteronism, loop diuretics may improve volume status, blood pressure, potassium and acidosis. Correction of hyperkalemia with oral agents (ZS-9, patiromir) that act as gastrointestinal cation exchangers may be useful. In the past, sodium polystyrene was used but reports of intestinal necrosis have limited its use. Presently patiromer and sodium zirconium cyclosilicate (ZS-9) are used for this purpose. Alkali therapy may be useful not only for the acidosis, but indirectly to help the potassium. In hyporeninemic hypoaldosteronism, exogenous mineralocorticoids are frequently not a first-line treatment because of the coexistent hypertension and volume overload. Patients with combined glucocorticoid and mineralocorticoid deficiency should receive both adrenal steroids in replacement dosages. Volume depletion should be avoided unless the patient is volume overexpanded or hypertensive. Infants with autosomal recessive or dominant PHA-1 should receive salt supplements in amounts sufficient to correct the volume depletion, hypotension, and other features of the syndrome and to allow normal growth. In contrast, patients with PHA-2 should receive thiazide diuretics along with dietary salt restriction.

In drug-induced disorders, obviously discontinuing the offending agent is ideal. In some cases, this may not always be feasible such as in patients with life-threatening disorders, for example, during TMP-SMX or pentamidine therapy in

AIDS patients with *Pneumocystis jiroveci* pneumonia. Based on the previous discussion, other mechanisms to increase distal potassium secretion or gastrointestinal exchangers can be tried.

Distinguishing the Types of Renal Tubular Acidosis. The contrasting findings and diagnostic features of the three types of RTA discussed in this chapter are summarized in Table 16.15.

Acidosis of Chronic Kidney Disease. Chronic kidney disease has long been known to be associated with acidosis.^{88,89} Acidosis is usually recognized with a GFR below 40 mL/min,⁹⁰ but is present in only some patients, not all. But recent evidence suggests that acid may be retained in at least some patients with CKD who do not have typical manifestations of acidosis such as reduced plasma HCO_3^- .^{91,92} Bone buffering may account for the relative stability of the plasma HCO_3^- .³ The metabolic acidosis when present is initially hyperchloremic but may convert to the high AG metabolic acidosis as GFR falls below 15 mL/min.^{37,93} The principal defect in net acid excretion in patients with reduced GFR is an inability to excrete NH_4^+ sufficiently to match net endogenous acid production. Endogenous acid production in adults is usually about 1 mEq per kg body weight (2-3 mEq/kg in children) but varies with diet such that low protein and high fruits and vegetables will lower the amount. Unlike patients with distal RTA, most patients with CKD can lower urine pH and excrete titratable acid normally.³⁷

The numerous consequences of chronic positive acid balance in CKD and certainly acidosis include progression of CKD at a faster rate⁹¹ and, dissolution of bone,³⁵ impaired hydroxylation of 25-hydroxycholecalciferol, and renal osteodystrophy^{37,94} sarcopenia from enhanced skeletal muscle protein degradation, loss of muscle strength, and insulin resistance. Acidosis has also been associated with increased mortality in CKD.^{95,96} The mechanisms whereby acidosis worsens progression of CKD are not totally understood but may include 1) increased fibrosis from activation of complement from increased interstitial NH_4^+ and 2) increased hormones such as endothelin, angiotensin II, and aldosterone.

Corresponding with the findings that acidosis worsens progression of CKD, several studies at this point have shown that treatment of acidosis can improve the course of CKD. Interestingly, patients with stage 2 and 3 CKD even without overt clinical metabolic acidosis, when given oral alkali progress more slowly (i.e., GFR declines less rapidly) and exhibit a significant reduction in net acid excretion.^{91,97}

Table 16.14 Treatment of Generalized Dysfunction of the Nephron With Hyperkalemia

Loop diuretic (furosemide, bumetanide)
Alkali therapy (Shohl's solution or NaHCO_3 tablets)
 K^+ binding resin
Low-potassium diet
Fludrocortisone (0.1-0.3 mg/day)
 Avoid in hypertension, volume expansion, heart failure
 Combine with loop diuretic
Avoid drugs associated with hyperkalemia, herbs and over-the-counter preparations containing potassium
In pseudohypoaldosteronism type 1, add NaCl tablets

Table 16.15 Contrasting Features and Diagnostic Studies in Renal Tubular Acidosis

Finding	Type of Renal Tubular Acidosis		
	Proximal	Classical Distal	Generalized Distal Dysfunction
Plasma $[\text{K}^+]$	Low	Low	High
Urine pH with acidosis	<5.5	>5.5	<5.5 or >5.5
Fanconi's lesion	Present with acquired PRTA	Absent	Absent
Fractional bicarbonate excretion	10%-15% during alkali therapy	2%-5%	5%-10%
Associated features	Fanconi's syndrome	Nephrocalcinosis/Hyperglobulinemia	Renal insufficiency

Moreover, a recent study in patients with stage 4 CKD found that a diet emphasizing fruits and vegetables had as beneficial an effect on slowing progression of CKD and compared favorably to NaHCO_3 therapy.⁹⁸ Note that neither approach, alkali supplementation, or additional fruits and vegetables in the diet require dietary protein restriction.

Alkali therapy not only slows progression of CKD but also helps to reverse the deleterious effects of chronic acidosis on bone and skeletal muscle.⁹⁷ An amount of alkali slightly in excess (1 to 2 mEq/kg/day) of dietary metabolic acid production, typically restores the $[\text{HCO}_3^-]$ to recommended levels (> 22 mEq/L; see below).³⁵ The clinical guidelines endorsed by the National Kidney Foundation Kidney Disease Outcomes Quality Initiative (KDOQI) recommend monitoring of total CO_2 in patients with chronic kidney disease with a goal of maintaining the $[\text{HCO}_3^-]$ above 22 mEq/L.⁹⁹ Alkali therapy may consist of NaHCO_3 tablets or modified Shohl's (Citric Acid – Sodium Citrate) Solution. Addition of alkali even in the form of NaHCO_3 does not lead to salt retention or worsened hypertension. There is some concern that plasma HCO_3^- could be raised to a higher level than desired, which has been associated in epidemiologic studies to worsen outcomes.⁸⁸

During the treatment of more advanced CKD patients, phosphate binders are usually prescribed. Sevelamer hydrochloride has been a widely used phosphate binder since it does not contain calcium; however, it is associated with an acid load.^{100,101} Use of sevelamer carbonate, the more recent formulation of sevelamer, avoids this issue.

As kidney disease progresses below a GFR of 15 mL/min, the non-AG acidosis typically evolves into the usual high AG acidosis of end-stage renal disease. An increased anion gap may be seen earlier when the anion gap is fully adjusted for albumin etc.¹⁰² The anions that account for the elevated anion gap include phosphate, sulfate, urate, etc.

High Anion Gap Acidoses. When the non-Cl⁻ anion of an acid load is not excreted rapidly, a high AG acidosis results, with a normal plasma Cl⁻. The normo-chloremic acidosis is maintained as long as the anion that was part of the original acid load remains in the blood. High AG acidosis is caused by the accumulation of organic acids such as lactate or ketoacids. This may occur if the anion does not undergo glomerular filtration (e.g., uremic acid anions), if the anion is filtered but is readily reabsorbed, or if, because of alteration in metabolic pathways (ketoacidosis, L-lactic acidosis), the anion cannot be utilized in the body sufficiently rapidly. Conceptually, an acid such as lactic acid or ketoacid will react with plasma HCO_3^- , “consuming” the HCO_3^- or converting it to CO_2 which will be exhaled, replacing that HCO_3^- anion with the acid anion such as lactate⁻ or ketoacid anion.

Theoretically, with a pure AG acidosis, the increment in the AG (ΔAG) above the normal value of approximately 10 mEq/L should conceptually equal the decrease in bicarbonate concentration (ΔHCO_3^-) below the normal value of 25 mEq/L. In some cases, the increment in AG may be the leading indicator of the existence of the high AG acidosis, not a decrement in pH or HCO_3^- . For example, with a mixed metabolic alkalosis and high AG acidosis, the HCO_3^- and pH may not fully reflect the metabolic acidosis, but the AG should. Such findings are not unusual when uremia or ketoacidosis leads to vomiting, for example.

Identification of the underlying cause of a high AG acidosis is facilitated by consideration of the clinical setting and associated laboratory values. The common causes are outlined in Table 16.16 and include 1) lactic acidosis (e.g., L-lactic acidosis and D-lactic acidosis), 2) ketoacidosis (e.g., diabetic, alcoholic, and starvation ketoacidoses), 3) toxin- or poison-induced acidosis (e.g., ethylene glycol, methyl alcohol, propylene glycol, or pyroglutamic acidosis), and 4) uremic acidosis.

Initial screening to differentiate the high AG acidoses should focus on 1) a history or other evidence of drug or toxin ingestion and ABG measurement to detect coexistent respiratory alkalosis (as with salicylates), 2) historical evidence of diabetes mellitus (diabetic ketoacidosis; DKA), 3) evidence of alcoholism or increased levels of β -hydroxybutyrate (alcoholic ketoacidosis; AKA), 4) observation for clinical signs of uremia and determination of the blood urea and creatinine

Table 16.16 Metabolic Acidosis With High Anion Gap

Conditions Associated with Type A Lactic Acidosis

- Hypovolemic shock
- Cholera
- Septic shock
- Cardiogenic shock
- Regional hypoperfusion
- Severe hypoxia
 - Severe asthma
 - Carbon monoxide poisoning
 - Severe anemia

Conditions Associated with Type B Lactic Acidosis

- Liver disease
- Diabetes mellitus
- Metformin**
- Catecholamine excess
- Thiamine deficiency
 - Intracellular inorganic phosphate depletion
 - Intravenous fructose, xylose, or sorbitol
- Alcohols and other ingested compounds metabolized by alcohol dehydrogenase
 - Ethanol, methanol, ethylene glycol, propylene glycol
- Mitochondrial toxins
 - Salicylates**, cyanide, 2,4-dinitrophenol, **Nonnucleoside anti-reverse transcriptase drugs**
- Other drugs (propofol, linezolid)
- Metastatic tumors (large tumors with regional hypoxemia or liver metastasis)

Seizures

- Inborn errors of metabolism

D-Lactic Acidosis

- Short bowel syndrome, ischemic bowel, Small-bowel obstruction

Ketoacidosis

- Diabetic
- Alcoholic

Intoxications

- Salicylates
- Ethylene glycol
- Methanol
- Pyroglutamic acid

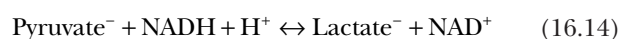
Uremia (Late Renal Failure)

levels (uremic acidosis), 5) inspection of the urine for oxalate crystals (ethylene glycol), and, finally, 6) recognition of the numerous settings in which lactic acid levels may be increased (hypotension, sepsis, cardiac failure, ischemic bowel, intestinal obstruction and bacterial overgrowth, leukemia, cancer, and exposure to certain drugs).

Lactic Acidosis

Physiology. Lactic acid can exist in two forms: L-lactic acid and D-lactic acid. In mammals, only the L stereoisomer is a product of metabolism and most lactic acidosis is therefore L-lactic acidosis. L-lactic acidosis is one of the most common forms of a high AG acidosis.

Lactate usually derives from pyruvate in the cytosolic reaction of lactate dehydrogenase:



Pyruvate is generated with the metabolism of glucose by cytosolic anaerobic glycolysis. Pyruvate in aerobic conditions will be further metabolized in mitochondria in the Krebs or tricarboxylic acid cycle. But in the cytosol, the equation above is a near equilibrium reaction such that the lactate to pyruvate ratio (usually about 10) is governed by the H^+ concentration or pH and the $\text{NADH}:\text{NAD}^+$ ratio, or the oxidation-reduction potential (or redox potential) of the cell.

After rearranging the mass action equation, the ratio of lactate concentration to pyruvate concentration may be expressed as

$$\frac{[\text{Lactate}^-]}{[\text{Pyruvate}^-]} = K_{\text{eq}} [\text{H}^+] \frac{[\text{NADH}]}{[\text{NAD}^+]} \quad (16.15)$$

The $\text{NADH}:\text{NAD}^+$ ratio is also involved in many other metabolic redox reactions.¹⁰³ Moreover, the steady-state concentrations of all these redox reactants are related to one another. Important in other considerations of acid-base pathophysiology are the redox pairs β -hydroxybutyrate–acetoacetate and ethanol–acetaldehyde, discussed below. In clinical practice, these considerations can be of practical importance in assessing diabetic and alcoholic ketoacidosis (discussed below).

Normally, the rates of lactate entry and exit from the blood are in balance, so that net lactate accumulation is zero. This dynamic aspect of lactate metabolism in which lactate is converted back to glucose in the liver is termed the *Cori cycle*. In conditions of lactic acidosis either net overproduction of lactic acid from glucose by some tissues or underutilization by other tissues results in net addition of L-lactic acid to the blood. Ischemia accelerates lactate production and simultaneously decreases lactate utilization.

The production of lactic acid has been estimated to be about 15 to 20 mEq/kg/day in normal humans.³⁵ This enormous quantity contrasts with total ECF buffer base stores of only about 10 to 15 mEq/kg, so that with enhanced production, lactic acid can accumulate. The quantitative aspects of normal lactate production and consumption in the Cori cycle demonstrate that the development of lactic acidosis can be the most rapid and devastating form of metabolic acidosis.^{103,104} The rate of lactic acid production can be increased by ischemia, seizures, extreme exercise, catecholamines, leukemia, and alkalosis.¹⁰³ The increase in production occurs principally through enhanced phosphofructokinase activity.

Decreased lactate consumption may also lead to L-lactic acidosis, and probably plays an important role in most clinical lactic acidosis. The principal organs for lactate removal during rest are the liver and kidneys. Both the liver and the kidneys and perhaps muscle have the capacity for increased lactate removal under the stress of increased lactate loads.¹⁰³ Hepatic utilization of lactate can be impeded by several factors: poor perfusion of the liver; defective active transport of lactate into cells; and inadequate metabolic conversion of lactate into pyruvate because of altered intracellular pH, redox state, or enzyme activity. Examples of states causing impaired hepatic lactate removal include primary diseases of the liver, enzymatic defects, tissue anoxia or ischemia, severe acidosis, and altered redox states, as occurs with alcohol intoxication, fructose consumption by fructose-intolerant individuals, or administration of nucleoside (analog) reverse transcriptase inhibitors (NRTIs) such as zidovudine and stavudine in patients with HIV infection^{103,105,106} or biguanides such as metformin.^{103,107,108} Deaths have been reported due to refractory lactic acidosis secondary to thiamine deficiency in patients receiving parenteral nutrition formulations without thiamine.¹⁰⁹ Thiamine is a cofactor for pyruvate dehydrogenase that catalyzes the oxidative decarboxylation of pyruvate to acetyl–coenzyme A under aerobic conditions. Pyruvate cannot be metabolized in this manner in the presence of thiamine deficiency, so that excess pyruvate is converted to hydrogen ions and lactate.

Diagnosis. Because lactic acid has a pK_a of 3.8, addition of lactic acid to the blood leads to a reduction in blood HCO_3^- concentration and an equivalent elevation in lactate concentration, which is associated with an increase in the AG. Lactate concentrations are mildly increased in various nonpathologic states (e.g., exercise), but the magnitude of the elevation is generally small. In practical terms, a lactate concentration greater than 4 mmol/L (normal is 0.67 to 1.8 mmol/L) is generally accepted as evidence that the metabolic acidosis is due to net lactic acid accumulation.

Clinical Spectrum. In the classical classification of the L-lactic acidoses (see Table 16.16), type A L-lactic acidosis is due to tissue hypoperfusion or acute hypoxia, whereas type B L-lactic acidosis is less common but associated with some common diseases, drugs and toxins, and hereditary and miscellaneous disorders.¹⁰³

Tissue underperfusion and acute tissue hypoxia are the most common causes of type A lactic acidosis. Severe arterial hypoxemia even in the absence of decreased perfusion can generate L-lactic acidosis. Inadequate cardiac output, of either the low-output or the high-output variety, is the common pathogenetic factor. Among the most common causes of L-lactic acidosis is bowel ischemia and infarction in patients in the medical intensive care unit. The prognosis is related directly to the increment in plasma L-lactate and the severity of the acidemia.^{103,104}

Numerous medical conditions (without tissue hypoxia) predispose to type B L-lactic acidosis (see Table 16.16). Hepatic failure reduces hepatic lactate metabolism, and leukemia increases lactate production. Catecholamines stimulate glycolysis and lactate production. Severe anemia, especially as a result of iron deficiency or methemoglobinemia, may cause lactic acidosis. Malignant cells produce more lactate than normal cells even under aerobic conditions. This phenomenon is magnified if the tumor expands rapidly and outstrips the blood supply. Therefore, exceptionally large

tumors may be associated with severe L-lactic acidosis. Seizures, extreme exertion, heat stroke, and tumor lysis syndrome may all cause L-lactic acidosis. Lactic acidosis with seizures is transient and self-limited, as the lactate that is produced can be rapidly metabolized.

Several drugs and toxins predispose to L-lactic acidosis (see Table 16.16). Of these, metformin and other biguanides (such as phenformin) are the most widely reported to have this effect.^{103,107,108} The occurrence of phenformin-induced lactic acidosis prompted the withdrawal of the drug from U.S. markets in 1977. Although much less frequent than phenformin-induced lactic acidosis, metformin-induced lactic acidosis is at higher risk in patients with CKD (and is contraindicated when the serum creatinine exceeds 1.4 mg/dL), and whenever there is hypoperfusion, hypotension, or chronic metabolic acidosis. Metformin should be discontinued several days before contrast dye administration. Although rare, metformin-induced lactic acidosis is the most frequent cause of lactic acidosis in diabetic patients and is associated with a mortality of up to 50%. Carbon monoxide poisoning frequently produces lactic acidosis by reduction of the oxygen-carrying capacity of hemoglobin. Cyanide binds cytochrome a and a_3 and blocks the flow of electrons to oxygen. In patients with HIV infection nucleoside analogs can induce toxic effects on mitochondria by inhibiting DNA polymerase- γ . Hyperlactatemia is common with NRTI therapy, especially stavudine and zidovudine, but the serum L-lactate level is usually only mildly elevated and compensated.^{103,106,110} This combination carries a high mortality.

Associated Clinical Features. Hyperventilation, abdominal pain, and disturbances in consciousness are frequently present, as are signs of inadequate cardiopulmonary function in type A L-lactic acidosis. Leukocytosis, hyperphosphatemia, and hyperuricemia are common, and hypoglycemia may occur.¹⁰³ Hyperkalemia may or may not accompany acute lactic acidosis.

Treatment of L-Lactic Acidosis: General Supportive Care. The overall mortality of patients with L-lactic acidosis is approximately 60%, but increases in those with coexisting hypotension.¹⁰³ The basic principle and only effective form of therapy for L-lactic acidosis is first to correct the underlying condition(s) initiating the disruption in normal lactate metabolism. In type A L-lactic acidosis, cessation of acid production by improvement of tissue oxygenation, restoration of the circulating fluid volume, improvement or augmentation of cardiac function, resection of ischemic tissue, and amelioration of sepsis are necessary in many cases. Septic shock requires control of the underlying infection and volume resuscitation in hypovolemic shock. Lactate levels are often followed as an indicator of tissue perfusion. Sodium bicarbonate administration is of little value in this setting.

Alkali Therapy. Alkali therapy is generally advocated only for acute, severe acidemia (pH of <7.1) to improve the detrimental hemodynamic consequences of the acidosis: decreased inotropy, venoconstriction, arterial vasodilation, and decreased catecholamine responsiveness. The only exception in milder acidosis may be with coexistent severe acute kidney injury.¹¹¹ However, in experimental models and clinical examples of lactic acidosis, NaHCO_3 therapy in large amounts may depress cardiac performance and exacerbate the acidemia. Paradoxically, bicarbonate therapy activates phosphofructokinase, which is regulated by intracellular pH,

thereby increasing lactate production. The use of alkali in states of moderate L-lactic acidemia is therefore controversial, and it is generally agreed that attempts to normalize the pH or HCO_3^- concentration by intravenous NaHCO_3 therapy is both potentially deleterious and practically impossible. Thus, raising the plasma HCO_3^- to approximately 15 mEq/L (not to the normal value) and the pH to between 7.2 to 7.25 is a reasonable goal to improve tissue pH. Constant infusion of hypertonic bicarbonate has many disadvantages and is discouraged. Fluid overload occurs rapidly with NaHCO_3 administration because of the massive amounts required in some cases. With rapid infusions of HCO_3^- , PCO_2 may increase, ionized calcium decrease, and hypernatremia develop. In addition, central venoconstriction and decreased cardiac output are common. The accumulation of lactic acid may be relentless and may necessitate administration of diuretics, ultrafiltration, or dialysis. Hemodialysis can simultaneously deliver HCO_3^- , remove lactate, remove excess ECF volume, and correct electrolyte abnormalities. The use of continuous renal replacement therapy (CRRT) as a means of lactate removal and simultaneous alkali addition is a promising adjunctive treatment in critically ill patients with L-lactic acidosis but has not consistently proved to be beneficial. It is important to note that CRRT also removes lactate in the dialysate and can confound interpretation of serial measurements of lactate concentrations.

If the underlying cause of the L-lactic acidosis can be remedied, blood lactate will be reconverted to HCO_3^- . HCO_3^- derived from lactate conversion and any new HCO_3^- generated by renal mechanisms during acidosis and from exogenous alkali therapy are additive and may result in an overshoot alkalosis.

Other Agents. Dichloroacetate (an activator of pyruvate dehydrogenase), methylene blue, THAM (0.3 mol/L tromethamine), and Tribonat (a mixture of THAM, acetate, NaHCO_3 , and phosphate) have all been advocated with experimental support in the past, but none has been proved to be effective clinically.^{112,113} Lactated Ringer's solution and lactate-containing peritoneal dialysis solutions should be avoided. These solutions contain racemic mixtures of both L- and D-lactate.

D-Lactic Acidosis. D-lactic acidosis is a rare cause of high AG acidosis. D-lactate can accumulate in some individuals as a by-product of metabolism by bacteria, which accumulate and overgrow in the gastrointestinal tract with jejunal bypass or short bowel syndrome. The diagnosis can be especially difficult since hospital laboratories routinely only measure L-lactic acid levels using stereospecific enzymatic assays, which do not detect D-lactic acid.

The manifestations of D-lactate acidosis are typically episodic encephalopathy and high AG acidosis in association with short bowel syndrome. Features include slurred speech, confusion, cognitive impairment, clumsiness, ataxia, hallucinations, and behavioral disturbances. D-Lactic acidosis has been described in patients with bowel obstruction, jejunal bypass, short bowel, or ischemic bowel disease. These disorders have in common ileus or stasis associated with overgrowth of flora in the gastrointestinal tract, which is exacerbated by a high-carbohydrate diet.¹⁰³ D-lactate acidosis occurs when fermentation by colonic bacteria in the intestine causes D-lactate to accumulate so that it can be absorbed into the circulation. The disorder should be suspected in patients with an unexplained AG acidosis

and some of the typical features noted previously. While results of specific testing are pending, the patient should not receive oral nutrition. Serum D-lactate levels of greater than 3 mmol/L confirm the diagnosis. Treatment with a low-carbohydrate diet and antibiotics (neomycin, vancomycin, or metronidazole) is often effective.^{114–117}

D lactate levels are also elevated in diabetic ketoacidosis¹¹⁸ and in some cases with infusions of drugs dissolved in propylene glycol.

Diabetic Ketoacidosis. Diabetic ketoacidosis (DKA) is due to increased fatty acid metabolism and the accumulation of ketoacid anions (acetoacetate and β -hydroxybutyrate) as a result of insulin deficiency or resistance, in association with elevated glucagon levels. There is also frequently increased secretion of catecholamines, cortisol, and growth hormone contributing to the pathogenesis. Insulin deficiency and/or resistance (as from high catecholamine levels) increases lipolysis releasing free fatty acids and glycerol. With high delivery of fatty acids to mitochondria in the setting of low insulin and high glucagon, acetyl-CoA entry into the Krebs cycle is limited and instead is converted to the ketoacid acetoacetic acid. Acetoacetic acid may then be reduced to beta-hydroxybutyric acid.

DKA is usually seen in insulin-dependent diabetes mellitus upon cessation of insulin therapy or during an intercurrent illness, such as an infection, gastroenteritis, pancreatitis, or myocardial infarction, which increases insulin requirements acutely. The accumulation of the anions of the ketoacids accounts for the increment in the AG, which is accompanied, most often, by evidence of hyperglycemia (glucose level of >300 mg/dL). In comparison to patients with AKA, described later, patients with DKA have metabolic profiles characterized by a higher plasma glucose level and lower β -hydroxybutyrate/acetoacetate and lactate/pyruvate ratios.^{117,119,120}

Atypical instances of DKA with euglycemia has been reported in patients with both type 1 and type 2 diabetes taking sodium glucose cotransporter 2 (SGLT2) inhibitors.¹²¹ These drugs are only approved in type 2 diabetes but have been used some in type 1 patients. The drugs are not only effective in lowering glucose but also have cardio- and possibly renal protective effects. Suppressed insulin and elevated glucagon are implicated in the pathogenesis of the ketoacidosis.

The extent of the acidosis and the rise in the anion gap in DKA will depend not only on the rate of net ketoacid production but also the loss of ketoacid anions in the urine and the rate of acid excretion by the kidneys. Patients with good kidney function that are well hydrated may excrete large amounts of the ketoacid anions in the urine and even present with a hyperchloremic metabolic acidosis rather than a high AG acidosis.³⁶ As discussed above, some portion of the usual elevation in AG may be D lactate.

In assessing the levels of ketoacids, coexistent lactic acidosis and the redox state discussed above can be important considerations. If lactate levels are increased as a result of lactic acidosis concurrently with ketone overproduction as a result of diabetic acidosis, the ketones exist primarily in the form of β -hydroxybutyrate. The results of certain tests for ketones that measure only acetoacetate (such as the nitroprusside reaction, e.g., Acetest tablets and reagent sticks), therefore may be misleadingly low or even negative despite high total ketone concentrations. Some hospital laboratories no longer use the nitroprusside reaction to estimate total ketones, and

measurement of β -hydroxybutyrate and acetoacetate are the preferred tests.

Treatment. The treatment of DKA has been well studied and recommendations standardized for most patients.^{122,123} The treatment usually requires volume repletion, insulin administration, and potassium repletion at the appropriate time. As discussed below, the acidosis may correct without specific treatment other than for the issues above.

Most, if not all, patients with DKA require correction of the volume depletion that almost invariably accompanies the osmotic diuresis of DKA. Several consensus algorithms have been published.^{122,123} Initially, fluid replacement is given as isotonic saline and later changed to 0.45% sodium chloride.^{117,119} Fluid therapy not only hopefully stabilizes cardiovascular function but also reduces stress hormone levels (including various vasoconstrictors) and improves renal function. Fluid therapy not only hopefully stabilizes cardiovascular function but also reduces stress hormone levels (including various vasoconstrictors) and improves renal function.

Total body K^+ depletion is usually present in DKA, although the K^+ level on admission may be elevated or normal. A normal or reduced K^+ value on admission indicates severe K^+ depletion and should be approached with caution. Administration of fluid, insulin, and alkali may cause the K^+ level to plummet. When urine output has been established, 20 to 30 mEq of potassium chloride should be administered in each liter of fluid as long as the K^+ value is less than 4.0 mEq/L. Equal caution should be exercised in the presence of hyperkalemia, especially if the patient has renal insufficiency, because the usual therapy does not always correct hyperkalemia. Never administer potassium chloride empirically.

For the acidosis per se, most experts only advise giving HCO_3^- if arterial pH is less than 6.9.^{122,123} There is no evidence to support the use of exogenous HCO_3^- administration when the pH is above 6.9. With extremely low pH, (pH < 6.9), giving HCO_3^- may improve cardiac contractility, lessen arterial vasodilatation, improve tissue perfusion, and drive K^+ into cells. HCO_3^- has several potential disadvantages: reducing ventilatory drive (and theoretically and paradoxically reducing central nervous system [CNS] pH), worsening ketogenesis, and inducing a post-treatment alkalosis when ketones are metabolized to HCO_3^- .

The AG should be followed closely during therapy because it is expected to decline as ketones are cleared from the plasma and precedes an increase in plasma HCO_3^- as the acidosis is repaired. It is not usually necessary to monitor blood ketone levels continuously.^{117,119,120}

As patients are treated with fluids, more ketoacid anions (beta-hydroxybutyrate and acetoacetate) may be excreted into the urine lowering the AG, but not necessarily changing the pH or HCO_3^- . However, the excreted anions represent loss of “potential HCO_3^- ” because the anions if retained could be metabolized to yield HCO_3^- . Therefore, the urinary excretion of ketoacid anions alone may not change the acidosis itself or the plasma HCO_3^- , but simply shift the acidosis to a hyperchloremic acidosis.¹²⁴ Almost all patients with DKA treated with isotonic saline (a hyperchloremic solution) will evolve to a hyperchloremic metabolic acidosis during treatment. With time, as plasma ketoacid anions are metabolized to HCO_3^- and as the kidneys excrete sufficient acid (simultaneously producing “new” HCO_3^-), the acidosis will correct to normal.

Alcoholic Ketoacidosis (AKA). Some patients with chronic alcoholism, especially binge drinkers who discontinue solid food intake while continuing alcohol consumption, develop the alcoholic form of ketoacidosis when alcohol ingestion is curtailed abruptly. Usually the onset of vomiting and abdominal pain with dehydration leads to cessation of alcohol consumption before the patient comes to the hospital.^{119,120} The metabolic acidosis may be severe but is accompanied by only modestly abnormal glucose levels, which are usually low but may be slightly elevated. Typically, insulin levels are low and levels of triglyceride, cortisol, epinephrine, glucagon, and growth hormone are increased.¹²⁵ The net result of this deranged metabolic state is ketosis. The acidosis is primarily due to elevated levels of ketones, which exist predominantly in the form of β -hydroxybutyrate because of the altered redox state induced by the metabolism of alcohol. The altered redox state will also produce higher lactate concentrations. Compared with patients with DKA, patients with AKA have lower plasma glucose concentrations and higher β -hydroxybutyrate/acetoacetate and lactate/pyruvate ratios.^{119,120} Qualitative ketone tests such as nitroprusside, which are more sensitive to acetoacetate, may be only trace positive or negative in AKA, despite markedly increased β -hydroxybutyrate levels. The metabolic acidosis may be severe but is accompanied by only modestly abnormal glucose levels, which are usually low but may be slightly elevated. Typically, insulin levels are low and levels of triglyceride, cortisol, epinephrine, glucagon, and growth hormone are increased.¹²⁵ The net result of this deranged metabolic state is ketosis. The acidosis is primarily due to elevated levels of ketones, which exist predominantly in the form of β -hydroxybutyrate because of the altered redox state induced by the metabolism of alcohol. The altered redox state will also produce higher lactate concentrations. Compared with patients with DKA, patients with AKA have lower plasma glucose concentrations and higher β -hydroxybutyrate/acetoacetate and lactate/pyruvate ratios.^{119,120} Qualitative ketone tests such as nitroprusside that are more sensitive to acetoacetate may be only trace positive or negative in AKA, despite markedly increased β -hydroxybutyrate levels.

This disorder is not rare and is underdiagnosed. The clinical presentation in AKA may be complex, and is often underdiagnosed. The typical high anion gap acidosis is often mixed with metabolic alkalosis (vomiting), respiratory alkalosis (alcoholic liver disease), lactic acidosis (hypoperfusion), and/or hyperchloremic acidosis (renal excretion of ketoacids). Finally, the elevation in the osmolar gap is usually accounted for by an increased blood alcohol level, but the differential diagnosis should always include ethylene glycol and/or methanol intoxication.

Treatment. Therapy includes intravenous glucose and saline administration, but insulin should be avoided. K^+ , PO_4^{3-} , Mg^{2+} , and vitamin supplementation (especially thiamine) are frequently necessary. Glucose in isotonic saline, not saline alone, is the mainstay of therapy. Because of superimposed starvation, patients with AKA often develop hypophosphatemia within 12 to 18 hours of admission. Treatment with glucose-containing intravenous fluids increases the risk of severe hypophosphatemia. Levels should be checked on admission and at 4, 6, 12, and 18 hours. Profound hypophosphatemia may provoke aspiration, platelet dysfunction, hemolysis, and rhabdomyolysis. Therefore, phosphate replacement should

be provided promptly when indicated. Hypokalemia and hypomagnesemia are also common and should not be overlooked.^{119,120}

Starvation Ketoacidosis. Fasting alone can increase ketoacid levels, although not usually above 10 mEq/L. Ketosis occurs within the first 24 to 48 hours of fasting, is accentuated by exercise and pregnancy, and is rapidly reversible by glucose or insulin administration. Frank ketoacidosis of more than a moderate extent is unusual. Low carbohydrate diets have also been implicated but ketogenic diets do not routinely cause frank acidosis.¹²⁶ Starvation-induced hypoinsulinemia and accentuated hepatic ketone production have been linked pathogenically.^{119,120}

Drug- and Toxin-Induced Acidosis: Salicylate. Intoxication with salicylates is more common in children than in adults and may result in the development of a high AG metabolic acidosis, but the acid-base abnormality most commonly associated with salicylate intoxication in adults is respiratory alkalosis due to direct stimulation of the respiratory center by salicylates.¹¹⁷ Adult patients with salicylate intoxication usually have pure respiratory alkalosis or mixed respiratory alkalosis–metabolic acidosis.¹¹⁷ Metabolic acidosis when it occurs is due to uncoupling of oxidative phosphorylation, and hence the usual high AG acidosis is mostly due to lactic acid and ketoacids. Only part of the increase in the AG is due to the increase in plasma salicylate concentration; for instance, a toxic salicylate level of 100 mg/dL would account for an increase in the AG of only 7 mEq/L. High ketone concentrations have been reported to be present in as many as 40% of adult salicylate-intoxicated patients, sometimes as a result of salicylate-induced hypoglycemia.¹²⁷ L-Lactic acid production is also often increased, partly as a direct drug effect¹¹⁷ and partly as a result of the hyperventilation induced by salicylate. Acidosis enhances the transit of salicylates into the central nervous system since the protonated form of salicylate preferentially crosses the blood-brain barrier. Pulmonary edema and acute lung injury may also occur during salicylate intoxication. Adult patients with salicylate intoxication usually have pure respiratory alkalosis or mixed respiratory alkalosis–metabolic acidosis.¹¹⁷ Metabolic acidosis when it occurs is due to uncoupling of oxidative phosphorylation, and hence the usual high AG acidosis is mostly due to lactic acid and ketoacids. Only part of the increase in the AG is due to the increase in plasma salicylate concentration; for instance, a toxic salicylate level of 100 mg/dL would account for an increase in the AG of only 7 mEq/L. High ketone concentrations have been reported to be present in as many as 40% of adult salicylate-intoxicated patients, sometimes as a result of salicylate-induced hypoglycemia.¹²⁷ L-Lactic acid production is also often increased, partly as a direct drug effect¹¹⁷ and partly as a result of the hyperventilation induced by salicylate. Acidosis enhances the transit of salicylates into the central nervous system since the protonated form of salicylate preferentially crosses the blood brain barrier. Pulmonary edema and acute lung injury may also occur during salicylate intoxication.

Treatment. General treatment should always consist of initial vigorous gastric lavage with isotonic saline followed by administration of activated charcoal via nasogastric tube. Treatment of the metabolic acidosis can be helpful because acidosis can enhance the entry of salicylate into the CNS. An alkaline diuresis will aid renal excretion of salicylate, so

NaHCO_3 is usually given cautiously, avoiding $\text{pH} > 7.6$ (and hypokalemia) with coexisting respiratory alkalosis. Acetazolamide is not recommended. Hemodialysis may be necessary in severe poisoning, especially if renal failure coexists; it is preferred in cases of severe intoxication and is superior to hemofiltration, which does not correct the acid-base abnormality.^{117,127}

Toxins: The Osmolar Gap in Toxin-Induced Acidosis. The so-called “osmolar gap” can indicate the presence of certain toxins. Under most physiologic conditions, Na^+ (and its accompanying anions), urea, and glucose generate much of the osmotic pressure of blood. Serum osmolality is calculated according to the following expression:

$$\text{Osmolality} = 2[\text{Na}^+] + \frac{\text{BUN}}{2.8} + \frac{\text{Glucose (mg/dL)}}{18} \quad (16.16)$$

The calculated and determined osmolalities usually agree within 10 mOsm/kg. When the measured osmolality exceeds the calculated osmolality by more than 10 mOsm/kg, one of two possibilities usually exists. First, the serum Na^+ may be spuriously low, as occurs with hyperlipidemia or hyperproteinemia (pseudohyponatremia). Second, osmolytes other than sodium salts, glucose, or urea may have accumulated in plasma. Examples are infused mannitol, radiocontrast media, or other solutes, including the alcohols, ethylene glycol, and acetone, which can increase the osmolality in plasma. For these examples, the difference between the osmolality and the measured osmolality is proportional to the concentration of the unmeasured solute. Such differences in these clinical circumstances have been referred to as the *osmolar gap*. In the presence of an appropriate clinical history and index of suspicion, the osmolar gap becomes a very reliable and helpful screening tool in assessing for toxin-associated high AG acidosis. An increase in the osmolar gap can be associated with AKA and lactic acidosis as discussed in the previous section, but in general, ethanol intoxication per se does not cause a high AG acidosis, although it can cause an elevated osmolar gap. Isopropyl alcohol (rubbing alcohol) poisoning similarly is not metabolized to a strong acid and does not elevate the AG, although the osmolar gap can be elevated and there can be a positive nitroprusside reaction from the metabolism to acetone.

Ethylene Glycol. Ingestion of ethylene glycol (EG), used in antifreeze, leads to a high AG metabolic acidosis^{117,128,129} in addition to severe CNS, cardiopulmonary, and renal damage.¹²⁵ A disparity between the measured and calculated blood osmolality (high osmolar gap) is often present, especially in the first few hours after ingestion. Typically over time, as EG is metabolized, the osmolar gap begins to fall and the anion gap begins to rise so that in advanced EG intoxication, the AG will be very high but the osmolar gap will close. The high AG is attributable to ethylene glycol metabolites, especially oxalic acid, glycolic acid, and other incompletely identified organic acids in addition to severe CNS, cardiopulmonary, and renal damage.¹²⁵ L-Lactic acid production also increases as a result of a toxic depression in the reaction rates of the citric acid cycle and altered intracellular redox state.¹²⁹ Recognition of oxalate crystals in the urine facilitates diagnosis. L-Lactic acid production also increases as a result of a toxic depression in the reaction rates of the citric acid cycle and altered intracellular redox state. Fluorescence of the urine by Wood's light (if the ingested ethylene glycol

contains a fluorescent vehicle) has been suggested as a diagnostic indicator but is neither specific nor sensitive.^{128,129}

The Wood's lamp also may detect the fluorescent vehicle on the shirt or blouse of the patient who has ingested the agent. Fluorescence of the urine by Wood's light (if the ingested ethylene glycol contains a fluorescent vehicle) has been suggested as a diagnostic indicator but is neither specific nor sensitive.^{128,129} Treatment includes prompt institution of osmotic diuresis, thiamine and pyridoxine supplementation, administration of 4-methylpyrazole (fomepizole),¹³⁰ or ethyl alcohol administration and dialysis.^{117,128,130} Do not induce vomiting. Intravenous fomepizole is the drug of choice. Both ethanol and fomepizole are competitive inhibitors of alcohol dehydrogenase and delay the conversion of ethylene glycol into its more toxic metabolites. Competitive inhibition of alcohol dehydrogenase with either fomepizole or ethyl alcohol is absolutely necessary in all patients to lessen toxicity, because ethanol and fomepizole compete for metabolic conversion of ethylene glycol and alter the cellular redox state. Fomepizole (initiated as a loading dose of 15 mg/kg, followed by 10 mg/kg every 12 hours), offers several advantages including set dosing and a predictable decline in ethylene glycol levels without the adverse effect of excessive obtundation, as seen with ethyl alcohol infusion. When these measures have been accomplished, hemodialysis may be initiated to remove the ethylene glycol metabolites. If intravenous ethanol is the only inhibitor of alcohol dehydrogenase available, its infusion should be increased during hemodialysis to allow maintenance of the blood alcohol level in the range of 100 to 150 mg/dL or more than 22 mmol/L. Ethanol can even be added to the dialysate bath. Importantly, hemodialysis can remove fomepizole. The usual indications for hemodialysis include 1) arterial pH of less than 7.3, 2) HCO_3^- concentration of less than 20 mEq/L, 3) osmolal gap of more than 10 mOsm/kg, and 4) oxalate crystalluria.¹²⁸

Methanol. Methanol has wide application in commercially available solvents and is used for industrial and automotive purposes. Sources include windshield wiper fluid, paint remover or thinner, deicing fluid, canned heating sources, varnish and shellac. Ingestion of methanol (wood alcohol) causes metabolic acidosis in addition to severe optic nerve and central nervous system manifestations resulting from its metabolism to formic acid from formaldehyde.^{117,128} Lactic acids and ketoacids as well as other unidentified organic acids may contribute to the acidosis. Because of the low molecular mass of methanol (32 Da), an osmolar gap is usually present early in the course but declines as the anion gap increases, the latter reflecting the metabolism of methanol. Therapy is generally similar to that for ethylene glycol intoxication, including general supportive measures, fomepizole administration, folate administration, and usually hemodialysis.^{125,130} Lactic acids and ketoacids as well as other unidentified organic acids may contribute to the acidosis. Because of the low molecular mass of methanol (32 Da), an osmolar gap is usually present early in the course but declines as the anion gap increases, the latter reflecting the metabolism of methanol. Therapy is generally similar to that for ethylene glycol intoxication, including general supportive measures, fomepizole administration, folate administration, and usually hemodialysis.^{125,130}

Paraldehyde. Intoxication with paraldehyde is very rare but is of historic interest. It is a result of the accumulation

of acetic acid, the metabolic product of the drug from acetaldehyde, and other organic acids. Unmetabolized paraldehyde is exhaled through the respiratory system.

Pyroglutamic Acid. Pyroglutamic acid, or 5-oxoproline, is an intermediate in the γ -glutamyl cycle for the synthesis of glutathione. Acetaminophen ingestion can, in rare cases, deplete intracellular glutathione. This results in increased formation of γ -glutamyl cysteine, which is metabolized to pyroglutamic acid.¹³¹ Accumulation of this intermediate, first appreciated in the rare patients with congenital glutathione synthetase deficiency, has been reported in critically ill patients taking acetaminophen. Many cases now appear to be in chronically ill, malnourished women taking acetaminophen chronically.¹³² Such patients have severe high AG acidosis and often alterations in mental status.^{131,132} All had elevated blood levels of pyroglutamic acid, which increased in proportion to the increase in the AG. Unfortunately, the assay is not widely available. It is conceivable that the heterozygote state for glutathione synthetase deficiency could predispose to pyroglutamic acidosis, because only a minority of critically ill patients receiving acetaminophen develop this form of metabolic acidosis.¹³¹

Propylene Glycol. Propylene glycol is used as a vehicle for some intravenous medications and some cosmetics and is metabolized to lactic acid in the liver by alcohol dehydrogenase. Patients can have an elevated osmolar gap, an elevated AG, and an acidosis.¹³³ Numerous intravenous preparations contain propylene glycol as the vehicle (lorazepam, diazepam, pentobarbital, phenytoin, nitroglycerin, and TMP-SMX). Propylene glycol may accumulate and cause a high AG, high osmolar gap acidosis in patients receiving continuous infusion or higher dosages of these agents, especially in the presence of chronic kidney disease, chronic liver disease, alcohol abuse, or pregnancy. The acidosis is the result of accumulation of L-lactic acid, D-lactic acid, and L-acetaldehyde. The acidosis typically abates with cessation of the offending agent and supportive care; occasionally dialysis is used and fomepizole can be considered.¹²⁵

Uremia. Advanced CKD usually eventually converts the non-gap metabolic acidosis discussed earlier to the typical high AG acidosis, or “uremic acidosis”^{37,89} reduced rate of NH_4^+ production and excretion because of cumulative and significant loss of renal mass³⁷ reduced rate of NH_4^+ production and excretion because of cumulative and significant loss of renal mass.³⁷ Usually, acidosis does not occur until a major portion of the total functional nephron population (>75%) has been compromised, because of the adaptation by surviving nephrons to increase ammoniogenesis. Eventually, however, there is a decrease in total renal ammonia excretion as renal mass is reduced to a level at which the GFR is 20 mL/min or less. PO_4^{3-} balance is maintained until late stage CKD as a result of both hyperparathyroidism, which decreases proximal PO_4^{3-} absorption, and an increase in plasma PO_4^{3-} as GFR declines. However, protein restriction and the administration of phosphate binders reduce the availability of PO_4^{3-} .

Treatment of Acidosis of Chronic Renal Failure. The uremic acidosis of renal failure can be treated with oral alkali replacement to maintain the HCO_3^- concentration above 22 mEq/L. This can be accomplished with relatively modest amounts of alkali (1.0 to 1.5 mEq/kg/day). Shohl's solution or sodium bicarbonate tablets (650-mg tablets)

are equally effective. It is assumed that alkali replacement serves to prevent the harmful effects of prolonged positive H^+ balance, especially progressive catabolism of muscle and loss of bone. Because sodium citrate (Shohl's solution) has been shown to enhance the absorption of aluminum from the gastrointestinal tract, it should never be administered to patients receiving aluminum-containing antacids because of the risk of aluminum intoxication. When hyperkalemia is present, furosemide (60-80 mg/day in divided doses usually) can be added. Occasionally a patient may require long-term treatment with potassium binding or exchange resins as discussed above.

Metabolic Alkalosis: Diagnosis of Simple and Mixed Forms of Metabolic Alkalosis. Metabolic alkalosis is a primary acid-base disturbance that is manifest in the most pure or simple form as alkalemia (elevated arterial pH) and an increase in PaCO_2 as a result of compensatory alveolar hypoventilation. Metabolic alkalosis is one of the more common acid-base disturbances in hospitalized patients and can be manifest as either a simple and a mixed acid-base disorder. A patient with a high plasma HCO_3^- concentration and a low plasma Cl^- concentration has either metabolic alkalosis or chronic respiratory acidosis. The arterial pH establishes the diagnosis, because it is increased in metabolic alkalosis and decreased in respiratory acidosis. Modest increases in the PaCO_2 are expected in metabolic alkalosis. A combination of the two disorders is not unusual, because many patients with chronic obstructive lung disease are treated with diuretics, which promote ECF contraction, hypokalemia, and metabolic alkalosis. Metabolic alkalosis is also frequently observed not as a pure or simple acid-base disturbance, but in association with other disorders such as respiratory acidosis, respiratory alkalosis, and metabolic acidosis (*mixed disorders*). *Mixed metabolic alkalosis–metabolic acidosis can be only be detected by laboratory parameters if the accompanying metabolic acidosis is a high AG acidosis.* The mixed disorder can be appreciated by comparison of the change in the AG above the normal value of 10 mEq/L ($\Delta\text{AG} = \text{patient's AG} - 10$) with the decrement in the $[\text{HCO}_3^-]$ below the normal value of 25 mEq/L ($\Delta\text{HCO}_3^- = 25 - \text{patient's HCO}_3^-$). In a pure high AG metabolic acidosis the increase in the AG (or ΔAG) should approximate the decrease in HCO_3^- (ΔHCO_3^-). A mixed metabolic alkalosis–high AG metabolic acidosis is recognized because the delta values are not similar. Often, there is no bicarbonate deficit, yet the AG is significantly elevated. Thus, in a patient with an AG of 20 but a near-normal HCO_3^- concentration, mixed metabolic alkalosis–metabolic acidosis should be strongly considered. Common examples include renal failure acidosis (uremic) with vomiting or DKA with vomiting.

Respiratory compensation for metabolic alkalosis is less predictable than that for metabolic acidosis. In general, the anticipated PCO_2 can be estimated by adding 15 to the patient's serum $[\text{HCO}_3^-]$ in the range of HCO_3^- from 25 to 40 mEq/L. Further elevation in PCO_2 is limited by hypoxemia and, to some extent, hypokalemia, which often accompany metabolic alkalosis. Nevertheless, if a patient has a PCO_2 of only 40 mm Hg while the $[\text{HCO}_3^-]$ is frankly elevated (e.g., 35 mEq/L) and the pH is in the alkalemic range, then respiratory compensation is inadequate and a mixed metabolic alkalosis–respiratory alkalosis exists.

In assessing a patient with metabolic alkalosis, two questions must be considered: 1) What is the source of alkali gain (or

acid loss) that *generated* the alkalosis? and 2) What renal mechanisms are operating to prevent urinary excretion of excess HCO_3^- , thereby *maintaining*, rather than correcting, the alkalosis? This second question is critical because under normal conditions and renal function, a patient should be able to excrete excess HCO_3^- just by reabsorbing less of the large amounts filtered. In the following discussion, the entities responsible for *generating* alkalosis are discussed individually and reference is made to the mechanisms necessary to sustain the increase in blood HCO_3^- concentration in each case. The general mechanisms responsible for the *maintenance of alkalosis* have been discussed in detail earlier in this chapter, and are often a result of the combined effects of a reduction in GFR as well as chloride, ECF volume, and potassium depletion (Fig. 16.9).

Hypokalemia is often an important component of the maintenance phase of metabolic alkalosis and has selective effects on 1) H^+ secretion and 2) ammonium excretion. The former is a result, in part, of stimulation of the $\text{H}^+-\text{K}^+-\text{ATPase}$ in type A intercalated cells of the collecting duct by hypokalemia. The latter, ammonium excretion, is a direct result of enhanced ammoniogenesis and ammonium transport (proximal convoluted tubule, TAL, medullary collecting duct) in response to hypokalemia. Finally, hyperaldosteronism (primary or secondary) participates in sustaining the alkalosis by increasing activity of the H^+-ATPase and $\text{H}^+-\text{K}^+-\text{ATPase}$ in type A intercalated cells as well as the ENaC and the $\text{Na}^+-\text{K}^+-\text{ATPase}$ in collecting duct principal cells. The net result of the latter process is to stimulate K^+ secretion through K^+ -selective channels in this same cell, which thus maintains the hypokalemia and alkalosis.¹⁶

Under normal circumstances, the kidneys display an impressive capacity to excrete HCO_3^- . Thus, *the development of metabolic alkalosis represents a failure of the kidneys to eliminate HCO_3^- at the normal capacity*. The kidneys retain, rather than excrete, the excess alkali and maintain the alkalosis if one of several mechanisms is operative (see Fig. 16.9):

1. Cl^- deficiency (ECF volume contraction) exists concurrently with K^+ deficiency to decrease GFR and/or enhance

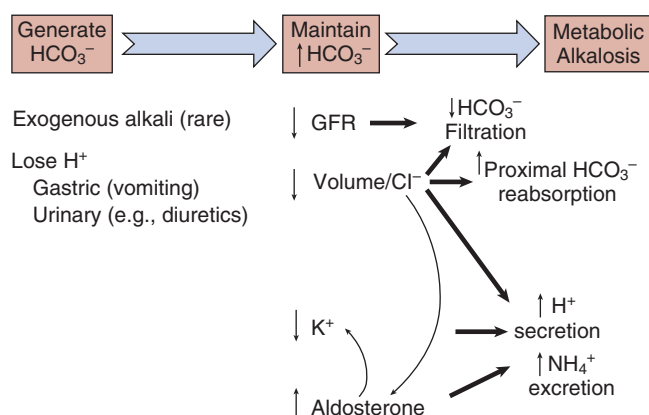


Fig. 16.9 Pathophysiologic basis of the generation and maintenance of chronic metabolic alkalosis. Paradoxical stimulation of bicarbonate absorption (H^+ secretion) and NH_4^+ production and excretion is the combined result of Cl^- deficiency (with reduction in GFR), K^+ deficiency, and secondary hyperaldosteronism. GFR, Glomerular filtration rate.

proximal and distal HCO_3^- absorption. This combination of disorders evokes secondary hyperreninemic hyperaldosteronism and stimulates H^+ secretion in the collecting duct. Hypokalemia independently stimulates ammoniogenesis and net acid excretion, thereby, adding additional or “new” bicarbonate to the systemic circulation. Repair of the alkalosis may be accomplished by saline and K^+ administration. These cases are sometimes referred to as chloride or saline responsive. This is usually indicated by a low urinary Cl^- once any effect of a diuretic has gone.

2. Increased mineralocorticoids (e.g., aldosterone) and hypokalemia are induced in other cases by autonomous factors unresponsive to increased ECF. The stimulation of distal H^+ secretion is then sufficient to reabsorb the increased filtered HCO_3^- load and to overcome the decreased proximal HCO_3^- reabsorption caused by ECF expansion. Repair of the alkalosis in this case rests with removal of the excess autonomous mineralocorticoid and potassium repletion; saline administration is ineffective. These cases are sometimes referred to as chloride resistant or saline resistant. Such conditions are usually indicated by a higher urinary Cl^- .

The various causes of metabolic alkalosis are summarized in Table 16.17. In attempting to establish the cause of metabolic alkalosis, one must assess the clinical circumstance (e.g., diuretics, vomiting) and the status of the ECF volume, blood pressure, serum K^+ concentration, and in some cases the renin-aldosterone system. Urine electrolyte determinations and urine screening for diuretics are helpful diagnostic tools (Table 16.18). A diagnostic approach to metabolic alkalosis is summarized in the flow diagram in Fig. 16.10 but the clinical circumstance may be the best starting point – history of gastric fluid losses, diuretics, hypertension, etc. In the absence of historical clues, the presence or absence of hypertension differentiates those conditions with primary sodium chloride retention (e.g., primary hyperaldosteronism, Cushing’s syndrome) versus those with excess losses (e.g., Bartter’s or Gitelman’s syndrome).

Exogenous Bicarbonate Loads. Long-term administration of alkali to individuals with normal renal function results in minimal, if any, alkalosis, since excess HCO_3^- is easily excreted normally. In patients with chronic renal insufficiency, however, overt alkalosis can develop after alkali administration, presumably because the capacity to excrete HCO_3^- is exceeded or because coexistent hemodynamic disturbances have caused enhanced HCO_3^- reabsorption.

Bicarbonate and Bicarbonate-Precursor Administration. The propensity of patients who have ECF contraction or renal disease plus alkali loads to develop alkalosis is exemplified by patients who receive oral or intravenous HCO_3^- , acetate loads in parenteral hyperalimentation solutions, sodium citrate loads (via regional anticoagulation, transfusions, plasma exchange anticoagulant, or infant formula), or antacids plus cation exchange resins. The use of trisodium citrate solution for regional anticoagulation has been reported to be a cause of metabolic alkalosis in patients receiving continuous renal replacement therapy.^{134–136} Citrate metabolism consumes a hydrogen ion and thereby generates HCO_3^- in liver and skeletal muscle.¹³⁶ Citrate metabolism consumes a hydrogen ion and thereby generates HCO_3^- in liver and skeletal muscle.

Table 16.17 Causes of Metabolic Alkalosis**Exogenous HCO_3^- Loads**

Acute alkali administration

Milk-alkali syndrome**Effective ECV Contraction, Normotension, K^+ Deficiency, and Secondary Hyperreninemic****Hyperaldosteronism**

Gastrointestinal origin

Vomiting**Gastric aspiration**

Congenital chloridorrhea

Villous adenoma

Renal origin

Diuretics (especially thiazides and loop diuretics)**Edematous states**

Post-hypercapnic state

Hypercalcemia-hypoparathyroidism

Recovery from lactic acidosis or ketoacidosis

Nonreabsorbable anions such as penicillin, carbenicillin

 Mg^{2+} deficiency **K^+ depletion**

Bartter's syndrome (loss of function mutations in thick ascending limb of Henle)

Gitelman's syndrome (loss-of-function of Na^+/Cl^- cotransporter - DCT)

Carbohydrate refeeding after starvation

ECV Expansion, Hypertension, K^+ Deficiency, and Hypermineralocorticoidism

Associated with high renin level

Renal artery stenosis

Accelerated hypertension

Renin-secreting tumor

Estrogen therapy

Associated with low renin level

Primary aldosteronism

Adrenal adenoma

Hyperplasia

Carcinoma

Glucocorticoid suppressible

Adrenal enzymatic defects

11 β -hydroxylase deficiency17 α -hydroxylase deficiency

Cushing's syndrome or disease

Ectopic corticotrophin

Adrenal carcinoma

Adrenal adenoma

Primary pituitary

Other

Licorice

Carbenoxolone

Smokeless (chewing) tobacco

Lydia Pinkham tablets

Gain-of-Function Mutation of ENaC with ECV Expansion, Hypertension, K^+ Deficiency, and Hyporeninemic**Hyperaldosteronism**

Liddle's syndrome

Table 16.18 Diagnosis of Metabolic Alkalosis**Saline-Responsive Alkalosis**Low Urinary $[\text{Cl}^-]$
(<10 mEq/L)

Normotensive

Vomiting

Nasogastric aspiration

Diuretic use (distant)

Post-hypercapnia

Villous adenoma

Bicarbonate treatment
of organic acidosis K^+ deficiency**Saline-Unresponsive Alkalosis**High or Normal Urinary $[\text{Cl}^-]$
(>15 - 20 mEq/L)

Hypertensive

Primary aldosteronism

Cushing's syndrome

Renal artery stenosis

Renal failure plus alkali therapy

Liddle's syndrome

Normotensive

 Mg^{2+} deficiencySevere K^+ deficiency

Bartter's syndrome

Gitelman's syndrome

Diuretic use (recent)

disorder is treated (e.g., improved systemic circulation in lactic acidosis or insulin in DKA). Any metabolic alkalosis will correct spontaneously unless maintained by volume / Cl^- depletion (perhaps caused in part by the acidosis) and/or K^+ deficiency.

Milk-Alkali Syndrome. Another cause of metabolic alkalosis historically was excessive ingestion of milk and antacids. Patients have hypercalcemia, metabolic alkalosis, and renal insufficiency. The syndrome is now seen mainly because of the use of calcium supplementation (e.g., for osteoporosis) with other alkali (e.g., calcium carbonate). Older women with poor dietary intake ("tea and toasters") are especially prone. In Asia, betel nut chewing is a cause because the nut is often wrapped in calcium hydroxide. Both hypercalcemia and vitamin D excess have been suggested to increase renal HCO_3^- reabsorption. Patients with these disorders are prone to develop nephrocalcinosis, renal insufficiency, and metabolic alkalosis.¹⁶ Renal insufficiency is a clear part of the syndrome and involved in the pathogenesis, limiting the excretion of HCO_3^- . Discontinuation of alkali ingestion is usually sufficient to repair the alkalosis.

Loss of Acid: Vomiting and Gastric Aspiration. Gastrointestinal loss of H^+ results in production of HCO_3^- in the body fluids. As discussed above, loss of an acid is the same as the gain of alkali in terms of pH. Increased H^+ loss through gastric secretions can be caused by vomiting due to physical or psychiatric reasons, nasogastric tube aspiration, or a gastric fistula (see Table 16.17).¹⁶

The fluid and sodium chloride loss in vomitus or in nasogastric suction results in ECF contraction with an increase in plasma renin activity, aldosterone, and other antinatriuretic factors.¹⁶ These factors decrease GFR and also enhance the capacity of the renal tubule to reabsorb HCO_3^- .¹⁵ During the active phase of vomiting, there is continued addition of HCO_3^- to plasma in exchange for Cl^- . The plasma HCO_3^- concentration increases to a level that exceeds the reabsorptive capacity of the proximal tubule. The excess sodium bicarbonate enters the distal tubule, where, under the influence of the increased level of aldosterone, K^+ and H^+ secretion is stimulated. Because of ECF contraction and hyponatremia,

Another setting whereby exogenous alkali can cause a temporary metabolic alkalosis is *after* treatment of an organic acidosis such as lactic acidosis or ketoacidosis. In such disorders, there are two sources of HCO_3^- , first administered HCO_3^- used to treat the acidosis, and second, metabolized organic anion which can be converted to HCO_3^- as the

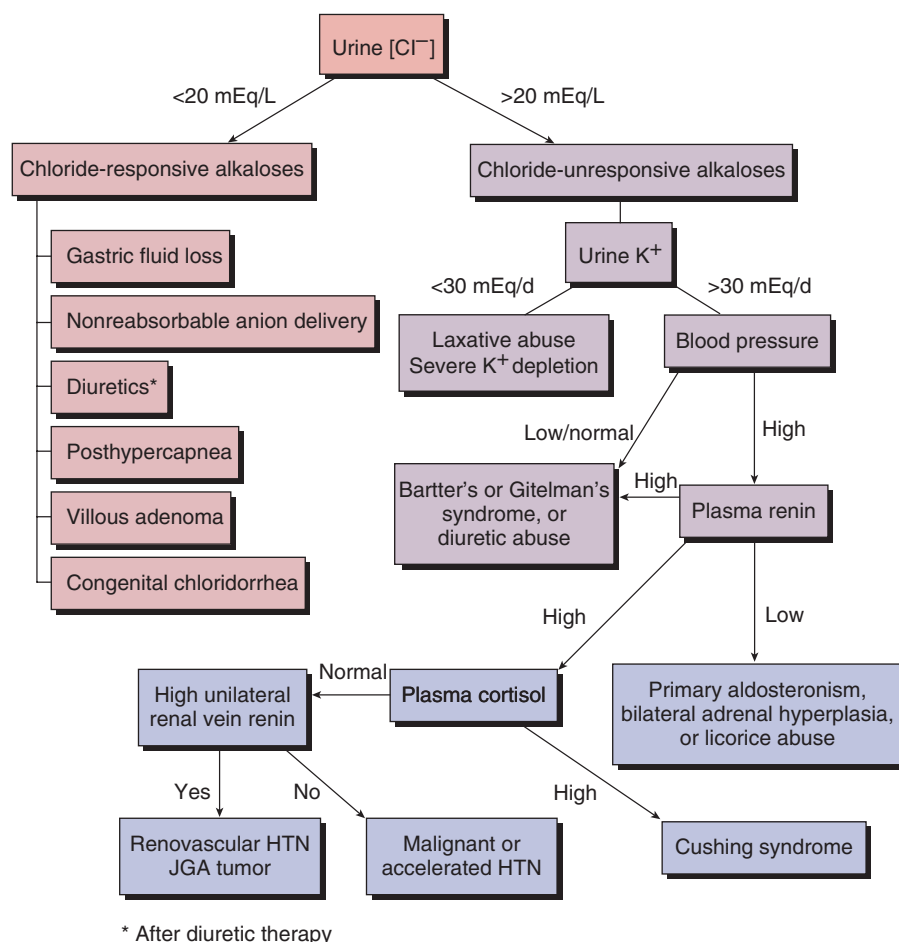


Fig. 16.10 Diagnostic algorithm for metabolic alkalosis, based on the spot urine Cl^- and K^+ concentrations. HTN, Hypertension; JGA, juxta-glomerular apparatus.

the kidney avidly conserves Cl^- . Consequently, in this disequilibrium state generated by active vomiting, the urine contains large quantities of K^+ , but has a low concentration of Cl^- . On cessation of vomiting, the plasma HCO_3^- concentration falls to the HCO_3^- reabsorptive capacity, which is markedly elevated by the continued effects of ECF contraction, hypokalemia, and hyperaldosteronism. The alkalosis is maintained at a slightly lower level than during the phase of active vomiting, and the urine is now relatively acidic with low concentrations of Na^+ , HCO_3^- , and Cl^- .

Correction of the ECF contraction with sodium chloride may be sufficient to reverse these events, with restoration of normal blood pH even without repair of K^+ deficits.¹⁵ Good clinical practice, however, dictates K^+ repletion as well.¹⁶ Good clinical practice, however, dictates K^+ repletion as well.¹⁶

Unusual Causes of Diarrhea: Villous Adenoma and Congenital Chloridorrhea. Most stool is alkaline and most forms of diarrhea cause metabolic acidosis. However, some unusual forms of diarrhea can cause metabolic alkalosis. Metabolic alkalosis has been described in cases of villous adenoma and is ascribed to high adenoma-derived K^+ secretory rates. K^+ and volume depletion likely cause the alkalosis. Congenital chloridorrhea is a rare autosomal recessive disorder associated with severe diarrhea, fecal acid loss, and HCO_3^- retention. The pathogenesis is loss of the normal ileal $\text{HCO}_3^-/\text{Cl}^-$ anion exchange

mechanism so that Cl^- cannot be normally reabsorbed. This is usually the result of mutations in the gene *SLC26A3*, also known as downregulated in adenoma (DRA).¹³⁷ The parallel Na^+/H^+ ion exchanger remains functional, which allows Na^+ to be reabsorbed and H^+ to be secreted. Subsequently, net H^+ and Cl^- exit in the stool, which causes Na^+ and HCO_3^- retention in the ECF.^{15,16} Alkalosis results and is sustained by concomitant ECF contraction with hyperaldosteronism and K^+ deficiency. Therapy consists of oral supplements of sodium and potassium chloride. The use of proton pump inhibitors has been advanced as a means of reducing chloride secretion by the parietal cells and thus reducing the diarrhea.¹³⁸

Renal Loss of Acid: Diuretics. Drugs that induce chloruresis without bicarbonaturia, such as thiazides and loop diuretics (furosemide, bumetanide, and torsemide), acutely diminish ECF volume without altering the total body HCO_3^- content. The HCO_3^- concentration in the blood and ECF increases. The PCO_2 does not increase commensurately, and a “contraction” alkalosis results.¹⁶ The degree of alkalosis is usually

small, however, because of cellular and non- HCO_3^- ECF buffering processes.^{15,16} Administration of diuretics long term tends to generate an alkalosis by increasing distal salt delivery, so that both K^+ and H^+ secretion are stimulated. Diuretics, by blocking Cl^- reabsorption in the distal tubule or by increasing H^+ pump activity, may also stimulate distal H^+ secretion and increase net acid excretion. Maintenance of alkalosis is ensured by the persistence of ECF contraction, secondary hyperaldosteronism, K^+ deficiency, enhanced ammonium production, stimulation of the apical H^+ - and H^+ - K^+ -ATPases, which persists for as long as diuretic administration continues. Repair of the alkalosis is achieved by cessation of diuretic administration and by providing Cl^- to normalize the ECF deficit.

Edematous States. In diseases associated with edema formation (CHF, nephrotic syndrome, cirrhosis), effective arterial blood volume is diminished, although total ECF is increased. Common to these diseases is often diminished renal plasma flow and GFR with limited distal Na^+ delivery. Net acid excretion is usually normal, and alkalosis does not develop, even with an enhanced proximal HCO_3^- reabsorptive capacity. However, the distal H^+ secretory mechanism is primed by hyperaldosteronism to excrete excessive net acid if GFR can be increased to enhance distal Na^+ delivery or if K^+ deficiency or diuretic administration supervenes.

Posthypercapnia. Prolonged CO_2 retention with chronic respiratory acidosis enhances renal HCO_3^- absorption and the generation of new HCO_3^- (increased net acid excretion). If the PCO_2 is returned to normal as with mechanical ventilation, metabolic alkalosis, caused by the persistently elevated HCO_3^- concentration, emerges. Normally there may be a brisk loss of urine HCO_3^- proportional to the change in PCO_2 , and a return to normal acid-base status. But often there are other factors that cause retention of excess HCO_3^- and therefore a metabolic alkalosis. Associated ECF contraction, secondary hyperaldosteronism, diuretics, or K^+ deficiency may help maintain the metabolic alkalosis.

Bartter's Syndrome. Bartter's syndrome and Gitelman's syndrome described next are distal nephron salt losing disorders characterized by hypokalemia, metabolic alkalosis, high renin and aldosterone, and often high prostaglandin E_2 .¹³⁹ As noted below, there are multiple genes involved resulting in heterogeneity in specific phenotypes.¹³⁹ Both classic Bartter's syndrome and antenatal Bartter's syndrome are inherited as autosomal recessive disorders and involve impaired TAL salt absorption, which results in salt wasting, volume depletion, and activation of the renin-angiotensin system.¹⁴⁰ These manifestations are the result of loss-of-function mutations of one of the genes that encode three transporters or regulatory proteins involved in NaCl absorption in the TAL.¹³⁹ The most prevalent disorder is the inheritance from both parents of mutations of the gene *NKCC2* that encodes the bumetanide-sensitive Na^+ - 2Cl^- - K^+ cotransporter on the apical membrane. Other mutations have also been described in rare families. For example, a mutation has been discovered in the gene *KCNJ1* that encodes the ATP-sensitive apical K^+ conductance channel (ROMK) that operates in parallel with the Na^+ - 2Cl^- - K^+ transporter to recycle K^+ . Both defects can be associated with classic Bartter's syndrome. A third mutation of the *CLCNKB* gene encoding the voltage-gated basolateral chloride channel (ClC-Kb) is associated only with classic Bartter's syndrome and is milder and rarely associated with

nephrocalcinosis. All three defects above have the same net effect, loss of Cl^- transport in the TAL.¹⁴¹ Antenatal Bartter's syndrome has been observed in consanguineous families in association with sensorineural deafness. The responsible gene, *BSND*, encodes the associated subunit, Barttin, that co-localizes with the ClC-Kb channel in both the TAL and K -secreting epithelial cells in the inner ear. Barttin appears to be necessary for the function of the voltage-gated chloride channel. Expression of ClC-Kb is lost when coexpressed with mutant Barttins. Thus, mutations in *BSND* represent a fourth category of patients with Bartter syndrome.¹⁴² With the exception of the deafness, the electrolyte and acid-base abnormalities are similar to the other forms of Bartter syndrome.

Two groups of investigators have reported features of Bartter's syndrome in patients with tetany who inherit autosomal dominant hypocalcemia as a result of activating mutations in the calcium-sensing receptor (CaSR).¹⁴³ Gain-of-function mutations of the gene encoding CaSR on the basolateral surface of the TAL cell inhibits the function of ROMK. Thus, mutations in CaSR appear to represent a fifth gene associated with Bartter's syndrome.¹³⁹ Acquired forms of Bartter's syndrome, usually characterized by hypokalemic metabolic alkalosis, hypomagnesemia, hypocalcemia and normal kidney function, have been reported in children and adults in association with aminoglycoside toxicity including gentamicin, amikacin, netilmicin, capreomycin, viomycin, colistin, and neomycin. Similarly, cisplatin and cyclosporine have also been implicated. Although the cellular mechanism of this association has not been clearly elucidated, *in vitro* studies suggest an effect on the CaSR.

Therefore, these defects, when considered collectively, lead to ECF contraction, hyperreninemic hyperaldosteronism, and increased delivery of Na^+ to the distal nephron and, thus, alkalosis and renal K^+ wasting and hypokalemia. Secondary overproduction of prostaglandins, juxtaglomerular apparatus hypertrophy, and vascular pressor unresponsiveness would then ensue. Most patients have hypercalciuria and normal serum magnesium levels, which distinguishes this disorder from Gitelman's syndrome.

Distinction from surreptitious vomiting, surreptitious diuretic use, and laxative abuse is necessary to make the diagnosis of Bartter's syndrome. The finding of a low urinary Cl^- concentration is helpful in identifying the vomiting patient. The urinary Cl^- concentration in Bartter's syndrome would be expected to be normal or increased, rather than depressed. Bartter-like manifestations have been reported in sporadic cases associated with chronic intermittent diuretic and laxative abuse, cystic fibrosis, and congenital chloride diarrhea.

The treatment of Bartter's syndrome will generally include K^+ supplementation and nonsteroidal antiinflammatory drugs (NSAIDs) to block prostaglandin production, and also amiloride or spironolactone to lessen distal K^+ secretion. ACE inhibitors have been used with limited success.

Gitelman's Syndrome. Patients with Gitelman's syndrome have a phenotype resembling that of Bartter's syndrome in that an autosomal recessive chloride-resistant metabolic alkalosis is associated with hypokalemia, normal to low blood pressure, volume depletion with secondary hyperreninemic hyperaldosteronism, and juxtaglomerular hyperplasia.^{141,144} However, hypocalciuria and symptomatic hypomagnesemia are usually useful in distinguishing Gitelman's syndrome from

Bartter's syndrome on clinical grounds.^{139,144} These unique features mimic the effect of long-term thiazide diuretic administration. In contrast, in most patients with Bartter's syndrome, the urine calcium level is elevated. However, hypocalciuria and symptomatic hypomagnesemia are usually useful in distinguishing Gitelman's syndrome from Bartter's syndrome on clinical grounds.^{139,144} These unique features mimic the effect of long-term thiazide diuretic administration. In contrast, in most patients with Bartter's syndrome, the urine calcium level is elevated.

A number of missense mutations in the gene *SLC12A3*, which encodes the thiazide-sensitive sodium-chloride cotransporter NCC in the distal convoluted tubule, have been described and account for the spectrum of clinical features, including the classical finding of hypocalciuria.¹⁴⁵ Symptoms may include salt craving, nocturia, cramps, and fatigue.¹⁴⁶

Treatment of Gitelman's syndrome, as of Bartter's syndrome, consists of potassium supplementation (KCl 40 mEq, three or four times daily, or more), but also magnesium supplementation in most patients. Amiloride (5 to 10 mg twice daily) is more effective than spironolactone. ACE inhibitors or ARBs in very low dose have been suggested as helpful in selected patients but may cause symptomatic hypotension. NSAIDs may be useful.

Nonreabsorbable Anions and Magnesium Deficiency. Administration of large amounts of nonreabsorbable anions, such as penicillin or carbenicillin, can enhance distal acidification and K^+ excretion by increasing the luminal potential difference attained or possibly by allowing Na^+ delivery to the CCT without Cl^- , which favors H^+ secretion without Cl^- -dependent HCO_3^- secretion.¹⁶ Mg^{2+} deficiency can also result in hypokalemic alkalosis, perhaps by causing K^+ deficiency or by enhancing distal acidification through stimulation of renin and hence aldosterone secretion.

Potassium Depletion. Isolated K^+ depletion can occasionally cause metabolic alkalosis, although generally of only modest severity. In most cases of metabolic alkalosis, K^+ depletion or hypokalemia is one of several contributing factors including volume or Cl^- depletion or mineralocorticoid excess. K^+ deficiency is known in various experimental models to cause 1) an intracellular acidosis, 2) activate H^+ secretion in proximal and distal nephron segments, and 3) increase ammonium production and excretion.²⁶ Activation of the renal H^+ , K^+ -ATPase in the collecting duct by chronic hypokalemia likely plays a role in maintenance of alkalosis associated with K^+ depletion. The alkalosis associated with severe K^+ depletion is resistant to salt administration and repair of the K^+ deficiency is necessary to correct the alkalosis.

Excess Mineralocorticoid Activity (see Table 16.17). As previously discussed, excess mineralocorticoids increase net acid excretion and tend to cause metabolic alkalosis in combination with volume expansion and hypertension. The degree of alkalosis is augmented by the simultaneous increase in K^+ excretion, which leads to K^+ deficiency and hypokalemia. Salt intake for sufficient distal Na^+ delivery is a prerequisite for the development of both the hypokalemia and the alkalosis. Hypertension develops partly as a result of ECF expansion from salt retention. The alkalosis is not progressive and is generally mild and not associated with edema, partly as a result of mechanisms referred to as "aldosterone escape". Volume expansion tends to antagonize the decrease in GFR through increases in natriuretic peptides (e.g. atrial natriuretic

peptide) and/or increase in tubule acidification induced by hypermineralocorticoidism and K^+ deficiency. Increased mineralocorticoid levels may be the result of autonomous primary adrenal overproduction of mineralocorticoids or of secondary aldosterone release by primary renal overproduction of renin. In both cases, the normal feedback by ECF on net mineralocorticoid production is disrupted and volume retention results in hypertension.

High Renin Levels, Secondary Hyperaldosteronism. States accompanied by inappropriately high renin levels may be associated with hyperaldosteronism and alkalosis. Renin levels are elevated because of primary elaboration of renin or, secondarily, by diminished effective circulating blood volume. Total ECF may not be diminished. Examples of high-renin hypertension include renovascular, accelerated, and malignant hypertension. Estrogens increase renin substrate and hence angiotensin II formation. Primary tumor overproduction of renin is another rare cause of hyperreninemic hyperaldosterone-induced metabolic alkalosis.¹⁶

Low Renin Levels, Primary Increased Mineralocorticoids. In some disorders, primary adrenal overproduction of mineralocorticoid suppresses renin elaboration. Hypertension occurs as the result of mineralocorticoid excess with volume overexpansion.

Primary Aldosteronism. Tumor involvement (adenoma or, rarely, carcinoma) or hyperplasia of the adrenal gland is associated with aldosterone overproduction. Mineralocorticoid administration or excess production (primary aldosteronism or Cushing's syndrome and adrenal cortical enzyme defects) increases net acid excretion and may result in metabolic alkalosis, which may be worsened by associated K^+ deficiency. ECF volume expansion from salt retention causes hypertension and antagonizes the reduction in GFR and/or increases tubule acidification induced by aldosterone and by K^+ deficiency. The kaliuresis may persist and cause continued K^+ depletion with polydipsia, inability to concentrate the urine, and polyuria.

Glucocorticoid-Remediable Hyperaldosteronism. Glucocorticoid-remediable hyperaldosteronism is an autosomal dominant form of hypertension, the features of which resemble those of primary aldosteronism (hypokalemic metabolic alkalosis and volume-dependent hypertension). In this disorder, however, glucocorticoid administration corrects the hypertension as well as the excessive excretion of 18-hydroxysteroid in the urine. Dluhy and associates have demonstrated that this disorder results from unequal crossing over between two genes located in close proximity on chromosome 8.¹⁴⁷ This region contains the glucocorticoid-responsive promoter region of the gene encoding 11 β -hydroxylase (*CYP11B1*) where it is joined to the structural portion of the *CYP11B2* gene encoding aldosterone synthase.¹⁴⁷ The chimeric gene produces excess amounts of aldosterone synthase, unresponsive to serum potassium or renin levels, but suppressed by glucocorticoid administration. Although a rare cause of primary aldosteronism, the syndrome is important to distinguish because treatment differs and it can be associated with severe hypertension, stroke, and accelerated hypertension during pregnancy. This region contains the glucocorticoid-responsive promoter region of the gene encoding 11 β -hydroxylase (*CYP11B1*) where it is joined to the structural portion of the *CYP11B2* gene encoding aldosterone synthase.¹⁴⁷ The chimeric gene produces excess amounts of aldosterone synthase, unresponsive to serum potassium or renin levels, but suppressed by glucocorticoid administration.

Although a rare cause of primary aldosteronism, the syndrome is important to distinguish because treatment differs and it can be associated with severe hypertension, stroke, and accelerated hypertension during pregnancy.

Cushing's Disease or Syndrome. Abnormally high glucocorticoid production caused by adrenal adenoma or carcinoma or ectopic corticotropin production causes metabolic alkalosis. The alkalosis may be ascribed to coexisting mineralocorticoid (deoxycorticosterone and corticosterone) hypersecretion or by direct effects of the glucocorticoids. Glucocorticoids also have the ability to enhance net acid secretion and NH_4^+ production by occupancy of mineralocorticoid receptors.

Liddle's Syndrome (Low Renin, Low Aldosterone). Liddle's syndrome is associated with severe hypertension often presenting in childhood, and is frequently accompanied by hypokalemic metabolic alkalosis. These features resemble those of primary hyperaldosteronism, but the renin and aldosterone levels are suppressed (pseudohyperaldosteronism). The defect is constitutive activation of the ENaC at the apical membrane of principal cells in the CCD. Liddle originally described patients with low renin and low aldosterone levels that did not respond to spironolactone. The defect in Liddle's syndrome is inherited as an autosomal dominant form of monogenic hypertension. This disorder has been attributed to an inherited gain of function abnormality in the genes that encode the β - or the γ -subunit of renal ENaC, resulting in certain abnormalities of the cytoplasmic tails of the β - or γ -subunit. The C termini contain PY amino acid motifs that are highly conserved, and essentially all mutations in Liddle's syndrome patients involve disruption or deletion of this motif. These PY motifs are important in regulating the number of sodium channels in the luminal membrane by binding to the WW domains of the Nedd4 (neural developmentally downregulated isoform 4)-like family of ubiquitin-protein ligases.¹⁴⁸ Disruption of the PY motif dramatically increases the surface localization of ENaC complex, because these channels are not internalized or degraded (Nedd4 pathway), but remain activated on the cell surface.¹⁴⁸ Persistent Na^+ absorption results in volume expansion, hypertension, hypokalemia, and metabolic alkalosis.

Apparent Mineralocorticoid Excess and Related Conditions. The syndrome of apparent mineralocorticoid excess is a rare autosomal recessive form of hypertension resulting from a defective enzyme 11-beta-hydroxysteroid dehydrogenase type 2 isoform (11-beta-HSD2), which is the kidney isoform of 11-beta-HSD.¹⁴⁹ This enzyme in the collecting duct normally converts cortisol to cortisone which does not activate the mineralocorticoid receptor, in contrast to cortisol which does. Thus, with a defective enzyme, circulating levels of cortisol will activate collecting duct mineralocorticoid receptors causing all the features of mineralocorticoid excess: volume expansion, low renin level, low aldosterone level, and a salt-sensitive form of hypertension, which may include metabolic alkalosis and hypokalemia. Ingestion of certain licorice products or chewing tobacco can also cause a typical pattern of hypermineralocorticoidism due to the presence of carbenoxolone or glycyrrhetic acid. These substances inhibit 11 β -hydroxysteroid dehydrogenase. The hypertension in these conditions responds to thiazides and spironolactone.

Symptoms. Symptoms of metabolic alkalosis can include changes in central and peripheral nervous system function similar to those in hypocalcemia: mental confusion,

obtundation, and a predisposition to seizures, as well as paresthesias, muscular cramping, and even tetany. Aggravation of arrhythmias and hypoxemia in chronic obstructive pulmonary disease is also a potential problem. Related electrolyte abnormalities, including hypokalemia and hypophosphatemia, are common, and patients may show symptoms of these deficiencies.

Treatment. The maintenance of metabolic alkalosis represents a failure of the kidney to excrete HCO_3^- sufficiently because of chloride or potassium deficiency or continuous mineralocorticoid activity or both. Treatment is primarily directed at correcting the underlying stimulus for HCO_3^- generation and at restoring the ability of the kidney to excrete the excess HCO_3^- . The diagnosis can be aided by assessing the urinary chloride concentration, systemic blood pressure, and the volume status of the patient (particularly the presence or absence of orthostasis) (see Fig. 16.10). Particularly helpful in the history is the presence or absence of vomiting, diuretic use, or alkali therapy. A high urine chloride level and hypertension suggest that mineralocorticoid excess is present. If primary aldosteronism is present, correction of the underlying cause (adenoma, bilateral hyperplasia, Cushing's syndrome) will reverse the alkalosis. Patients with bilateral adrenal hyperplasia may respond to spironolactone. Normotensive patients with a high urine chloride concentration may have Bartter's or Gitelman's syndrome, if diuretic use or vomiting can be excluded. A low urine chloride level and relative hypotension suggests a chloride-responsive metabolic alkalosis such as vomiting or nasogastric suction. $[\text{H}^+]$ loss by the stomach or kidneys can be mitigated by the use of proton pump inhibitors or the discontinuation of diuretics. The second aspect of treatment is to remove the factors that sustain HCO_3^- reabsorption, such as ECF volume contraction or K^+ deficiency.

Patients with CHF or unexplained volume overexpansion represent special challenges in the critical care setting. Patients with a low urine chloride concentration, usually indicative of a "chloride-responsive" form of metabolic alkalosis, may not tolerate normal saline infusion. Renal HCO_3^- loss can be accelerated by administration of acetazolamide (250 to 500 mg intravenously), a carbonic anhydrase inhibitor, if associated conditions that preclude infusion of saline (e.g., elevated pulmonary capillary wedge pressure, or evidence of CHF) are present.¹⁶ Acetazolamide is usually very effective in patients with adequate renal function, but can exacerbate urinary K^+ losses. Hypokalemia should be expected following acetazolamide IV in alkalotic patients and must be treated promptly. Dilute hydrochloric acid (0.1 normal HCl) is also effective but must be infused slowly in a central line because of severe hemolysis and is difficult to titrate. If 0.1 N HCl is used, the goal should not be to restore the pH to normal, but to reduce the pH to approximately 7.50. Patients receiving continuous renal replacement therapy in the intensive care unit typically develop metabolic alkalosis when high-bicarbonate dialysate is used or when citrate regional anticoagulation is employed. Therapy should include reduction of alkali loads via dialysis by reducing the bicarbonate concentration in the dialysate when possible. If not effective, 0.1 N HCl IV may be necessary in this setting.



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KEY REFERENCES

- Adrogué HJ, Gennari FJ, Galla JH, et al. Assessing acid-base disorders. *Kidney Int.* 2009;76(12):1239–1247.
- Bidani A, Tuazon D, Heming T. Regulation of whole body acid-base balance. In: DuBose TD, Hamm LL, eds. *Acid-Base and Electrolyte Disorders: A Companion to Brenner and Rector's The Kidney*. Philadelphia: Saunders; 2002:1–21.
- Lemann J Jr, Bushinsky DA, Hamm LL. Bone buffering of acid and base in humans. *Am J Physiol Renal Physiol.* 2003;285(5):F811–F832.
- Madias N, Adrogué H. Respiratory alkalosis. In: DuBose TD, Hamm LL, eds. *Acid-Base and Electrolyte Disorders: A Companion to Brenner and Rector's The Kidney*. Philadelphia: Saunders; 2002:147–164.
- Madias NE. Renal acidification responses to respiratory acid-base disorders. *J Nephrol.* 2010;23(suppl 16):S85–S91.
- DuBose TD Jr, Good DW. Effects of diuretics on renal acid-base transport. *Semin Nephrol.* 1988;8(3):282–294.
- Goraya N, Simoni J, Sager LN, et al. Acid retention in chronic kidney disease is inversely related to GFR. *Am J Physiol Renal Physiol.* 2018;314(5):F985–F991.
- Nath KA, Hostetter MK, Hostetter TH. Pathophysiology of chronic tubulo-interstitial disease in rats. Interactions of dietary acid load, ammonia, and complement component C3. *J Clin Invest.* 1985;76(2):667–675.
- Scialla JJ. The balance of the evidence on acid-base homeostasis and progression of chronic kidney disease. *Kidney Int.* 2015;88(1):9–11.
- Gennari FJ. Pathophysiology of metabolic alkalosis: a new classification based on the centrality of stimulated collecting duct ion transport. *Am J Kidney Dis.* 2011;58(4):626–636.
- DuBose TD. Metabolic alkalosis. In: Gilbert S, Weiner DE, eds. *National Kidney Foundation Primer on Kidney Diseases*. 6th ed. Philadelphia: Elsevier; 2013:137–143.
- Hamm LL. Mixed acid-base disorders. In: Kokko JP, Tannen R, eds. *Fluid and Electrolytes*. 3rd ed. Philadelphia: Saunders; 1996.
- Hamm LL, Hering-Smith KS, Nakhoul NL. Acid-base and potassium homeostasis. *Semin Nephrol.* 2013;33(3):257–264.
- Emmett M. Diagnosis of simple and mixed disorders. In: DuBose TD, Hamm LL, eds. *Acid-Base and Electrolyte Disorders: A Companion to Brenner and Rector's The Kidney*. Philadelphia: Saunders; 2002:41–53.
- Oh MS, Carroll HJ. The anion gap. *N Engl J Med.* 1977;297(15):814–817.
- Gabow PA, Kaehny WD, Fennessey PV, et al. Diagnostic importance of an increased serum anion gap. *N Engl J Med.* 1980;303(15):854–858.
- DuBose TD, Goode AK. Renal tubular acidosis. In: Coffma TM, Falk RJ, Molitoris BA, et al, eds. *Schrier's Diseases of The Kidney*. Philadelphia: Lippincott Williams & Wilkins; 2013:587–614.
- DuBose TD, Alpern RJ. Renal Tubular Acidosis. *The Metabolic and Molecular Bases of Inherited Disease*. New York: McGraw-Hill; 2001.
- DuBose TD, McDonald GA. Renal tubular acidosis. In: DuBose TD, Hamm LL, eds. *Acid-Base and Electrolyte Disorders: A Companion to Brenner and Rector's The Kidney*. Philadelphia: Saunders; 2002:189–206.
- Adrogué HJ, Wilson H, Boyd AE 3rd, et al. Plasma acid-base patterns in diabetic ketoacidosis. *N Engl J Med.* 1982;307(26):1603–1610.
- Kraut JA, Kurtz I. Metabolic acidosis of CKD: diagnosis, clinical characteristics, and treatment. *Am J Kidney Dis.* 2005;45(6):978–993.
- Battle D, Haque SK. Genetic causes and mechanisms of distal renal tubular acidosis. *Nephrol Dial Transplant.* 2012;27(10):3691–3704.
- Laing CM, Toye AM, Capasso G, et al. Renal tubular acidosis: developments in our understanding of the molecular basis. *Int J Biochem Cell Biol.* 2005;37(6):1151–1161.
- Kleyman TR, Roberts C, Ling BN. A mechanism for pentamidine-induced hyperkalemia: inhibition of distal nephron sodium transport. *Ann Intern Med.* 1995;122(2):103–106.
- Hoorn EJ, Walsh SB, McCormick JA, et al. The calcineurin inhibitor tacrolimus activates the renal sodium chloride cotransporter to cause hypertension. *Nat Med.* 2011;17(10):1304–1309.
- Wilson FH, Disse-Nicodeme S, Choate KA, et al. Human hypertension caused by mutations in WNK kinases. *Science.* 2001;293(5532):1107–1112.
- Hadchouel J, Ellison DH, Gamba G. Regulation of renal electrolyte transport by WNK and SPAK-OSR1 kinases. *Annu Rev Physiol.* 2016;78:367–389.
- Murthy M, Kurz T, O'Shaughnessy KM. WNK signalling pathways in blood pressure regulation. *Cell Mol Life Sci.* 2017;74(7):1261–1280.
- Boyden LM, Choi M, Choate KA, et al. Mutations in kelch-like 3 and cullin 3 cause hypertension and electrolyte abnormalities. *Nature.* 2012;482(7383):98–102.
- Kraut JA, Madias NE. Metabolic acidosis of CKD: an update. *Am J Kidney Dis.* 2016;67(2):307–317.
- Nagami GT, Hamm LL. Regulation of acid-base balance in chronic kidney disease. *Adv Chronic Kidney Dis.* 2017;24(5):274–279.
- Chen W, Abramowitz MK. Epidemiology of acid-base derangements in CKD. *Adv Chronic Kidney Dis.* 2017;24(5):280–288.
- Wesson DE, Simoni J, Broglio K, et al. Acid retention accompanies reduced GFR in humans and increases plasma levels of endothelin and aldosterone. *Am J Physiol Renal Physiol.* 2011;300(4):F830–F837.
- Vallet M, Metzger M, Haymann JP, et al. Urinary ammonia and long-term outcomes in chronic kidney disease. *Kidney Int.* 2015;88(1):137–145.
- Abramowitz MK, Melamed ML, Bauer C, et al. Effects of oral sodium bicarbonate in patients with CKD. *Clin J Am Soc Nephrol.* 2013.
- Gauthier P, Simon EE, Lemann J Jr. Acidosis of chronic renal failure. In: DuBose TD, Hamm LL, eds. *Acid-Base and Electrolyte Disorders*. Philadelphia: Saunders; 2002:207–216.
- Menon V, Tighiouart H, Vaughn NS, et al. Serum bicarbonate and long-term outcomes in CKD. *Am J Kidney Dis.* 2010;56(5):907–914.
- Navaneethan SD, Schold JD, Arrigain S, et al. Serum bicarbonate and mortality in stage 3 and stage 4 chronic kidney disease. *Clin J Am Soc Nephrol.* 2011;6(10):2395–2402.
- Loniewski I, Wesson DE. Bicarbonate therapy for prevention of chronic kidney disease progression. *Kidney Int.* 2014;85(3):529–535.
- Goraya N, Simoni J, Jo CH, et al. A comparison of treating metabolic acidosis in CKD stage 4 hypertensive kidney disease with fruits and vegetables or sodium bicarbonate. *Clin J Am Soc Nephrol.* 2013;8(3):371–381.
- Laski ME, Wesson DE. Lactic acidosis. In: DuBose TD, Hamm LL, eds. *Acid-Base and Electrolyte Disorders: A Companion to Brenner and Rector's The Kidney*. Philadelphia: Saunders; 2002.
- Whitney GM, Szerlip HM. Acid-base disorders in the critical care setting. In: DuBose TD, Hamm LL, eds. *Acid-Base and Electrolyte Disorders: A Companion to Brenner and Rector's The Kidney*. Philadelphia: Saunders; 2002:165–187.
- Halperin ML, Kamel KS, Cherney DZ. Ketoacidosis. In: DuBose TD, Hamm LL, eds. *Acid-Base and Electrolyte Disorders: A Companion to Brenner and Rector's The Kidney*. Philadelphia: Saunders; 2002:67–82.
- Peters AL, Buschur EO, Buse JB, et al. Euglycemic diabetic ketoacidosis: a potential complication of treatment with sodium-glucose cotransporter 2 inhibition. *Diabetes Care.* 2015;38(9):1687–1693.
- Kraut JA, Kurtz I. Toxic alcohol ingestions: clinical features, diagnosis, and management. *Clin J Am Soc Nephrol.* 2008;3(1):208–225.
- Brent J. Fomepizole for ethylene glycol and methanol poisoning. *N Engl J Med.* 2009;360(21):2216–2223.
- Seyberth HW, Weber S, Komhoff M. Bartter's and Gitelman's syndrome. *Curr Opin Pediatr.* 2017;29(2):179–186.
- Shaer AJ. Inherited primary renal tubular hypokalemic alkalosis: a review of Gitelman and Bartter syndromes. *Am J Med Sci.* 2001;322(6):316–332.
- Funder JW. Apparent mineralocorticoid excess. *J Steroid Biochem Mol Biol.* 2017;165(Pt A):151–153.

REFERENCES

- Adrogué HJ, Gennari FJ, Galla JH, et al. Assessing acid-base disorders. *Kidney Int.* 2009;76(12):1239–1247.
- Bidani A, Tuazon D, Heming T. Regulation of whole body acid-base balance. In: DuBose TD, Hamm LL, eds. *Acid-Base and Electrolyte Disorders: A Companion to Brenner and Rector's The Kidney*. Philadelphia: Saunders; 2002:1–21.
- Lemann J Jr, Bushinsky DA, Hamm LL. Bone buffering of acid and base in humans. *Am J Physiol Renal Physiol.* 2003;285(5):F811–F832.
- Madias N, Adrogué H. Respiratory alkalosis. In: DuBose TD, Hamm LL, eds. *Acid-Base and Electrolyte Disorders: A Companion to Brenner and Rector's The Kidney*. Philadelphia: Saunders; 2002:147–164.
- Madias NE. Renal acidification responses to respiratory acid-base disorders. *J Nephrol.* 2010;23(suppl 16):S85–S91.
- Bidani A, DuBose TD. Acid-base regulation: cellular and whole body. In: Arieff AI, DeFronzo RA, eds. *Fluid, Electrolyte, and Acid-Base Disorders*. New York: Churchill; 1995:69.
- DuBose TD Jr, Good DW. Effects of diuretics on renal acid-base transport. *Semin Nephrol.* 1988;8(3):282–294.
- Krapf R. Mechanisms of adaptation to chronic respiratory acidosis in the rabbit proximal tubule. *J Clin Invest.* 1989;83(3):890–896.
- Schwartz GJ, Al Awqati Q. Carbon dioxide causes exocytosis of vesicles containing H⁺ pumps in isolated perfused proximal and collecting tubules. *J Clin Invest.* 1985;75(5):1638–1644.
- Gennari FJ, Maddox DA. Renal regulation of acid-base homeostasis. In: Seldin DW, Giebisch G, eds. *The Kidney: Physiology and Pathophysiology*. Philadelphia: Lippincott Williams & Wilkins; 2000:2015–2054.
- Albert MS, Dell RB, Winters RW. Quantitative displacement of acid-base equilibrium in metabolic acidosis. *Ann Intern Med.* 1967;66(2):312–322.
- Goraya N, Simoni J, Sager LN, et al. Acid retention in chronic kidney disease is inversely related to GFR. *Am J Physiol Renal Physiol.* 2018;314(5):F985–F991.
- Nath KA, Hostetter MK, Hostetter TH. Pathophysiology of chronic tubulo-interstitial disease in rats. Interactions of dietary acid load, ammonia, and complement component C3. *J Clin Invest.* 1985;76(2):667–675.
- Scialla JJ. The balance of the evidence on acid-base homeostasis and progression of chronic kidney disease. *Kidney Int.* 2015;88(1):9–11.
- Gennari FJ. Pathophysiology of metabolic alkalosis: a new classification based on the centrality of stimulated collecting duct ion transport. *Am J Kidney Dis.* 2011;58(4):626–636.
- DuBose TD. Metabolic alkalosis. In: Gilbert S, Weiner DE, eds. *National Kidney Foundation Primer on Kidney Diseases*. 6th ed. Philadelphia: Elsevier; 2013:137–143.
- DuBose TD Jr, Codina J, Burges A, et al. Regulation of H(+)-K(+)-ATPase expression in kidney 582. *Am J Physiol.* 1995;269(4 Pt 2):F500–F507.
- Wesson DE. Na/H exchange and H-K ATPase increase distal tubule acidification in chronic alkalosis. *Kidney Int.* 1998;53(4):945–951.
- Wesson DE, Dolson GM. Endothelin-1 increases rat distal tubule acidification in vivo. *Am J Physiol.* 1997;273(4 Pt 2):F586–F594.
- Wesson DE. Combined K⁺ and Cl⁻ repletion corrects augmented H⁺ secretion by distal tubules in chronic alkalosis 1651. *Am J Physiol.* 1994;266(4 Pt 2):F592–F603.
- Wesson DE. Combined K⁺ and Cl⁻ repletion corrects augmented H⁺ secretion by distal tubules in chronic alkalosis. *Am J Physiol.* 1994;266(4 Pt 2):F592–F603.
- Guntupalli J, Onuigbo M, Wall S, et al. Adaptation to low-K⁺ media increases H(+)-K(+)-ATPase but not H(+)-ATPase-mediated pH_i recovery in OMCD1 cells. *Am J Physiol.* 1997;273(2 Pt 1):t71.
- Wall SM, Mehta P, DuBose TD Jr. Dietary K⁺ restriction upregulates total and Sch-28080-sensitive bicarbonate absorption in rat tMCD. *Am J Physiol.* 1998;275(4 Pt 2):F543–F549.
- Krapf R, Alpern RJ, Seldin DW. Clinical syndromes of metabolic acidosis. In: Seldin DW, Giebisch G, eds. *The Kidney*. Philadelphia: Lippincott Williams & Wilkins; 2000:2055–2072.
- Hamm LL. Mixed acid-base disorders. In: Kokko JP, Tannen R, eds. *Fluid and Electrolytes*. 3rd ed. Philadelphia: Saunders; 1996.
- Hamm LL, Hering-Smith KS, Nakhoul NL. Acid-base and potassium homeostasis. *Semin Nephrol.* 2013;33(3):257–264.
- Emmett M. Diagnosis of simple and mixed disorders. In: DuBose TD, Hamm LL, eds. *Acid-Base and Electrolyte Disorders: A Companion to Brenner and Rector's The Kidney*. Philadelphia: Saunders; 2002:41–53.
- Oh MS, Carroll HJ. The anion gap. *N Engl J Med.* 1977;297(15):814–817.
- Feldman M, Soni N, Dickson B. Influence of hypoalbuminemia or hyperalbuminemia on the serum anion gap. *J Lab Clin Med.* 2005;146(6):317–320.
- Gabow PA, Kaehny WD, Fennessey PV, et al. Diagnostic importance of an increased serum anion gap. *N Engl J Med.* 1980;303(15):854–858.
- Brower RG, Matthay MA, Morris A, et al. Ventilation with lower tidal volumes as compared with traditional tidal volumes for acute lung injury and the acute respiratory distress syndrome. *N Engl J Med.* 2000;342(18):1301–1308.
- DuBose TD, Goode AK. Renal tubular acidosis. In: Coffma TM, Falk RJ, Molitoris BA, et al, eds. *Schrier's Diseases of the Kidney*. Philadelphia: Lippincott Williams & Wilkins; 2013:587–614.
- DuBose TD, Alpern RJ. *Renal Tubular Acidosis. The Metabolic and Molecular Bases of Inherited Disease*. New York: McGraw-Hill; 2001.
- Battle DC, Hizon M, Cohen E, et al. The use of the urinary anion gap in the diagnosis of hyperchloremic metabolic acidosis. *N Engl J Med.* 1988;318(10):594–599.
- DuBose TD, McDonald GA. Renal tubular acidosis. In: DuBose TD, Hamm LL, eds. *Acid-Base and Electrolyte Disorders: A Companion to Brenner and Rector's The Kidney*. Philadelphia: Saunders; 2002:189–206.
- Adrogué HJ, Wilson H, Boyd AE 3rd, et al. Plasma acid-base patterns in diabetic ketoacidosis. *N Engl J Med.* 1982;307(26):1603–1610.
- Kraut JA, Kurtz I. Metabolic acidosis of CKD: diagnosis, clinical characteristics, and treatment. *Am J Kidney Dis.* 2005;45(6):978–993.
- Morris RC Jr, Nigon K, Reed EB. Evidence that the severity of depletion of inorganic phosphate determines the severity of the disturbance of adenine nucleotide metabolism in the liver and renal cortex of the fructose-loaded rat. *J Clin Invest.* 1978;61(1):209–220.
- DuBose TD Jr. Hydrogen ion secretion by the collecting duct as a determinant of the urine to blood PCO₂ gradient in alkaline urine. *J Clin Invest.* 1982;69(1):145–156.
- Sly WS, Whyte MP, Sundaram V, et al. Carbonic anhydrase II deficiency in 12 families with the autosomal recessive syndrome of osteopetrosis with renal tubular acidosis and cerebral calcification. *N Engl J Med.* 1985;313(3):139–145.
- Morris RC Jr. Renal tubular acidosis. Mechanisms, classification and implications. *N Engl J Med.* 1969;281(25):1405–1413.
- DuBose TD Jr, Cafilisch CR. Validation of the difference in urine and blood carbon dioxide tension during bicarbonate loading as an index of distal nephron acidification in experimental models of distal renal tubular acidosis. *J Clin Invest.* 1985;75(4):1116–1123.
- Battle D, Haque SK. Genetic causes and mechanisms of distal renal tubular acidosis. *Nephrol Dial Transplant.* 2012;27(10):3691–3704.
- Bonilla-Felix M. Primary distal renal tubular acidosis as a result of a gradient defect. *Am J Kidney Dis.* 1996;27(3):428–430.
- DuBose TD Jr, Lucci MS, Hogg RJ, et al. Comparison of acidification parameters in superficial and deep nephrons of the rat. *Am J Physiol.* 1983;244(5):F497–F503.
- Bruce LJ, Cope DL, Jones GK, et al. Familial distal renal tubular acidosis is associated with mutations in the red cell anion exchanger (Band 3, AE1) gene. *J Clin Invest.* 1997;100(7):1693–1707.
- Jarolim P, Shayakul C, Prabakaran D, et al. Autosomal dominant distal renal tubular acidosis is associated in three families with heterozygosity for the R589H mutation in the AE1 (band 3) Cl⁻/HCO₃⁻ exchanger. *J Biol Chem.* 1998;273(11):6380–6388.
- Karet FE, Finberg KE, Nelson RD, et al. Mutations in the gene encoding B1 subunit of H⁺-ATPase cause renal tubular acidosis with sensorineural deafness. *Nat Genet.* 1999;21(1):84–90.
- Karet FE, Finberg KE, Nayir A, et al. Localization of a gene for autosomal recessive distal renal tubular acidosis with normal hearing (rdRTA2) to 7q33-34. *Am J Hum Genet.* 1999;65(6):1656–1665.
- Smith AN, Finberg KE, Wagner CA, et al. Molecular cloning and characterization of Atp6n1b: a novel fourth murine vacuolar H⁺-ATPase a-subunit gene. *J Biol Chem.* 2001;276(45):42382–42388.
- Smith AN, Skaug J, Choate KA, et al. Mutations in ATP6N1B, encoding a new kidney vacuolar proton pump 116-kD subunit, cause recessive distal renal tubular acidosis with preserved hearing. *Nat Genet.* 2000;26(1):71–75.
- Kaitwatcharachai C, Vasuvattakul S, Yenchitsomanus P, et al. Distal renal tubular acidosis and high urine carbon dioxide tension in a patient with southeast Asian ovalocytosis. *Am J Kidney Dis.* 1999;33(6):1147–1152.
- DuBose TD Jr, Good DW. Role of the thick ascending limb and inner medullary collecting duct in the regulation of urinary acidification. *Semin Nephrol.* 1991;11(2):120–128.

54. DuBose TD Jr, Good DW, Hamm LL, et al. Ammonium transport in the kidney: new physiological concepts and their clinical implications. *J Am Soc Nephrol*. 1991;1(11):1193–1203.
55. Laing CM, Toye AM, Capasso G, et al. Renal tubular acidosis: developments in our understanding of the molecular basis. *Int J Biochem Cell Biol*. 2005;37(6):1151–1161.
56. Frick KK, Krieger NS, Nehrke K, et al. Metabolic acidosis increases intracellular calcium in bone cells through activation of the proton receptor OGR1. *J Bone Miner Res*. 2009;24(2):305–313.
57. Pessler F, Emery H, Dai L, et al. The spectrum of renal tubular acidosis in paediatric Sjogren syndrome. *Rheumatology (Oxford)*. 2006;45(1):85–91.
58. Morris RC Jr, Sebastian A. Alkali therapy in renal tubular acidosis: who needs it? *J Am Soc Nephrol*. 2002;13(8):2186–2188.
59. Wrong O, Henderson JE, Kaye M. Distal renal tubular acidosis: alkali heals osteomalacia and increases net production of 1,25-dihydroxyvitamin D. *Nephron Physiol*. 2005;101(3):p72–p76.
60. DuBose TD Jr, Good DW. Effects of chronic Cl depletion alkalosis on proximal tubule transport and renal production of ammonium. *Am J Physiol*. 1995;269(4 Pt 2):F508–F514.
61. DuBose TD Jr, Good DW. Chronic hyperkalemia impairs ammonium transport and accumulation in the inner medulla of the rat. *J Clin Invest*. 1992;90(4):1443–1449.
62. Watts BA III, Good DW. Effects of ammonium on intracellular pH in rat medullary thick ascending limb: mechanisms of apical membrane NH₄⁺ transport. *J Gen Physiol*. 1994;103(5):917–936.
63. DuBose TD Jr, Caflisch CR. Effect of selective aldosterone deficiency on acidification in nephron segments of the rat inner medulla. *J Clin Invest*. 1988;82(5):1624–1632.
64. Antonipillai I, Wang Y, Horton R. Tumor necrosis factor and interleukin-1 may regulate renin secretion. *Endocrinology*. 1990;126(1):273–278.
65. Caliskan Y, Kalayoglu-Besik S, Sargin D, et al. Cyclosporine-associated hyperkalemia: report of four allogeneic blood stem-cell transplant cases. *Transplantation*. 2003;75(7):1069–1072.
66. Schlanger LE, Kleyman TR, Ling BN. K(+)–sparing diuretic actions of trimethoprim: inhibition of Na⁺ channels in A6 distal nephron cells. *Kidney Int*. 1994;45(4):1070–1076.
67. Kleyman TR, Roberts C, Ling BN. A mechanism for pentamidine-induced hyperkalemia: inhibition of distal nephron sodium transport. *Ann Intern Med*. 1995;122(2):103–106.
68. Velazquez H, Perazella MA, Wright FS, et al. Renal mechanism of trimethoprim-induced hyperkalemia. *Ann Intern Med*. 1993;119(4):296–301.
69. Lea JP, Sands JM, McMahon SJ, et al. Evidence that the inhibition of Na⁺/K(+)–ATPase activity by FK506 involves calcineurin. *Kidney Int*. 1994;46(3):647–652.
70. Hoorn EJ, Walsh SB, McCormick JA, et al. The calcineurin inhibitor tacrolimus activates the renal sodium chloride cotransporter to cause hypertension. *Nat Med*. 2011;17(10):1304–1309.
71. Caramelo C, Bello E, Ruiz E, et al. Hyperkalemia in patients infected with the human immunodeficiency virus: involvement of a systemic mechanism. *Kidney Int*. 1999;56(1):198–205.
72. Geller DS, Rodriguez-Soriano J, Boado AV, et al. Mutations in the mineralocorticoid receptor gene cause renal-specific pseudohypoaldosteronism type I. *Hypertension*. 1998;32(3):21.
73. Chang SS, Grunder S, Hanukoglu A, et al. Mutations in subunits of the epithelial sodium channel cause salt wasting with hyperkalemic acidosis, pseudohypoaldosteronism type I. *Nat Genet*. 1996;12(3):248–253.
74. Grunder S, Firsov D, Chang SS, et al. A mutation causing pseudohypoaldosteronism type I identifies a conserved glycine that is involved in the gating of the epithelial sodium channel. *EMBO J*. 1997;16(5):899–907.
75. Viemann M, Peter M, Lopez-Siguero JP, et al. Evidence for genetic heterogeneity of pseudohypoaldosteronism type I: identification of a novel mutation in the human mineralocorticoid receptor in one sporadic case and no mutations in two autosomal dominant kindreds. *J Clin Endocrinol Metab*. 2001;86(5):2056–2059.
76. Adachi M, Tachibana K, Asakura Y, et al. Compound heterozygous mutations in the gamma subunit gene of ENaC (1627delG and 1570-IG→A) in one sporadic Japanese patient with a systemic form of pseudohypoaldosteronism type I. *J Clin Endocrinol Metab*. 2001;86(1):9–12.
77. Thomas CP, Zhou J, Liu KZ, et al. Systemic pseudohypoaldosteronism from deletion of the promoter region of the human Beta epithelial Na⁺ channel subunit. *Am J Respir Cell Mol Biol*. 2002;27(3):314–319.
78. Barker PM, Nguyen MS, Gatz JT, et al. Role of gammaENaC subunit in lung liquid clearance and electrolyte balance in newborn mice. Insights into perinatal adaptation and pseudohypoaldosteronism. *J Clin Invest*. 1998;102(8):1634–1640.
79. Achard JM, Disse-Nicodeme S, Fiquet-Kempf B, et al. Phenotypic and genetic heterogeneity of familial hyperkalemic hypertension (Gordon syndrome). *Clin Exp Pharmacol Physiol*. 2001;28(12):1048–1052.
80. Wilson FH, Disse-Nicodeme S, Choate KA, et al. Human hypertension caused by mutations in WNK kinases. *Science*. 2001;293(5532):1107–1112.
81. Wilson FH, Kahle KT, Sabath E, et al. Molecular pathogenesis of inherited hypertension with hyperkalemia: the Na-Cl cotransporter is inhibited by wild-type but not mutant WNK4. *Proc Natl Acad Sci USA*. 2003.
82. Hadchouel J, Ellison DH, Gamba G. Regulation of renal electrolyte transport by WNK and SPAK-OSR1 kinases. *Annu Rev Physiol*. 2016;78:367–389.
83. Murthy M, Kurz T, O'Shaughnessy KM. WNK signalling pathways in blood pressure regulation. *Cell Mol Life Sci*. 2017;74(7):1261–1280.
84. Boyden LM, Choi M, Choate KA, et al. Mutations in kelch-like 3 and cullin 3 cause hypertension and electrolyte abnormalities. *Nature*. 2012;482(7383):98–102.
85. Kahle KT, Macgregor GG, Wilson FH, et al. Paracellular Cl[−] permeability is regulated by WNK4 kinase: insight into normal physiology and hypertension. *Proc Natl Acad Sci USA*. 2004;101(41):14877–14882.
86. Kahle KT, Wilson FH, Leng Q, et al. WNK4 regulates the balance between renal NaCl reabsorption and K⁺ secretion. *Nat Genet*. 2003;35(4):372–376.
87. Grimm PR, Coleman R, Delpire E, et al. Constitutively active SPAK causes hyperkalemia by activating NCC and remodeling distal tubules. *J Am Soc Nephrol*. 2017;28(9):2597–2606.
88. Kraut JA, Madias NE. Metabolic acidosis of CKD: an update. *Am J Kidney Dis*. 2016;67(2):307–317.
89. Nagami GT, Hamm LL. Regulation of acid-base balance in chronic kidney disease. *Adv Chronic Kidney Dis*. 2017;24(5):274–279.
90. Chen W, Abramowitz MK. Epidemiology of acid-base derangements in CKD. *Adv Chronic Kidney Dis*. 2017;24(5):280–288.
91. Wesson DE, Simoni J, Broglio K, et al. Acid retention accompanies reduced GFR in humans and increases plasma levels of endothelin and aldosterone. *Am J Physiol Renal Physiol*. 2011;300(4):F830–F837.
92. Vallet M, Metzger M, Haymann JP, et al. Urinary ammonia and long-term outcomes in chronic kidney disease. *Kidney Int*. 2015;88(1):137–145.
93. Abramowitz MK, Melamed ML, Bauer C, et al. Effects of oral sodium bicarbonate in patients with CKD. *Clin J Am Soc Nephrol*. 2013.
94. Gauthier P, Simon EE, Lemann J Jr. Acidosis of chronic renal failure. In: DuBose TD, Hamm LL, eds. *Acid-Base and Electrolyte Disorders*. Philadelphia: Saunders; 2002:207–216.
95. Menon V, Tighiouart H, Vaughn NS, et al. Serum bicarbonate and long-term outcomes in CKD. *Am J Kidney Dis*. 2010;56(5):907–914.
96. Navaneethan SD, Schold JD, Arrigain S, et al. Serum bicarbonate and mortality in stage 3 and stage 4 chronic kidney disease. *Clin J Am Soc Nephrol*. 2011;6(10):2395–2402.
97. Loniewski I, Wesson DE. Bicarbonate therapy for prevention of chronic kidney disease progression. *Kidney Int*. 2014;85(3):529–535.
98. Goraya N, Simoni J, Jo CH, et al. A comparison of treating metabolic acidosis in CKD stage 4 hypertensive kidney disease with fruits and vegetables or sodium bicarbonate. *Clin J Am Soc Nephrol*. 2013;8(3):371–381.
99. Kidney Disease Improving Global Outcomes Chronic Kidney Disease Work G. KDIGO Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. *Kidney Int*. 2013;(suppl 3):1–150.
100. Sonikian MA, Pani IT, Iliopoulos AN, et al. Metabolic acidosis aggravation and hyperkalemia in hemodialysis patients treated by sevelamer hydrochloride. *Ren Fail*. 2005;27(2):143–147.
101. Wrong OM, Harland CE. Sevelamer-induced acidosis. *Kidney Int*. 2005;67:776–777, author reply 7. United States.
102. Abramowitz MK, Hostetter TH, Melamed ML. The serum anion gap is altered in early kidney disease and associates with mortality. *Kidney Int*. 2012;82(6):701–709.

103. Laski ME, Wesson DE. Lactic acidosis. In: DuBose TD, Hamm LL, eds. *Acid-Base and Electrolyte Disorders: A Companion to Brenner and Rector's The Kidney*. Philadelphia: Saunders; 2002.
104. Luft FC. Lactic acidosis update for critical care clinicians. *J Am Soc Nephrol*. 2001;12(suppl 17):S15–S19.
105. John M, Mallal S. Hyperlactatemia syndromes in people with HIV infection. *Curr Opin Infect Dis*. 2002;15(1):23–29.
106. Cote HC, Brumme ZL, Craib KJ, et al. Changes in mitochondrial DNA as a marker of nucleoside toxicity in HIV-infected patients. *N Engl J Med*. 2002;346(11):811–820.
107. Lalau JD, Race JM. Lactic acidosis in metformin therapy. *Drugs*. 1999;58(suppl 1):55–60, discussion 75–82.
108. Calabrese AT, Coley KC, DaPos SV, et al. Evaluation of prescribing practices: risk of lactic acidosis with metformin therapy. *Arch Intern Med*. 2002;162(4):434–437.
109. Romanski SA, McMahon MM. Metabolic acidosis and thiamine deficiency. *Mayo Clin Proc*. 1999;74(3):259–263.
110. Gerard Y, Maulin L, Yazdanpanah Y, et al. Symptomatic hyperlactatemia: an emerging complication of antiretroviral therapy. *AIDS*. 2000;14(17):2723–2730.
111. Jaber S, Paugam C, Futier E, et al. Sodium bicarbonate therapy for patients with severe metabolic acidemia in the intensive care unit (BICAR-ICU): a multicentre, open-label, randomised controlled, phase 3 trial. *Lancet*. 2018;392(10141):31–40.
112. Stacpoole PW, Wright EC, Baumgartner TG, et al. A controlled clinical trial of dichloroacetate for treatment of lactic acidosis in adults. The Dichloroacetate-Lactic Acidosis Study Group. *N Engl J Med*. 1992;327(22):1564–1569.
113. Bjerneroth G. Alkaline buffers for correction of metabolic acidosis during cardiopulmonary resuscitation with focus on Tribonat—a review. *Resuscitation*. 1998;37(3):161–171.
114. Uchida H, Yamamoto H, Kisaki Y, et al. D-lactic acidosis in short-bowel syndrome managed with antibiotics and probiotics. *J Pediatr Surg*. 2004;39(4):634–636.
115. Jorens PG, Demey HE, Schepens PJ, et al. Unusual D-lactic acid acidosis from propylene glycol metabolism in overdose. *J Toxicol Clin Toxicol*. 2004;42(2):163–169.
116. Lalive PH, Hadengue A, Mensi N, et al. Recurrent encephalopathy after small bowel resection. Implication of D-lactate. *Rev Neurol (Paris)*. 2001;157(6–7):679–681.
117. Whitney GM, Szerlip HM. Acid-base disorders in the critical care setting. In: DuBose TD, Hamm LL, eds. *Acid-Base and Electrolyte Disorders: A Companion to Brenner and Rector's The Kidney*. Philadelphia: Saunders; 2002:165–187.
118. Bo J, Li W, Chen Z, et al. D-lactate: a novel contributor to metabolic acidosis and high anion gap in diabetic ketoacidosis. *Clin Chem*. 2013;59:1406–1407.
119. Halperin ML, Kamel KS, Cherney DZ. Ketoacidosis. In: DuBose TD, Hamm LL, eds. *Acid-Base and Electrolyte Disorders: A Companion to Brenner and Rector's The Kidney*. Philadelphia: Saunders; 2002:67–82.
120. Umpierrez GE, DiGirolamo M, Tuvlin JA, et al. Differences in metabolic and hormonal milieu in diabetic- and alcohol-induced ketoacidosis. *J Crit Care*. 2000;15(2):52–59.
121. Peters AL, Buschur EO, Buse JB, et al. Euglycemic diabetic ketoacidosis: a potential complication of treatment with sodium-glucose cotransporter 2 inhibition. *Diabetes Care*. 2015;38(9):1687–1693.
122. Savage MW, Dhatariya KK, Kilvert A, et al. Joint British Diabetes Societies guideline for the management of diabetic ketoacidosis. *Diabet Med*. 2011;28(5):508–515.
123. Kitabchi AE, Umpierrez GE, Miles JM, et al. Hyperglycemic crises in adult patients with diabetes. *Diabetes Care*. 2009;32(7):1335–1343.
124. Oh MS, Carroll HJ, Uribarri J. Mechanism of normochloremic and hyperchloremic acidosis in diabetic ketoacidosis. *Nephron*. 1990;54(1):1–6.
125. Kraut JA, Kurtz I. Toxic alcohol ingestions: clinical features, diagnosis, and management. *Clin J Am Soc Nephrol*. 2008;3(1):208–225.
126. Gomez-Arbelaiz D, Crujeiras AB, Castro AI, et al. Acid-base safety during the course of a very low-calorie-ketogenic diet. *Endocrine*. 2017;58(1):81–90.
127. Proudfoot AT, Krenzelok EP, Brent J, et al. Does urine alkalization increase salicylate elimination? If so, why? *Toxicol Rev*. 2003;22(3):129–136.
128. Brent J. Fomepizole for ethylene glycol and methanol poisoning. *N Engl J Med*. 2009;360(21):2216–2223.
129. Fraser AD. Clinical toxicologic implications of ethylene glycol and glycolic acid poisoning. *Ther Drug Monit*. 2002;24(2):232–238.
130. Velez LI, Shepherd G, Lee YC, et al. Ethylene glycol ingestion treated only with fomepizole. *J Med Toxicol*. 2007;3(3):125–128.
131. Mizock BA, Belyaev S, Mecher C. Unexplained metabolic acidosis in critically ill patients: the role of pyroglutamic acid. *Intensive Care Med*. 2004;30(3):502–505.
132. Emmett M. Acetaminophen toxicity and 5-oxoproline (pyroglutamic acid): a tale of two cycles, one an ATP-depleting futile cycle and the other a useful cycle. *Clin J Am Soc Nephrol*. 2014;9(1):191–200.
133. Wilson KC, Reardon C, Theodore AC, et al. Propylene glycol toxicity: a severe iatrogenic illness in ICU patients receiving IV benzodiazepines: a case series and prospective, observational pilot study. *Chest*. 2005;128(3):1674–1681.
134. Gupta M, Wadhwa NK, Bukovsky R. Regional citrate anticoagulation for continuous venovenous hemodiafiltration using calcium-containing dialysate. *Am J Kidney Dis*. 2004;43(1):67–73.
135. Meier-Kriesche HU, Gitomer J, Finkel K, et al. Increased total to ionized calcium ratio during continuous venovenous hemodialysis with regional citrate anticoagulation. *Crit Care Med*. 2001;29(4):748–752.
136. Slowinski T, Morgera S, Joannidis M, et al. Safety and efficacy of regional citrate anticoagulation in continuous venovenous hemodialysis in the presence of liver failure: the Liver Citrate Anticoagulation Threshold (L-CAT) observational study. *Crit Care*. 2015;19:349.
137. Hoglund P, Haila S, Socha J, et al. Mutations of the Down-regulated in adenoma (DRA) gene cause congenital chloride diarrhoea. *Nat Genet*. 1996;14(3):316–319.
138. Aichbichler BW, Zerr CH, Santa Ana CA, et al. Proton-pump inhibition of gastric chloride secretion in congenital chloridorrhea. *N Engl J Med*. 1997;336(2):106–109.
139. Seyberth HW, Weber S, Komhoff M. Bartter's and Gitelman's syndrome. *Curr Opin Pediatr*. 2017;29(2):179–186.
140. Simon DB, Karet FE, Rodriguez-Soriano J, et al. Genetic heterogeneity of Bartter's syndrome revealed by mutations in the K⁺ channel, ROMK. *Nat Genet*. 1996;14(2):152–156.
141. Hebert SC, Gullans SR. The molecular basis of inherited hypokalemic alkalosis: Bartter's and Gitelman's syndromes. *Am J Physiol*. 1996;271(5 Pt 2):F957–F959.
142. Simon DB, Karet FE, Rodriguez-Soriano J, et al. Genetic heterogeneity of Bartter's syndrome revealed by mutations in the K⁺ channel, ROMK. *Nat Genet*. 1996;14(2):152–156.
143. Watanabe S, Fukumoto S, Chang H, et al. Association between activating mutations of calcium-sensing receptor and Bartter's syndrome. *Lancet*. 2002;360(9334):692–694.
144. Shaer AJ. Inherited primary renal tubular hypokalemic alkalosis: a review of Gitelman and Bartter syndromes. *Am J Med Sci*. 2001;322(6):316–332.
145. Monkawa T, Kurihara I, Kobayashi K, et al. Novel mutations in thiazide-sensitive Na-Cl cotransporter gene of patients with Gitelman's syndrome. *J Am Soc Nephrol*. 2000;11(1):65–70.
146. Cruz DN, Shaer AJ, Bia MJ, et al. Gitelman's syndrome revisited: an evaluation of symptoms and health-related quality of life. *Kidney Int*. 2001;59(2):710–717.
147. Dluhy RG, Anderson B, Harlin B, et al. Glucocorticoid-remediable aldosteronism is associated with severe hypertension in early childhood. *J Pediatr*. 2001;138(5):715–720.
148. Kamynina E, Staub O. Concerted action of ENaC, Nedd4-2, and Sgk1 in transepithelial Na⁺ transport. *Am J Physiol Renal Physiol*. 2002;283(3):F377–F387.
149. Funder JW. Apparent mineralocorticoid excess. *J Steroid Biochem Mol Biol*. 2017;165(Pt A):151–153.